
Clinical Study Protocol

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A Phase III, Multicentre, Randomised, Double-blind, Chronic-dosing, Parallel-group, Placebo-controlled Extension Study to Evaluate the Long-term Efficacy and Safety of Tozorakimab in Participants with Chronic Obstructive Pulmonary Disease (COPD) with a History of Exacerbations (PROSPERO)

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This CSP has been subject to a peer review according to AstraZeneca Standard procedures. The Clinical Study Protocol is publicly registered and the results are disclosed and/or published according to the AstraZeneca Global Standard - Bioethics and in compliance with prevailing laws and regulations.

Version Scope: Global

Title: Long-term Efficacy and Safety of Tozorakimab in Participants with Chronic Obstructive Pulmonary Disease with a History of Exacerbations

Study Phase: III

Acronym: PROSPERO

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SUMMARY OF CHANGES TABLE

DOCUMENT HISTORY	
Document	Date
CSP Version 4.0	01 July 2025
CSP Version 3.0	26 March 2024
CSP Version 2.0	27 July 2023
Original Protocol	09 August 2022

CSP Version 4.0, 01 July 2025

This modification is considered to be substantial based on the criteria set forth in Regulation (EU) No 536/2014 of the European Parliament.

Overall Rationale for the Amendment:

The current protocol amendment is considered substantial as a key secondary objective was updated. In addition, the amendment specifies that by the time of primary reporting of this study, dosing will be discontinued and participants in the treatment period will be asked to continue to the E/D visit 12 weeks after their final dose of IP and complete the study.

Summary of Changes:

List of Substantial Modifications

Section Number and Name	Description of Change	Brief Rationale
1.1 Synopsis (Objectives and Endpoints), 3 Objectives and Endpoints, 9.1.2 Type I Error Control, and 9.4.2.2 Secondary Endpoints	A key secondary endpoint of annualised rate of COPD exacerbations requiring hospitalisation and/or ER/ED visits has been added.	The change reflects the clinical importance of these events, thereby providing a comprehensive assessment of exacerbations.

1.1 Synopsis (Overall Design), 1.3 Schedule of Activities footnote 'a', 4.1 Overall Design, 4.4 End of Study Definition, and 7.1.1 Procedures for Discontinuation of Investigational Product	Added text that to ensure consistency with reporting PROSPERO and its predecessor studies, some participants may not complete the full 28-week treatment period. By the time of PROSPERO's primary reporting, IP dosing will be discontinued for all participants. As a result, some participants may conclude the study prior to finishing the full treatment duration. Participants will be made aware of this during the consent process. The participants who do not complete the full treatment period will be asked to attend the E/D visit, scheduled 12 weeks after their final IP dose, to complete their participation.	To ensure that the primary reporting of the predecessor studies and PROSPERO occur within a similar timeframe to allow concurrent regulatory submissions including all data to last participant last dose.
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List of Non-substantial Modifications

Section Number and Name	Description of Change	Brief Rationale
Title page	Updated error in AstraZeneca AB and AstraZeneca K.K. address and included Regulatory Agency Identifier Number.	Correction of typographical error and CSP template requirement.
1.3.1 Schedule of Activities for the 52-week Treatment Period, 1.3.2 Schedule of Activities for the 28-week Treatment Period and 9.5 Interim Analysis	Updated error in crosslinks.	Correction of typographical error.
10 Appendix A Regulatory, Ethical, and Study Oversight Considerations, A1 Regulatory and Ethical Considerations Regulatory Reporting Requirements for SAEs	Added text ‘In the European Union, the Sponsor will comply with safety reporting requirements and procedures as described in the European Clinical Trials Regulation (EU) No 536/2014. All Suspected Unexpected Serious Adverse Reactions (SUSARs) to investigational medicinal product will be reported to the EudraVigilance database within the required regulatory timelines.’	Updated based on EU-CTR guidelines for safety reporting.
Appendix D Actions Required in Cases of Increases in Liver Biochemistry and Evaluation of Hy’s Law D6 Laboratory Tests Table for Hy’s law lab kit for central laboratories	Deleted ‘IgM and IgG anti-HSV’ and added ‘IgG anti-HSV and HSV 1/2 IgM testing or HSV PCR’ and corresponding footnote.	Updated due to change in the methodology across AstraZeneca studies for testing Hy’s Law.
Appendix L Protocol Version History	Moved the summary of changes for CSP v3 26 March 2024 to the appendix.	CSP template requirement.

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1 **PROTOCOL SUMMARY**

1.1 **Synopsis**

Protocol Title: A Phase III, Multicentre, Randomised, Double-blind, Chronic-dosing, Parallel-group, Placebo-controlled Extension Study to Evaluate the Long-term Efficacy and Safety of Tozorakimab in Participants with Chronic Obstructive Pulmonary Disease (COPD) with a History of Exacerbations (PROSPERO)

Title: Long-term Efficacy and Safety of Tozorakimab in Participants with Chronic Obstructive Pulmonary Disease with a History of Exacerbations

Rationale:

This is a Phase III, multicentre, randomised, double-blind, chronic-dosing, parallel-group, placebo-controlled extension study to evaluate the long-term efficacy and safety of tozorakimab versus placebo on top of standard of care (SoC) in participants with COPD with a history of exacerbations (PROSPERO) who completed the 52-week double-blind treatment period of D9180C00003 (OBERON) or D9180C00004 (TITANIA) studies (hereafter referred to as ‘predecessor studies’). Eligible participants that are willing to continue treatment must have completed the treatment period and not been prematurely discontinued from investigational product (IP) in the predecessor studies.

This extension study is primarily designed to evaluate the efficacy of tozorakimab on severe COPD exacerbations and to characterise the long-term safety profile of tozorakimab in participants with COPD treated for longer than 1 year.

Objectives and Endpoints:

All data related to the efficacy endpoints will be combined with the 52 weeks of data obtained from the predecessor studies as part of an integrated analysis. As with the primary and key secondary objectives, other secondary objectives will also be analysed in the population of former or current smokers as well as in former smokers.

Objective	Endpoint/Estimand
Primary	
To evaluate the long-term effect of tozorakimab as an add on to SoC compared with SoC plus placebo on the rate of severe COPD exacerbations in former smokers.	Endpoint: Annualised rate of severe COPD exacerbations. Summary measure: Rate ratio. Population: Full analysis set, former smokers. Intercurrent events: Treatment policy using data regardless of treatment discontinuation during study and imputing data post re-randomisation or post-withdrawal from the study based upon reason for withdrawal.

Objective	Endpoint/Estimand
Key Secondary	
To evaluate the long-term effect of tozorakimab as an add on to SoC compared with SoC plus placebo on the rate of severe COPD exacerbations in former or current smokers.	Endpoint: Annualised rate of severe COPD exacerbations. Summary measure: Rate ratio. Population: Full analysis set, former or current smokers. Intercurrent events: Treatment policy using data regardless of treatment discontinuation during study and imputing data post re-randomisation or post-withdrawal from the study based upon reason for withdrawal.
To evaluate the long-term effect of tozorakimab as an add on to SoC compared with SoC plus placebo on the time to first severe COPD exacerbation.	Endpoint: Time to first severe COPD exacerbation. Summary measure: Hazard ratio. Population: Full analysis set, former smokers. Further key secondary analyses in former or current smokers. Intercurrent events: Treatment policy using data regardless of treatment discontinuation during study and imputing data post re-randomisation or post-withdrawal from the study based upon reason for withdrawal.
To evaluate the long-term effect of tozorakimab as an add on to SoC compared to with SoC plus placebo on COPD related health care utilisation.	Endpoint: Annualised rate of COPD exacerbations requiring hospitalisation and/or ER/ED visits. Summary measure: Rate ratio Population: Full analysis set, former smokers Further key secondary analyses in former or current smokers. Intercurrent events: Treatment policy using data regardless of treatment discontinuation during study and imputing data post re-randomisation or post-withdrawal from the study based upon reason for withdrawal.
Secondary	
To evaluate the long-term effect of tozorakimab as an add on to SoC compared with SoC plus placebo on the time to first moderate to severe COPD exacerbation.	Endpoint: Time to first moderate-to-severe COPD exacerbation. Summary measure: Hazard ratio Population: Full analysis set, former smokers. Further supportive analysis in former or current smokers. Intercurrent events: While on treatment. Data collected after the occurrence of a relevant intercurrent event (discontinuation from randomised treatment allowing for dosing interval, initiation of prohibited biologic medication) will be excluded.
To evaluate the long-term effect of tozorakimab as an add on to SoC compared with SoC plus placebo on the rate of moderate to severe COPD exacerbations.	Endpoint: Annualised rate of moderate to severe COPD exacerbations. Summary measure: Rate ratio Population: Full analysis set, former smokers. Further supportive analysis in former or current smokers. Intercurrent events: While on treatment. Data collected after the occurrence of a relevant intercurrent event (discontinuation from randomised treatment allowing for dosing interval, initiation of prohibited biologic medication) will be excluded.

Objective	Endpoint/Estimand
To evaluate the effect of tozorakimab as an add on to SoC compared with SoC plus placebo on time to all-cause death.	Endpoint: Time to all-cause death. Summary measure: Hazard ratio Population: Full analysis set, former smokers. Further supportive analysis in former or current smokers. Intercurrent events: Treatment policy using data regardless of treatment discontinuation during study and imputing data post re-randomisation or post-withdrawal from the study based upon reason for withdrawal.
To evaluate the PK and immunogenicity of tozorakimab.	- Trough serum concentrations of tozorakimab over the treatment period. - Presence of antidrug antibodies.
Safety	
To assess the long-term safety and tolerability of tozorakimab as an add on to SoC compared with SoC plus placebo.	AEs, vital signs, laboratory assessments, and ECGs. Assessments related to AEs include: - Occurrence/frequency - Relationship to IP as assessed by Investigator - Intensity - Seriousness - Death - AEs leading to discontinuation of IP - Other significant AEs

AE, adverse event(s); COPD, chronic obstructive pulmonary disease; ECG, electrocardiogram; ED, Emergency department; ER, Emergency room; IP, investigational product; PK, pharmacokinetic(s); SoC, standard of care.

For exploratory objectives and endpoints, refer to Section 3 of the clinical study protocol (CSP).

Overall Design:

This is a Phase III, multicentre, randomised, double-blind, chronic dosing, parallel-group, placebo-controlled extension study to evaluate the long-term efficacy and safety of tozorakimab in participants with COPD with a history of COPD exacerbations (PROSPERO). Pooled analysis with data from the predecessor studies will be performed to assess efficacy starting from randomisation in the predecessor studies, allowing for collection of greater numbers of severe COPD exacerbation events, given the low rates of hospitalisation due to COPD since the severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) pandemic (Alqahtani et al 2021). Given that tozorakimab is expected to be a long-term, chronic maintenance treatment, this study will also characterise the long-term safety profile of tozorakimab in COPD patients treated for longer than 1 year.

The placebo cohort is included to provide a control arm, so as to enable a continued comparison of long-term efficacy, in particular the comparison of time to first severe exacerbations and to act as a comparator to contextualise safety observations.

Participants that are willing to continue in the study and who have completed the study treatment period in one of the predecessor studies and have not been prematurely discontinued from IP may be eligible to enrol into this long-term extension (LTE) study (PROSPERO) if they fulfil the inclusion/exclusion criteria.

Visit 1 of PROSPERO will be conducted on the same day as the last scheduled visit of the treatment period in the predecessor study (Visit 15, Week 52). If this is not feasible, Visit 1 of PROSPERO may be conducted up to 12 weeks after the last dose of IP in the predecessor study (scheduled at Visit 14, Week 48), provided that the participant has remained in predecessor study follow-up during this time.

Participants who meet the eligibility criteria for PROSPERO will exit the predecessor study and receive the first dose in PROSPERO at Visit 1, otherwise, participants will continue in the predecessor study follow-up period until completion (refer to the respective predecessor study protocol).

IP will be administered via accessorised pre-filled syringe (APFS) every 4 weeks. Starting from Visit 5, participants will have an option for self-administration of IP by the participant or their caregiver.

At the end of the treatment period participants will enter 8-week post-treatment follow-up period with the final follow-up visit performed 12 weeks after the last IP dose.

The primary reporting of the study may take place once both predecessor studies have been unblinded and reported without waiting for all participants to complete treatment in this study. To ensure consistency with reporting PROSPERO and its predecessor studies, some participants may not complete the full 28-week treatment period. By the time of PROSPERO's primary reporting, IP dosing will be discontinued for all participants. As a result, some participants may conclude the study prior to finishing the full treatment duration. Participants will be made aware of this during the consent process. The participants who do not complete the full treatment period will be asked to attend the E/D visit, scheduled 12 weeks after their final IP dose, to complete their participation.

Disclosure Statement:

This is a randomised, double-blind, chronic dosing, parallel-group, placebo-controlled, extension study with 3 treatment arms that will evaluate the long-term efficacy and safety of tozorakimab in participants with COPD with a history of COPD exacerbations.

Number of Participants:

The total sample size will be determined by the number of participants completing the predecessor studies but is anticipated to be up to 2120.

Intervention Groups and Duration:

Participants will be assigned to treatment using randomisation and trial supply management (RTSM).

Under CSP Version 1.0, 50% of participants from the placebo arm in the predecessor studies will be re-randomised to active treatment, equally to tozorakimab 300 mg every 4 weeks (Q4W) or tozorakimab 300 mg every 8 weeks (Q8W), and 50% of the participants from the placebo arm will continue to receive placebo alongside SoC. All participants from active treatment arms in predecessor studies will remain on the same treatment. Tozorakimab 300 mg Q8W will have alternating matching placebo between active tozorakimab administrations. The duration of the treatment period is 52 weeks, with the last dose of IP administered at Week 48.

Under CSP Version 2.0, 50% of participants from the placebo arm in the predecessor studies will be re-randomised to tozorakimab 300 mg Q4W, and 50% of the participants from the placebo arm will continue to receive placebo alongside SoC. Participants from tozorakimab 300 mg Q4W in predecessor studies will remain on 300 mg Q4W and participants from tozorakimab 300 mg Q8W in predecessor studies will switch to tozorakimab 300 mg Q4W at the point of entry to PROSPERO. The duration of the treatment period is 28 weeks, with the last dose of IP administered at Week 24. Those participants who are already randomised under CSP Version 1.0 can continue to complete the study with the same treatment regimen as assigned and for a treatment period of 52 weeks after the amendment.

All study intervention will be supplied via blinded APFS. The re-randomisation of PROSPERO from the predecessor studies will allow more participants to receive the potentially effective experimental treatment rather than extended SoC, while retaining a comparator arm.

The primary database lock will occur after the readout of the predecessor studies. Once the blind is broken in the predecessor studies, then for those participants who had been on active treatment in the predecessor studies, the Sponsor will become unblinded to regimen. However, study conduct and blinding at the site and for the participants will remain unchanged. The study team in direct contact with sites or with influence over management of participants or recording of endpoints in PROSPERO will remain blinded and be separate from the team involved in reporting of the predecessor studies.

Independent Data Monitoring Committee:

An independent data monitoring committee (IDMC) will be established to oversee the safety of the study participants and to make appropriate recommendations regarding further study conduct based on the available data, including regular review of safety data and at a planned interim analysis where the opportunity to report on the study early for overwhelming efficacy may be considered. The committee will operate in accordance with an IDMC charter.

Adjudication Committee:

An Independent Adjudication Committee will provide an independent, external, systematic, and unbiased assessment of heart failure events (worsening and new onset), major acute cardiovascular events (MACE), and all adverse events (AEs) with fatal outcome.

Statistical Methods:

Sample Size Estimation

The number of participants entering PROSPERO is determined by the number of participants who complete the treatment period in the predecessor studies (OBERON and TITANIA) and who have not been prematurely discontinued from IP. As such, the sample size entering PROSPERO is anticipated to be up to 2120 participants, although, this is dependent upon the number of participants recruited to the tozorakimab Q8W arm and the proportion of participants who complete the predecessor study and consent to enter this extension, so the actual number may be either higher or lower. All participants randomised and dosed in the predecessor studies will be used in integrated analyses.

Primary Endpoint Analysis

The primary endpoint is the annualised rate of severe COPD exacerbations. This will be assessed for each dose of tozorakimab versus placebo primarily in the population of former smokers, although confirmatory hypothesis tests will only be conducted on the comparison of tozorakimab 300 mg Q4W to placebo.

Analyses will be conducted in the full analysis set (FAS), using all available data regardless of treatment or study discontinuation, treatment switching, or change in smoking status, aside from data after re-randomisation from placebo to active therapy. The use of this population will provide an estimate of the treatment policy estimand.

The rate ratio of severe COPD exacerbations in each tozorakimab dose regimen group relative to the placebo group will be derived using a negative binomial model. The response variable in the model will be the annualised rate of severe COPD exacerbations. The model will include treatment, predecessor study, region, maintenance inhaled therapy, and baseline COPD exacerbation history as categorical covariates, and baseline post-bronchodilator (BD) percent predicted forced expiratory volume in 1 second (FEV₁) and log blood eosinophil count as continuous covariates. The logarithm of the participant's corresponding follow-up time will be used as an offset variable in the model. For the analysis in the overall population, smoking status will also be included as a covariate.

The estimated treatment effect (ie, rate ratios) relative to the placebo comparator will be displayed along with the associated 2-sided 95% confidence interval (CI) and p-values. In addition, the model adjusted exacerbation rate in each treatment group will be presented.

Data after withdrawal from study or re-randomisation from placebo to active will be imputed using Dropout Reason-based Multiple Imputation.

Interim Analysis

An IDMC will oversee a futility analysis in the predecessor studies to assess whether the studies can be stopped for futility, and may oversee an interim analysis in this study and the predecessor studies based upon the primary endpoint in each study to assess whether the

Under CSP Version 1.0: The placebo arm will be re-randomised in a 1:1:2 ratio (tozorakimab Q4W, tozorakimab Q8W, and matching placebo, respectively).

Under CSP Version 2.0: the placebo arm will be re-randomised in a 1:1 ratio (tozorakimab Q4W and matching placebo, respectively) and participants in the tozorakimab 300 mg Q8W arm will switch to tozorakimab 300 mg Q4W. Those participants who are already randomised under CSP Version 1.0 can continue to complete the study with the same treatment regimen as assigned and for a treatment period of 52 weeks after this amendment. CSP, clinical study protocol; FU, follow-up; LTE, long-term extension; O/T, OBERON or TITANIA predecessor studies; Q4W, every 4 weeks; Q8W, every 8 weeks; SC, subcutaneous(ly); V, Visit.

1.3 Schedule of Activities

1.3.1 Schedule of Activities for the 52-Week Treatment Period

The Schedule of Activities for the 52-week treatment period applies to participants who are enrolled under CSP Version 1.0 and is provided in [Table 1](#) (see Section [4.1.2](#) for more details).

Starting from Week 16 (Visit 5 of PROSPERO), and every other visit going forward, the participants who have opted in for virtual visits, completed training, and been deemed suitable for self-administration/home visits will have an opportunity to conduct their visits as a remote telemedicine option with at-home IP self-administration or caregiver administration (refer to Section [6.2.2.3](#) for details). At-home self-administration is optional and available at Visit 5, Visit 7, Visit 9, Visit 11, and Visit 13. If the study visit is not planned to be conducted via telemedicine, then this visit should be conducted onsite.

Table 1 Schedule of Activities for the 52-Week Treatment Period

Study Stage	Enrol	Intervention (TREATMENT)														FU	E/D Followed by Phone FU ^a	CSP Section
Week	W0	W4	W8	W12	W16	W20	W24	W28	W32	W36	W40	W44	W48	W52	W60	12 weeks post last IP dose		
Visit	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12	V13	V14	V15			
Optional Telemed (T) Visit					T		T		T		T		T					
Visit Window	Same day as pred study V15 ^b	± 5 days														± 7 days		
General Assessments																		
ICF PROSPERO ^c	X																4.1.1	
Eligibility criteria	X																5.1, 5.2	
Randomisation transaction	X																4.1.1	
Clinical Assessments																		
Full physical exam																X	8.2.1	
Abbreviated physical exam	(X) ^d (> 4 wk)	X	X	X		X		X		X		X		X	X		8.2.1	
Weight	(X) ^d (> 4 wk)	X	X	X		X		X		X		X		X		X	8.2.1	
Concomitant medication	(X) ^e (≥ 1 day)	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	6.5	
Maintenance inhaled therapy use	(X) ^e (≥ 1 day)	X	X	X	X	X	X	X	X	X	X	X	X	X		X	6.5.2	
Smoking status	(X) ^e (≥ 1 day)	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	8.2.2	
Supplemental O ₂ status	(X) ^e (≥ 1 day)	X	X	X	X	X	X	X	X	X	X	X	X	X		X	8.2.1	
COPD exacerbation evaluation	(X) ^e (≥ 1 day)	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	8.1.1	

Table 1 Schedule of Activities for the 52-Week Treatment Period

Study Stage	Enrol	Intervention (TREATMENT)														FU	E/D Followed by Phone FU ^a	CSP Section
Week	W0	W4	W8	W12	W16	W20	W24	W28	W32	W36	W40	W44	W48	W52	W60	12 weeks post last IP dose		
Visit	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12	V13	V14	V15			
Optional Telemed (T) Visit					T		T		T		T		T					
Visit Window	Same day as pred study V15 ^b	± 5 days														± 7 days		
HRU	(X) ^e (≥ 1 day)	X	X	X	X	X	X	X	X	X	X	X	X	X		X	8.7	
CT ^{f, g}														X		X	8.1.4	
Pre-BD spirometry ^g								X						X		X	8.1.3, 6.5.3	
Post-BD spirometry ^{g, h}								X						X			8.1.3, 6.5.3	
Safety Assessments																		
12-lead ECG	(X) ^d (> 4 wk)			X				X						X	X	X	8.2.4	
Vital signs	(X) ^d (> 4 wk)	X	X	X		X		X		X		X		X	X	X	8.2.3	
SpO ₂	(X) ^d (> 4 wk)	X	X	X		X		X		X		X		X	X	X	8.2.3	
AE, SAE	(X) ^e (≥ 1 day)	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	8.3	
Echo- cardiography ^{g, i}														X		X	8.2.5	
Laboratory Assessments																		
Clinical chemistry (including plasma fibrinogen)	(X) ^d (> 4 wk)			X				X						X	X	X	8.2.6	

Table 1 Schedule of Activities for the 52-Week Treatment Period

Study Stage	Enrol	Intervention (TREATMENT)														FU	E/D Followed by Phone FU ^a	CSP Section
Week	W0	W4	W8	W12	W16	W20	W24	W28	W32	W36	W40	W44	W48	W52	W60	12 weeks post last IP dose		
Visit	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12	V13	V14	V15			
Optional Telemed (T) Visit					T		T		T		T		T					
Visit Window	Same day as pred study V15 ^b	± 5 days														± 7 days		
NT-proBNP	(X) ^d (> 4 wk)			X				X						X	X	X	8.2.6	
Haematology	(X) ^d (> 4 wk)			X				X						X	X	X	8.2.6	
Urinalysis	(X) ^d (> 4 wk)			X				X						X	X	X	8.2.6	
FSH (if applicable) ^j	(X)																8.2.6	
Serum pregnancy test (FOCBP only) ^k	X																8.2.6	
Urine pregnancy test (dipstick, FOCBP only) ^l	(X) ^e (≥ 1 day)	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	8.2.6	
PK ^m	(X) ^d (> 4 wk)			X				X						X	X	X	8.4.1	
Serum ADA ^m	(X) ^d (> 4 wk)			X				X						X	X	X	8.4.2	
IP Administration																		
IP administration ⁿ	X	X	X	X	X	X	X	X	X	X	X	X	X				6.2.2.2	
Training for IP self- administration ^o		X ^o	X ^o	X ^o													6.2.2.3	

- ^a The E/D Visit should be performed 12 weeks post-last dose of IP and should be performed onsite. By the time of primary reporting, IP dosing will be discontinued for all participants. As a result, some participants may conclude the study prior to finishing the full treatment duration. The participants who do not complete the full treatment period will be asked to attend the E/D visit 12 weeks post-last dose of IP. Refer to Section 7.1.1 for full details of procedures for IP discontinuation.
- ^b Visit 1 of PROSPERO will occur on the same day of the predecessor study Week 52 (Visit 15); Visit 1 in PROSPERO may, under certain circumstances, be rescheduled for a later date, but not later than 12 weeks after the last dose of IP in the predecessor study, provided the participant has remained in follow-up in that study (refer to Section 4.1.1 for details).
- ^c The ICF for PROSPERO may be provided at Week 48 (Visit 14) of the predecessor studies.
- ^d These safety-related assessments must be repeated if Visit 1 of PROSPERO is performed beyond 4 weeks from the predecessor study Visit 15 (Week 52) (Section 4.1.1).
- ^e These assessments will not be repeated if Visit 1 occurs on the same day as Visit 15 (Week 52) in the predecessor study (Section 4.1.1).
- ^f The follow-up CT (optional) will be performed only in participants with a qualified baseline CT in the predecessor study (as confirmed by the imaging vendor). For participants who discontinue IP early, it is highly recommended that CT is performed at the E/D Visit instead of at Visit 14. However, it should not be performed if less than 24 weeks have elapsed since PROSPERO enrolment. The allowance for CT is ± 2 weeks.
- ^g These procedures have to be completed as part of Visit 15 (Week 52) in the predecessor study and will not be repeated at Visit 1 in PROSPERO.
- ^h Post-BD spirometry assessment is optional and can be skipped if it cannot be tolerated by the participant or otherwise considered not feasible.
- ⁱ Echocardiography is to be performed in participants with an echocardiogram in a predecessor study or this study. During the study, additional echocardiography assessments must be performed as soon as practical if new onset or worsening of heart failure is suspected, and may also be performed for other reasons as clinically indicated.
- ^j FSH test performed only for female participants to confirm postmenopausal status in females aged < 50 years who have been amenorrhoeic for ≥ 12 months.
- ^k The sample must be collected before administration of IP. Participants can be dosed at Visit 1 based on a negative urine pregnancy test.
- ^l A serum pregnancy test should be used to confirm any positive urine pregnancy tests. Participant FOCBP who are doing optional telemedicine visits with IP self-administration (starting from Visit 5, Week 16) will be provided with urine pregnancy dipstick tests and advised to perform the test prior to IP self-administration. If the urine pregnancy test result is positive, the participant should contact the site and should not administer IP. A positive urine test result must be confirmed with a serum beta-human chorionic gonadotropin.
- ^m Samples should be collected before administration of IP.
- ⁿ IP should be administered after all other assessments have been completed at the scheduled visit. APFS will be used at the site as well as in potential at-home self or caregiver administration.
- ^o Training for optional IP self-administration should be performed for applicable participants before initiation of IP self-administration. Training should be performed at least once, and more trainings may be required at the discretion of the Investigator.

ADA, antidrug antibodies; AE, adverse event(s); APFS, accessorised pre-filled syringe; BD, bronchodilator; COPD, chronic obstructive pulmonary disease; CSP, clinical study protocol; CT, computed tomography; ECG, electrocardiogram; E/D, early discontinuation visit; Enrol, enrolment; exam, examination; FOCBP, female(s) of childbearing potential; FSH, follicle-stimulating hormone; FU, Follow-up Visit; HRU, healthcare resource utilisation; ICF, informed consent form; IP, investigational product; NT-proBNP, N-terminal pro-brain natriuretic peptide; PK, pharmacokinetic(s); Pred, predecessor; SAE, serious adverse event; SpO₂, oxygen saturation; Telemed (T), Telemedicine Visit (remote contact with the participants using telecommunications technology, including phone calls, virtual or video visits, and mobile health devices); V, Visit; W, Week; wk, weeks; X, to be performed; (X), to be repeated or performed if applicable.

1.3.2 Schedule of Activities for the 28-Week Treatment Period

The Schedule of Activities for the 28-week treatment period applies to participants who are enrolled under CSP Version 2.0 and is provided in [Table 2](#) (see Section [4.1.2](#) for more details).

Starting from Week 16 (Visit 5 of PROSPERO), and every other visit going forward, the participants who have opted in for virtual visits, completed training, and been deemed suitable for self-administration/home visits will have an opportunity to conduct their visits as a remote telemedicine option with at-home IP self-administration or caregiver administration (refer to Section [6.2.2.3](#) for details). At-home self-administration is optional and available at Visit 5 and Visit 7. If the study visit is not planned to be conducted via telemedicine, then this visit should be conducted onsite.

Table 2 Schedule of Activities for the 28-Week Treatment Period

Study Stage	Enrol	Intervention (TREATMENT)							FU	E/D Followed by Phone FU ^a	CSP Section
Week	W0	W4	W8	W12	W16	W20	W24	W28	W36	12 weeks post last IP dose	
Visit	V1	V2	V3	V4	V5	V6	V7	V8	V9		
Optional Telemed (T) Visit					T		T				
Visit Window	Same day as pred study V15 ^b	± 5 days							± 7 days		
General Assessments											
ICF PROSPERO ^c	X										4.1.1
Eligibility criteria	X										5.1, 5.2
Randomisation transaction	X										4.1.1
Clinical Assessments											
Full physical exam										X	8.2.1
Abbreviated physical exam	(X) ^d (> 4 wk)	X	X	X		X		X	X		8.2.1
Weight	(X) ^d (> 4 wk)	X	X	X		X		X		X	8.2.1
Concomitant medication	(X) ^e (≥ 1 day)	X	X	X	X	X	X	X	X	X	6.5
Maintenance inhaled therapy use	(X) ^e (≥ 1 day)	X	X	X	X	X	X	X		X	6.5.2
Smoking status	(X) ^e (≥ 1 day)	X	X	X	X	X	X	X	X	X	8.2.2
Supplemental O ₂ status	(X) ^e (≥ 1 day)	X	X	X	X	X	X	X		X	8.2.1
COPD exacerbation evaluation	(X) ^e (≥ 1 day)	X	X	X	X	X	X	X	X	X	8.1.1
HRU	(X) ^e (≥ 1 day)	X	X	X	X	X	X	X		X	8.7
CT ^{f, g}								X		X	8.1.4

Table 2 Schedule of Activities for the 28-Week Treatment Period

Study Stage	Enrol	Intervention (TREATMENT)							FU	E/D Followed by Phone FU ^a	CSP Section
Week	W0	W4	W8	W12	W16	W20	W24	W28	W36	12 weeks post last IP dose	
Visit	V1	V2	V3	V4	V5	V6	V7	V8	V9		
Optional Telemed (T) Visit					T		T				
Visit Window	Same day as pred study V15 ^b	± 5 days							± 7 days		
Pre-BD spirometry ^g								X		X	8.1.3, 6.5.3
Post-BD spirometry ^{g, h}								X			8.1.3, 6.5.3
Safety Assessments											
12-lead ECG	(X) ^d (> 4 wk)			X				X	X	X	8.2.4
Vital signs	(X) ^d (> 4 wk)	X	X	X		X		X	X	X	8.2.3
SpO ₂	(X) ^d (> 4 wk)	X	X	X		X		X	X	X	8.2.3
AE, SAE	(X) ^e (≥ 1 day)	X	X	X	X	X	X	X	X	X	8.3
Echocardiography ^{g, i}								X		X	8.2.5
Laboratory Assessments											
Clinical chemistry (including plasma fibrinogen)	(X) ^d (> 4 wk)			X				X	X	X	8.2.6
NT-proBNP	(X) ^d (> 4 wk)			X				X	X	X	8.2.6
Haematology	(X) ^d (> 4 wk)			X				X	X	X	8.2.6
Urinalysis	(X) ^d (> 4 wk)			X				X	X	X	8.2.6
FSH (if applicable) ^j	(X)										8.2.6

Table 2 Schedule of Activities for the 28-Week Treatment Period

Study Stage	Enrol	Intervention (TREATMENT)							FU	E/D Followed by Phone FU ^a	CSP Section
Week	W0	W4	W8	W12	W16	W20	W24	W28	W36	12 weeks post last IP dose	
Visit	V1	V2	V3	V4	V5	V6	V7	V8	V9		
Optional Telemed (T) Visit					T		T				
Visit Window	Same day as pred study V15 ^b	± 5 days								± 7 days	
Serum pregnancy test (FOCBP only) ^k	X										8.2.6
Urine pregnancy test (dipstick, FOCBP only) ^l	(X) ^e (≥ 1 day)	X	X	X	X	X	X	X	X	X	8.2.6
PK ^m	(X) ^d (> 4 wk)			X				X	X	X	8.4.1
Serum ADA ^m	(X) ^d (> 4 wk)			X				X	X	X	8.4.2
IP Administration											
IP administration ⁿ	X	X	X	X	X	X	X				6.2.2.2
Training for IP self-administration ^o		X ^o	X ^o	X ^o							6.2.2.3

- ^a The E/D Visit should be performed 12 weeks post-last dose of IP and should be performed onsite. By the time of primary reporting, IP dosing will be discontinued for all participants. As a result, some participants may conclude the study prior to finishing the full treatment duration. The participants who do not complete the full treatment period will be asked to attend the E/D visit 12 weeks post-last dose of IP. Refer to Section 7.1.1 for full details of procedures for IP discontinuation.
- ^b Visit 1 of PROSPERO will occur on the same day of the predecessor study Week 52 (Visit 15); Visit 1 in PROSPERO may, under certain circumstances, be rescheduled for a later date, but not later than 12 weeks after the last dose of IP in the predecessor study, provided the participant has remained in follow-up in that study (refer to Section 4.1.1 for details).
- ^c The ICF for PROSPERO may be provided at Week 48 (Visit 14) of the predecessor studies.
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- ^e These assessments will not be repeated if Visit 1 occurs on the same day as Visit 15 (Week 52) in the predecessor study (Section 4.1.1).
- ^f The follow-up CT (optional) will be performed only in participants with a qualified baseline CT in the predecessor study (as confirmed by the imaging vendor). For participants who discontinue IP early, it is highly recommended that CT is performed at the E/D Visit instead of at Visit 8. However, it should not be performed if less than 24 weeks have elapsed since PROSPERO enrolment. The allowance for CT is ± 2 weeks.
- ^g These procedures have to be completed as part of Visit 15 (Week 52) in the predecessor study and will not be repeated at Visit 1 in PROSPERO.

- ^h Post-BD spirometry assessment is optional and can be skipped if it cannot be tolerated by the participant or otherwise considered not feasible.
- ⁱ Echocardiography is to be performed in participants with an echocardiogram in a predecessor study or this study. During the study, additional echocardiography assessments must be performed as soon as practical if new onset or worsening of heart failure is suspected, and may also be performed for other reasons as clinically indicated.
- ^j FSH test performed only for female participants to confirm post-menopausal status in females aged < 50 years who have been amenorrhoeic for ≥ 12 months.
- ^k The sample must be collected before administration of IP. Participants can be dosed at Visit 1 based on a negative urine pregnancy test.
- ^l A serum pregnancy test should be used to confirm any positive urine pregnancy tests. Participant FOCBP who are doing optional telemedicine visits with IP self-administration (starting from Visit 5, Week 16) will be provided with urine pregnancy dipstick tests and advised to perform the test prior to IP self-administration. If the urine pregnancy test result is positive, the participant should contact the site and should not administer IP. A positive urine test result must be confirmed with a serum beta-human chorionic gonadotropin.
- ^m Samples should be collected before administration of IP.
- ⁿ IP should be administered after all other assessments have been completed at the scheduled visit. APFS will be used at the site as well as in potential at-home self or caregiver administration.
- ^o Training for optional IP self-administration should be performed for applicable participants before initiation of IP self-administration. Training should be performed at least once, and more trainings may be required at the discretion of the Investigator.

ADA, antidrug antibodies; AE, adverse event(s); APFS, accessorised pre-filled syringe; BD, bronchodilator; COPD, chronic obstructive pulmonary disease; CSP, clinical study protocol; CT, computed tomography; ECG, electrocardiogram; E/D, early discontinuation visit; Enrol, enrolment; exam, examination; FOCBP, female(s) of childbearing potential; FSH, follicle-stimulating hormone; FU, Follow-up Visit; HRU, healthcare resource utilisation; ICF, informed consent form; IP, investigational product; NT-proBNP, N-terminal pro-brain natriuretic peptide; PK, pharmacokinetic(s); Pred, predecessor; SAE, serious adverse event; SpO₂, oxygen saturation; Telemed (T), Telemedicine Visit (remote contact with the participants using telecommunications technology, including phone calls, virtual or video visits, and mobile health devices); V, Visit; W, Week; wk, weeks; X, to be performed; (X), to be repeated or performed if applicable.

2 INTRODUCTION

Tozorakimab is a human immunoglobulin G1 mAb that potently and specifically binds to IL-33 and blocks its signalling. Tozorakimab is being developed for the treatment of asthma, COPD, diabetic kidney disease, and severe viral lung infection. A Phase II study has been conducted in atopic dermatitis, but development in this indication has since been terminated. The mechanism of action of tozorakimab makes it a potential treatment option for the high unmet need in patients with COPD experiencing COPD symptom burden and at risk for exacerbations.

2.1 Study Rationale

Tozorakimab is hypothesised to have a beneficial effect in patients with COPD by reducing the frequency and severity of COPD exacerbations and improving COPD symptoms and health-related quality of life. Furthermore, a subsequent improvement is hypothesised for lung function as well. Neutralisation of IL-33 activity with tozorakimab has potential to disrupt the cycle of inflammatory structural damage in lungs of patients with COPD.

Two pivotal Phase III studies in COPD started in early 2022 (Study D9180C00003 [OBERON] and Study D9180C00004 [TITANIA]). Both studies enrol adults with symptomatic COPD and history of COPD exacerbations; each study will recruit up to 424 participants in the 300 mg Q4W and placebo treatment arms and approximately 212 participants in the 300 mg Q8W arm (refer to Section 9.2 for details) and treat them for a period of 52 weeks. The primary endpoint of both studies is the annualised rate of moderate to severe COPD exacerbations, analysed in the Primary Population of former smokers, followed by an analysis in the Overall Population of former or current smokers. Both studies have identical protocols but are conducted in different countries.

The purpose of this Phase III study (PROSPERO) is to continue the evaluation of long-term efficacy and safety of tozorakimab in adult participants with symptomatic COPD completing the predecessor studies OBERON or TITANIA.

It is expected that this study, in conjunction with the predecessor studies, will provide sufficient severe exacerbation events to be able to demonstrate efficacy of tozorakimab versus placebo on top of SoC in reducing severe COPD exacerbations (time to first hospitalisation and death due to COPD). Pooled analysis with data from the predecessor studies will be performed to assess efficacy starting from randomisation in the predecessor studies, allowing for collection of greater numbers of severe COPD exacerbation events, given that rates of hospitalisation have been low since the SARS-CoV-2 pandemic ([Alqahtani et al 2021](#)). This study will also assess the long-term safety profile of tozorakimab in COPD patients to allow for further characterisation of the benefit-risk profile. The placebo cohort is included to provide a control arm, so as to enable a continued comparison of long-term efficacy, in particular the comparison of severe exacerbations rates, and to act as a comparator to contextualise safety observations. However, the size of the placebo cohort will be

substantially reduced comparing to the predecessor studies: 50% of placebo participants will be re-randomised to active treatment (refer to Section 4.1 for details).

2.2 Background

Chronic obstructive pulmonary disease is the fourth leading cause of death in the world and is projected to be the third leading cause of death worldwide by 2030 ([Adeloye et al 2015](#)). Chronic obstructive pulmonary disease is characterised by persistent respiratory symptoms and airflow limitation (post-BD FEV₁/FVC < 0.70) that is due to airway and/or alveolar abnormalities, usually caused by significant exposure to noxious particles or gases (where tobacco smoke is dominating), and influenced by host factors including abnormal lung development. Significant comorbidities may have an impact on morbidity and mortality ([GOLD 2022](#)). Chronic obstructive pulmonary disease is a life-threatening respiratory condition, and the disease course is characterised by a long period of disability, which results in considerable loss of health-related quality of life and restrictions in daily life ([van Manen et al 2003](#)).

Chronic obstructive pulmonary disease is not fully reversible, usually progressive, and associated with an enhanced chronic inflammatory response in the lung. There is an increasing evidence base demonstrating that the overall symptomatic burden has a substantial detrimental impact on health-related quality of life and also contributes to increased risk of exacerbations and a worse disease prognosis ([Miravittles and Ribera 2017](#)).

Acute exacerbations of COPD are episodes of symptom worsening that have significant adverse consequences for patients ([Wedzicha and Seemungal 2007](#)). Greater frequency of exacerbations is associated with accelerated lung function decline, health-related quality of life impairment, and increased mortality ([Donaldson et al 2002](#), [Seemungal et al 1998](#), [Soler-Cataluna et al 2005](#)). Furthermore, as the incidence of COPD increases, exacerbations place a greater burden on health care systems, accounting for more than 10 million unscheduled attendances per year in the United States ([Mannino and Braman 2007](#)). The direct costs of COPD treatment in the United States are greater than \$32 billion per year, with exacerbations estimated to account for 50% to 75% of these healthcare costs ([Celli et al 2004](#), [Guarascio et al 2013](#), [Toy et al 2010](#)). Exacerbations are also important outcome measures in COPD, with acute treatment targeting accelerated recovery, whereas long-term maintenance inhaled therapy is designed to prevent and reduce their frequency and severity ([Ritchie and Wedzicha 2020](#)).

Despite adequate treatment with optimised maintenance inhaled therapy, approximately 30% to 40% of patients continue to have moderate or severe exacerbations ([Müllerová et al 2017](#), [Vestbo et al 2017](#)). Severe exacerbations of COPD worsen the course of COPD disease and involve a rapid decline in health status, disease progression, and high mortality, with each severe COPD exacerbation increasing the risk of further exacerbations ([Wedzicha et al 2013](#)),

re-hospitalisation (Myers et al 2020), and death (Suisse 2012). Even maximal triple therapy (LABA + LAMA + ICS) may still be insufficient to prevent all such events (Rabe et al 2020); hence, a substantial unmet medical need remains.

Tozorakimab is a fully human IgG1 mAb that binds to IL-33 and potently and specifically blocks all forms of IL-33 to prevent their signalling (Kearley et al 2015). By binding to the reduced form of IL-33 (IL-33^{red}), tozorakimab potently inhibits ST2-dependent inflammation responses in several primary human cells and in an allergen-driven in vivo model of COPD. The oxidised form of IL-33 (IL-33^{ox}), which cannot bind ST2 (Cohen et al 2015), signals via the RAGE/EGFR pathway (Scott et al 2022, Strickson et al 2022). Tozorakimab cannot bind IL-33^{ox} but can prevent the oxidation of IL-33 and IL-33^{ox}-dependent signalling via the RAGE/EGFR complex and mimic the action of soluble ST2. Inhibition of IL-33^{ox} signalling by tozorakimab can improve airway epithelial repair functions and reverse airway epithelial dysfunction in respiratory disease, including mucous hypersecretion (Strickson et al 2022, Scott et al 2022).

Neutralisation of IL-33 may disrupt the cycle of inflammatory damage to allow epithelial recovery. Recent clinical data reveal a significant beneficial effect of inhibiting IL-33 on reducing moderate-to-severe COPD exacerbations on top of optimised inhaled therapy (Rabe et al 2021).

A detailed description of the chemistry, pharmacology, mechanism of action, and safety of tozorakimab is provided in the IB.

2.3 Benefit/Risk Assessment

In order to evaluate the clinical benefit/risk balance for tozorakimab, nonclinical and clinical data were taken into consideration, as well as review of the available information for mAbs that are in development for the treatment of COPD.

More detailed information about the known and expected benefits and potential risks of tozorakimab may be found in the IB.

2.3.1 Risk Assessment

Potential risks of tozorakimab are the following (refer to the IB for further details):

- The IL-33/ST2 axis is involved in the host response to pathogens, however, the relationship between IL-33 suppression and serious infection has not been fully established. Therefore, serious infections, including opportunistic infections, are considered a theoretical risk of tozorakimab. Risk minimisation measures include exclusion of participants in a predecessor study with conditions predisposing to infections (eg, known primary or secondary immune deficiency syndromes) or recent severe infections. No significant increase in the incidence of serious or opportunistic infections

with tozorakimab has been observed in the Phase I study or Phase II studies reported to date in atopic dermatitis and asthma.

Note: The Sponsor is not aware of any current evidence of a direct link between IL-33, or IL33 suppression, and contracting SARS-CoV-2 and early stages- of COVID-19 illness post-infection. It has been hypothesised that IL-33 suppression might be beneficial in the later, hyperinflammatory phase of severe COVID-19 illness ([Stanczak et al 2021](#)). Any additional infection control measures in accordance with COVID-19-related local and national guidelines should also be followed.

- Progression of heart failure is considered a theoretical potential risk of tozorakimab. Preclinical data exist for both protective and detrimental effect of IL-33/ST2 pathway in animal models of heart failure and endothelial dysfunction ([Parikh et al 2016](#)). Individuals that express an IL-33 loss-of-function variant (rs146597587-C) do not appear to have an increased risk of developing heart failure in 3 separate cohorts (unpublished observations of the Sponsor), and homozygotes for this rare IL-33 variant do not show reduced life expectancy nor reduced fertility ([Smith et al 2017](#)). There were no cases of heart failure in the completed Phase I study of tozorakimab Q2W or the final analyses of the Phase II studies evaluating the use of tozorakimab for atopic dermatitis and asthma. Mitigation of progression of heart failure progression risk includes exclusion of participants with any unstable cardiovascular disease, heart failure of NYHA class III-IV, and/or LVEF < 30% in the predecessor studies, as well as echocardiography monitoring in participants with heart failure of NYHA class I-II and/or high risk, ECG and NT-proBNP monitoring throughout the study.
- Serious hypersensitivity (including Type 1 to 4 hypersensitivity reaction): As mAbs are foreign proteins, they have the potential to provoke hypersensitivity reactions. In order to mitigate the risk of serious hypersensitivity reactions, participants are excluded from the predecessor study if they have a history of severe allergic reaction requiring hospitalisation.
- Injection-site reactions are a potential risk based on the SC route of administration and the nature of tozorakimab as an exogenous protein substance. Mitigations include instructions to the site personnel on injection technique, rotation of injection sites, and routine injection site assessment at each visit.
- An incidental finding of an intestinal AE (colitis) was observed in the long-term toxicity preclinical study of tozorakimab. To mitigate any potential risks, participants with recent history of active inflammatory bowel diseases are excluded from the predecessor study. Gastrointestinal AEs are subject to close safety monitoring during the study.

Risks of the study procedures are as follows:

- Forced spirometry may generate aerosols with risk of spreading respiratory infectious disease. The risk would be to staff and other participants nearby, rather than the participant undergoing the procedure. Usual infection control measures should be followed during procedures, including use of filters during forced spirometry, per the instructions of the central vendor. Any additional infection control measures in accordance with COVID-19-related local and national guidelines should also be followed.

- The ionising radiation from low dose CT may cause a very small increased risk of developing cancer in participants who also may have an elevated risk of cancer due to smoking history. Participants will be informed of the risks associated with CT before entering the study, and not every participant in the study will receive a CT (Section 8.1.4). In keeping with the ALARA concept, (As Low as Reasonably Achievable) the trial recommends using optimised CT scan protocols, using the lowest possible radiation dose to each participant in order to provide acceptable image quality for the quantitative measurements performed in the study. For more detailed information related to CT radiation dose, imaging risks, and best imaging practices, please refer to the study Radiology Manual.
- Risk minimisation measures include the exclusion of participants in a predecessor study and this study with allergy or reaction to any component of the IP formulation, active or recent malignancy, unstable cardiovascular comorbidities, inflammatory bowel disease, and active significant infection, in conjunction with the performance of routine pharmacovigilance activities. Additionally, an unblinded IDMC and a blinded Medical Monitor will perform regular evaluations of safety data from this study (Section 9.6). Additionally, an IDMC will perform regular evaluations of safety data throughout the predecessor studies.

2.3.2 Benefit Assessment

The purpose of this Phase III study is to evaluate the long-term efficacy and safety of tozorakimab in participants with symptomatic COPD who completed the treatment period in predecessor studies OBERON or TITANIA. Participants will be required to continue on stable doses of their COPD maintenance therapy throughout the study to minimise the risk of disease worsening during the study and will receive medically indicated treatment for COPD exacerbations. For participants randomised to tozorakimab, there is a potential benefit in terms of reduction in COPD exacerbations, improvement in COPD symptoms and health-related quality of life. A recent Phase II clinical trial showed that treatment with itepekimab (formerly REGN3500/SAR440340), a competitor mAb binding IL-33 with a very similar mechanism of action to tozorakimab, resulted in a marked reduction of COPD exacerbations as well as a significant improvement of FEV₁ in former smokers (Rabe et al 2021). While the clinical effect of tozorakimab in patients with COPD is yet to be determined, it is conceivable that treatment with tozorakimab will lead to similar improvement. However, it is possible that participants will not receive any individual benefit from participating in this study.

2.3.3 Overall Benefit:Risk Conclusion

As of May 2023, an estimated 1500 participants have been dosed with tozorakimab across multiple indications. Safety data collected to date, including data from the Phase II studies of tozorakimab in asthma and atopic dermatitis, support the favourable safety profile of tozorakimab for all dose regimens evaluated in the PROSPERO study.

Taking into account the measures taken to minimise risk to participants in this study, the benefit/risk assessment for tozorakimab in participants with symptomatic COPD and a history of exacerbations appears favourable.

More detailed information about the known and expected benefits and risks and safety profile of tozorakimab can be found in the IB.

3 OBJECTIVES AND ENDPOINTS

All data related to the efficacy endpoints will be combined together with the 52 weeks of data obtained from the predecessor studies as part of an integrated analysis.

Table 3 Objectives and Endpoints

Objective	Endpoint/Estimand
Primary	
To evaluate the long-term effect of tozorakimab as an add on to SoC compared with SoC plus placebo on the rate of severe COPD exacerbations in former smokers.	Endpoint: Annualised rate of severe COPD exacerbations. Summary measure: Rate ratio. Population: Full analysis set, former smokers. Intercurrent events: Treatment policy using data regardless of treatment discontinuation during study and imputing data post re-randomisation or post-withdrawal from the study based upon reason for withdrawal.
Key Secondary	
To evaluate the long-term effect of tozorakimab as an add on to SoC compared with SoC plus placebo on the rate of severe COPD exacerbations in former or current smokers.	Endpoint: Annualised rate of severe COPD exacerbations. Summary measure: Rate ratio. Population: Full analysis set, former or current smokers. Intercurrent events: Treatment policy using data regardless of treatment discontinuation during study and imputing data post re-randomisation or post-withdrawal from the study based upon reason for withdrawal.
To evaluate the long-term effect of tozorakimab as an add on to SoC compared with SoC plus placebo on the time to first severe COPD exacerbation.	Endpoint: Time to first severe COPD exacerbation. Summary measure: Hazard ratio. Population: Full analysis set, former smokers. Further key secondary analyses in former or current smokers. Intercurrent events: Treatment policy using data regardless of treatment discontinuation during study and imputing data post re-randomisation or post-withdrawal from the study based upon reason for withdrawal.
To evaluate the long-term effect of tozorakimab as an add on to SoC compared to with SoC plus placebo on COPD related health care utilisation.	Endpoint: Annualised rate of COPD exacerbations requiring hospitalisation and/or ER/ED visits. Summary measure: Rate ratio Population: Full analysis set, former smokers Further key secondary analyses in former or current smokers. Intercurrent events: Treatment policy using data regardless of treatment discontinuation during study and imputing data post re-randomisation or post-withdrawal from the study based upon reason for withdrawal.

Table 3 Objectives and Endpoints

Objective	Endpoint/Estimand
Secondary	
To evaluate the long-term effect of tozorakimab as an add on to SoC compared with SoC plus placebo on the time to first moderate to severe COPD exacerbation.	Endpoint: Time to first moderate-to-severe COPD exacerbation. Summary measure: Hazard ratio Population: Full analysis set, former smokers. Further supportive analysis in former or current smokers. Intercurrent events: While on treatment. Data collected after the occurrence of a relevant intercurrent event (discontinuation from randomised treatment allowing for dosing interval, initiation of prohibited biologic medication) will be excluded.
To evaluate the long-term effect of tozorakimab as an add on to SoC compared with SoC plus placebo on the rate of moderate to severe COPD exacerbations.	Endpoint: Annualised rate of moderate to severe COPD exacerbations. Summary measure: Rate ratio Population: Full analysis set, former smokers. Further supportive analysis in former or current smokers. Intercurrent events: While on treatment. Data collected after the occurrence of a relevant intercurrent event (discontinuation from randomised treatment allowing for dosing interval, initiation of prohibited biologic medication) will be excluded.
To evaluate the effect of tozorakimab as an add on to SoC compared with SoC plus placebo on time to all-cause death.	Endpoint: Time to all-cause death. Summary measure: Hazard ratio Population: Full analysis set, former smokers. Further supportive analysis in former or current smokers. Intercurrent events: Treatment policy using data regardless of treatment discontinuation during study and imputing data post re-randomisation or post-withdrawal from the study based upon reason for withdrawal.
To evaluate the PK and immunogenicity of tozorakimab.	• Trough serum concentrations of tozorakimab over the treatment period. • Presence of antidrug antibodies.
Safety	
To assess the long-term safety and tolerability of tozorakimab as an add on to SoC compared with SoC plus placebo.	AEs, vital signs, laboratory assessments, and ECGs. Assessments related to AEs include: • Occurrence/frequency. • Relationship to IP as assessed by Investigator. • Intensity. • Seriousness. • Death. • AEs leading to discontinuation of IP. • Other significant AEs.

Table 3 Objectives and Endpoints

Objective	Endpoint/Estimand
Exploratory^a	
To evaluate the long-term effect of tozorakimab as an add on to SoC compared with SoC plus placebo on measures of lung function.	• Change from baseline over the study period in trough FEV₁ (mL). • Change from baseline over the study period in post-BD FEV₁ (mL). • Rate of decline in trough FEV₁ (mL) over the study period. • Rate of decline in post-BD FEV₁ (mL) over the study period.
To evaluate the long-term effect of tozorakimab as an add on to SoC compared with SoC plus placebo on mucous plugging, airway structure and function, and emphysema progression.	Change from baseline in: • Airway wall and lumen dimensions (percent wall area, wall area, wall thickness, and lumen area) • ATI and E/I • Disease probability and parametric response mapping (fEmph and fSAD) • Volume-adjusted PD15 • Mucous score
To evaluate the effect of tozorakimab as an add on to SoC compared with SoC plus placebo on overall and COPD-related healthcare resource utilisation.	Overall and COPD-related HRU data: • Annualised rate of hospitalisations. • Annualised rate of ER/ED visits. • Annualised rate of hospitalisations and ER/ED visits combined. • Number of days in hospital, as measured by total number of days in hospital, general ward, ICU, and CCU. • Rate of hospital readmissions.
To evaluate the effect of tozorakimab as an add on to SoC compared with SoC plus placebo on time to respiratory-related death.	Time to respiratory-related death.

AE, adverse event(s); ATI, air trapping index; BD, bronchodilator; CCU, critical care unit; COPD, chronic obstructive pulmonary disease; CSR, clinical study report; CT, computed tomography; ECG, electrocardiogram; ED, emergency department; E/I, expiratory/inspiratory density ratio; ER, emergency room; FEV₁, forced expiratory volume in 1 second; ICU, intensive care unit; HRU, healthcare resource utilisation; IP, investigational product; PD, 15th percentile in the CT lung density histogram; PK, pharmacokinetic(s); SoC, standard of care.

Note: As with the primary and secondary objectives, other exploratory objectives will also be analysed in the population of former or current smokers as well as in former smokers.

^a Exploratory endpoints may be reported outside of the CSR. qCT endpoints will be reported outside of the CSR using integrated analyses in order for sufficient images to be available.

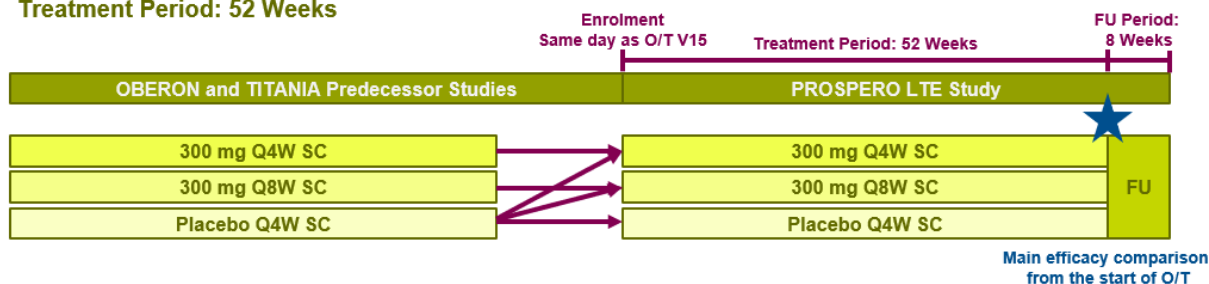
4 STUDY DESIGN

4.1 Overall Design

Figure 2 Study Design

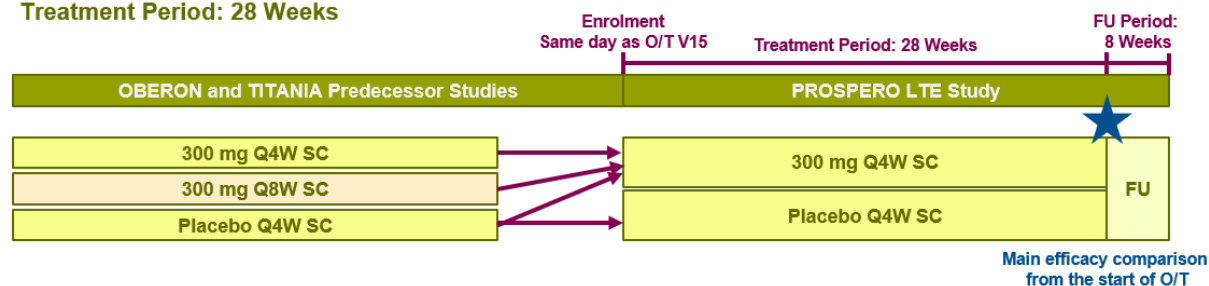
CSP Version 1

Treatment Period: 52 Weeks



CSP Version 2

Treatment Period: 28 Weeks



Note: Refer to Section 4.1.1 if Visit 1 of PROSPERO is not conducted on the same day as Visit 15 of the predecessor studies.

Under CSP Version 1.0: The placebo arm will be re-randomised in a 1:1:2 ratio (tozorakimab Q4W, tozorakimab Q8W, and matching placebo, respectively).

Under CSP Version 2.0: the placebo arm will be re-randomised in a 1:1 ratio (tozorakimab Q4W and matching placebo, respectively) and participants in the tozorakimab 300 mg Q8W arm will switch to tozorakimab 300 mg Q4W. Those participants who are already randomised under CSP Version 1.0 can continue to complete the study with the same treatment regimen as assigned and for a treatment period of 52 weeks after this amendment.

CSP, clinical study protocol; FU, follow-up; LTE, long-term extension; O/T, OBERON or TITANIA predecessor studies; Q4W, every 4 weeks; Q8W, every 8 weeks; SC, subcutaneous(ly); V, Visit.

This is a Phase III, multicentre, randomised, double-blind, chronic-dosing, parallel-group, placebo-controlled extension study to evaluate the long-term efficacy and safety of tozorakimab 300 mg Q4W and 300 mg Q8W administered SC, in participants with COPD with history of COPD exacerbations.

Eligible participants must have completed the treatment period and have not prematurely discontinued from IP in the predecessor studies OBERON or TITANIA and fulfil the inclusion/exclusion criteria. Participants will be encouraged to continue on the same stable COPD maintenance therapy throughout the study.

It is anticipated that PROSPERO will enrol up to 2120 participants.

The study design is updated in this amendment; participants will therefore be enrolled in the study prior to the amendment under CSP version 1.0 or under this version (Version 2.0).

CSP Version 1.0

Participants previously randomised to tozorakimab in either of the predecessor studies will remain on the same active treatment and dosing regimen in PROSPERO. Participants previously randomised to the placebo arm in the predecessor study will be re-randomised in a 1:1:2 ratio to receive either tozorakimab 300 mg Q4W, tozorakimab 300mg Q8W, or placebo on top of SoC.

The study will consist of an Enrolment Visit, preferably on the same day as Visit 15 of the predecessor study, and a 52-week treatment period in which IP will be administered Q4W. The last dose of IP will be administered at Week 48 and the end of treatment assessments will be performed at Week 52. Participants will enter the 8-week post-treatment follow-up, with the final follow-up visit at Week 60 (12 weeks after the last dose of IP).

CSP Version 2.0

Participants previously randomised to the placebo arm in the predecessor studies will be re-randomised in a 1:1 ratio to receive either tozorakimab 300 mg Q4W or placebo on top of SoC in PROSPERO. Participants previously randomised to tozorakimab 300 mg Q4W in the predecessor studies will remain on tozorakimab 300 mg Q4W. Participants previously randomised to tozorakimab 300 mg Q8W in the predecessor studies will switch to tozorakimab 300 mg Q4W at the point of entry to PROSPERO. Those participants who are already randomised under CSP Version 1.0 can continue to complete the study with the same treatment regimen as assigned and for a treatment period of 52 weeks after this amendment.

The study will consist of an Enrolment Visit, preferably on the same day as Visit 15 of the predecessor study, and a 28-week treatment period in which IP will be administered Q4W. The last dose of IP will be administered at Week 24 and the end of treatment assessments will be performed at Week 28. Participants will enter the 8-week post-treatment follow-up, with the final follow-up visit at Week 36 (12 weeks after the last dose of IP).

The placebo cohort is included to provide a control arm, so as to enable a continued comparison of long-term efficacy, in particular the comparison of severe exacerbations rates, and to act as a comparator to contextualise safety observations.

The Sponsor reserves the right to extend the PROSPERO study or to terminate it early depending on the overall development programme.

The primary reporting of the study may occur once both predecessor studies have been unblinded and reported without waiting for all participants to complete treatment in this study. To ensure consistency with reporting PROSPERO and its predecessor studies, some participants may not complete the full 28-week treatment period. By the time of PROSPERO's primary reporting, IP dosing will be discontinued for all participants. As a result, some participants may conclude the study prior to finishing the full treatment duration. Participants will be made aware of this during the consent process. The participants who do not complete the full treatment period will be asked to attend the E/D visit, scheduled 12 weeks after their final IP dose, to complete their participation.

Once the blind is broken in the predecessor studies, the Sponsor will be unblinded to which participants received active tozorakimab in those studies. As such until the primary analysis of PROSPERO, the study team in direct contact with sites or with influence over management of participants or recording of endpoints in PROSPERO will remain blinded and be separate from the team involved in reporting of the predecessor studies. The study conduct and blinding at the site and for the participants will remain unchanged.

Independent Data Monitoring Committee

An IDMC will be established to oversee the safety of the study participants and make appropriate recommendations regarding further study conduct based on the available data, including regular review of safety data and at planned interim analyses where the opportunity to report on the study early for overwhelming efficacy may be considered. The committee will operate in accordance with an IDMC charter.

Adjudication Committee

An Independent Adjudication Committee will provide an independent, external, systematic, and unbiased assessment of heart failure events (worsening and new onset), MACE, and all AEs with fatal outcome.

4.1.1 Enrolment

Visit 1 of PROSPERO will be conducted on the same day as the last scheduled visit of the treatment period in the predecessor study (Visit 15, Week 52). If this is not feasible, Visit 1 of PROSPERO may be conducted up to 12 weeks after the last dose of IP in the predecessor study (scheduled at Visit 14, Week 48), provided that the participant has remained in predecessor study follow-up during this time. In these circumstances, the Sponsor Medical Monitor should be contacted. Visit 1 of PROSPERO will not, under any circumstances, occur later than the time for Visit 16 (Week 60) of the predecessor study.

In order to provide participants time to consider participation in this study and to minimise the risk of an interrupted dosing regimen when participants transition between the predecessor study and this study, participants may be provided with the ICF at Week 48 (Visit 14) of the

predecessor study and will be asked to sign the ICF of PROSPERO study prior to any study-specific procedures for PROSPERO being performed.

Procedures of Visit 15 (Week 52) of the predecessor study should be completed in the predecessor study before initiating PROSPERO. If Visit 1 of PROSPERO is not performed on the same day as the predecessor study Visit 15 (Week 52), any new information about AEs, SAEs, COPD exacerbations, concomitant medications, smoking, and supplemental O₂ status, and HRU will be collected, and urine pregnancy test (FOCBP only) will be repeated prior to dosing. In cases when Visit 1 of PROSPERO is delayed beyond 4 weeks after Visit 15 of the predecessor study, safety-related assessments (physical exam, ECG, vital signs, SpO₂, safety laboratory tests, PK/ADA, etc) will also be repeated prior to dosing as described in the SoA (Section 1.3).

Before randomisation and first IP administration in PROSPERO, all relevant data, including, but not limited to, AEs, SAEs, concomitant medication, HRU, and COPD exacerbations will be collected as part of the predecessor study and should be recorded and up to date in the predecessor study. Information about AEs, SAEs, concomitant medications, and COPD exacerbations ongoing at time of entry to PROSPERO will be recorded in the PROSPERO study eCRF.

The study information, such as medical history and demographics, are to be generally collected in the predecessor studies and will be additionally collected in PROSPERO as needed.

Participants who do meet the eligibility criteria for PROSPERO and are willing to enter this study will exit the predecessor study and receive the first dose in PROSPERO at Visit 1; otherwise, participants will continue in the predecessor study follow-up period until completion (refer to the respective predecessor study protocol).

The participants who have signed the ICF and meet the eligibility criteria should be registered in a RTSM system on Visit 1. The participant will keep the same e-code as in the predecessor study.

Randomisation and first IP dosing in PROSPERO must occur on the same day.

The complete list of Visit 1 procedures is provided in the SoA (Section 1.3).

4.1.2 Treatment Period

The study design is updated in this amendment; participants will therefore be enrolled in the study prior to the amendment under CSP Version 1.0 or under this version (Version 2.0) (Figure 2).

CSP Version 1.0

Participants previously randomised to the placebo arm in the predecessor study will be re-randomised in a 1:1:2 ratio to receive either tozorakimab 300 mg Q4W, tozorakimab 300 mg Q8W, or placebo on top of SoC. The treatment period is 52 weeks.

CSP Version 2.0

Participants previously randomised to the placebo arm in the predecessor studies will be re-randomised in a 1:1 ratio to receive either tozorakimab 300 mg Q4W or placebo on top of SoC in PROSPERO. Participants previously randomised to tozorakimab 300 mg Q4W in the predecessor studies will remain on tozorakimab 300 mg Q4W. Participants previously randomised to tozorakimab 300 mg Q8W in the predecessor studies will switch to tozorakimab 300 mg Q4W at the point of entry to PROSPERO. The treatment period is 28 weeks. Those participants who are already randomised under CSP Version 1.0 can continue to complete the study with the same treatment regimen as assigned and for a treatment period of 52 weeks after this amendment.

Treatment allocation for PROSPERO will be predefined in RTSM system for all participants randomised in the predecessor studies.

Participants will attend scheduled clinic visits Q4W to complete protocol-specific assessments listed in the SoA (Section 1.3). Unscheduled visits may be initiated as needed, and assessments performed at these visits are at the discretion of the Investigator.

During treatment period, participants and/or caregivers will first receive trainings for self-administration of IP, if applicable (Section 6.2.2.3). Starting from Visit 5, participants may attend study visits remotely with self-administration of IP at every other visit (see Section 1.3 for details).

The Investigator must evaluate the participant to ensure the participant/caregiver is capable of performing self-administration of IP (refer to separate self-administration instructions). At-home self-administration is optional: If the study visit is not planned to be conducted via telemedicine, then this visit should be conducted onsite.

Participants will be encouraged to continue on the same COPD maintenance therapy throughout the study (Section 6.5.2). In case of a COPD worsening/exacerbation (Section 8.1.1), participants should be evaluated at the study site at an unscheduled visit, when feasible, or a scheduled visit, if the worsening happens to fall within a scheduled visit window.

Participants who prematurely discontinue IP should return to the study site and complete the E/D visit procedures 12 weeks after the last IP dose, or as soon as feasible after 12 weeks have passed from the last dose of IP (refer to Section 1.3 and Section 7.1.1). In cases when the

participant is not able or not willing to return to the site for the E/D visits 12 weeks after the last IP dose, E/D visit assessments can be performed earlier than this, at a time point feasible for the participant.

Discontinued participants will be encouraged to remain in the study and be followed up to the end of treatment period via telemedicine visits (refer to Section 7.1.1 for details).

Participants who complete the study treatment period will attend a follow-up visit 12 weeks after last dose (refer to Section 1.3).

Completion or E/D of IP will be registered via RTSM for each participant.

4.1.3 Study Conduct Mitigation During Study Disruptions Due to Cases of Civil Crisis, Natural Disaster, or Public Health Crisis

The guidance given below supersedes instructions provided elsewhere in this CSP and should be implemented only during cases of civil crisis, natural disaster, or public health crisis (eg, during quarantines and resulting site closures, regional travel restrictions, and considerations if site personnel or study participants become infected with SARS-CoV-2 or similar pandemic infection) which would prevent the conduct of study-related activities at study sites, thereby compromising the study site staff or the participant's ability to conduct the study. The Investigator or designee should contact the study Sponsor to discuss whether the mitigation plans below should be implemented.

To ensure continuity of the clinical study during a civil crisis, natural disaster, or public health crisis, changes may be implemented to ensure the safety of study participants, maintain compliance with GCP, and minimise risks to study integrity.

Where allowable by local health authorities, ethics committees, healthcare provider guidelines (eg, hospital policies), or local government, these changes may include the following options:

- Obtaining consent/reconsent for the mitigation procedures (note, in the case of verbal consent/reconsent, the ICF should be signed as soon as feasible [ie, at the participant's next visit]).
- Home visit or visit in remote location: performed by a site qualified HCP or HCP provided by a TPV.
- Telemedicine visit: remote contact with the participants using telecommunications technology including phone calls, virtual or video visits, and mobile health devices.
- At-home IP self-administration: performed by a site qualified HCP, HCP provided by a TPV, if possible, or the participant/caregiver. Additional information related to the visit can be obtained via telemedicine.

Investigators, blinded Sponsors and investigator site staff, and study participants should remain blinded throughout the course of the study including the aforementioned mitigation procedures. For further details on study conduct during civil crisis, natural disaster, or public health crisis, refer to [Appendix H](#).

4.2 Scientific Rationale for Study Design

4.2.1 Rationale for Selection of Participant Population

The study population for the predecessor studies were identified based on tozorakimab mechanism of action, unmet need in patients experiencing COPD symptom burden and at risk for exacerbations, and outcomes of recent Phase II study of itepekimab, with a similar MoA to tozorakimab, which resulted in a marked reduction of COPD exacerbations as well as a significant improvement of FEV₁ in former smokers ([Rabe et al 2021](#)).

Interleukin-33 has been implicated in the pathogenesis of goblet cell hypertrophy and mucous hypersecretion ([Allinne et al 2019](#), [Schmitz et al 2005](#)). A large proportion of patients with high levels of mucins and mucous plugging have a higher cough and sputum burden ([Okajima et al 2020](#), [Radicioni et al 2021](#)) and risk of exacerbations ([Kim et al 2015](#), [Kim et al 2019](#), [Lahousse et al 2017](#), [Stott-Miller et al 2020](#)). For that reason, a population of patients with symptomatic COPD and at risk for exacerbations, with a CAT score of 10 or more, with a score 2 or more from sputum and phlegm, has been selected for the predecessor studies.

The rationale for the selection of participants for PROSPERO is that they have completed the predecessor study treatment period and not been prematurely discontinued from IP.

4.2.2 Rationale for Selection of Primary Endpoint

Severe exacerbations of COPD worsen the course of COPD disease and involve a rapid decline in health status, disease progression and high mortality, with each severe COPD exacerbation increasing the risk of further exacerbations ([Wedzicha et al 2013](#)), re-hospitalisation ([Myers et al 2020](#)) and death ([Suisse et al 2012](#)).

Annualised rate of severe COPD exacerbations is a well-accepted and clinically meaningful endpoint to evaluate whether tozorakimab plus SoC versus placebo plus SoC can reduce the risk of severe COPD exacerbations.

4.2.3 Rationale for Selection of Comparator

The comparator in this study is placebo. Both tozorakimab and placebo will be administered on top of optimised SoC maintenance inhaled therapy; therefore, the treatment of the placebo group will correspond to current SoC for patients with COPD. The use of placebo will allow for a double-blinded treatment period and to provide a control arm, so as to enable a continued comparison of long-term efficacy, in particular the comparison of severe exacerbations rates, and to act as a comparator to contextualise safety observations.

Placebo was used as the comparator during the OBERON/TITANIA predecessor studies and will continue as the comparator in this LTE study. Fifty percent of the participants in the predecessor placebo arm will be assigned to active treatment and 50% of the participants in the predecessor placebo arm will continue to receive placebo. The re-randomisation of PROSPERO from the predecessor studies will allow more participants to receive the potentially effective experimental treatment rather than extended SoC, while retaining a comparator arm.

4.2.4 Rationale for Selection of Study Duration

The treatment period was selected to maximise the number of events of severe COPD exacerbation by the time of unblinding of the predecessor studies and to give a predictable duration to participants given the uncertainty in recruitment time of the predecessor studies. The primary reporting of the study may take place once both predecessor studies have been unblinded and reported without waiting for all participants to complete treatment in this study to enable the timely completion of regulatory submission.

4.2.5 Participant Input into Design

Patient feedback on the study design of the predecessor studies was obtained from a panel of 5 patients with COPD and 1 caregiver of a patient with COPD within the Sponsor Patient Partnership Program, an internal working group that enables patient feedback on clinical trial study designs. During the discussion, patients shared which aspects of the proposed study they found appealing or concerning and provided their feedback on visit frequency, study procedures, and use of digital solutions. Patient feedback received during design of predecessor studies had a significant influence on the design of the PROSPERO study. Site visit frequency and number of study assessments were reduced. In addition, APFS was introduced to the PROSPERO study to allow for optional at home administration, further reducing the number of clinic visits and burden for the patient.

4.3 Justification for Dose

The present study is a LTE study that will enable participants to continue on their IP regimen or be re-randomised from predecessor studies OBERON or TITANIA. The 2 dose regimens of SC tozorakimab 300 mg Q4W and 300 mg Q8W were initially proposed in the predecessor studies. These dose regimens were selected based on an integrative PK/PD model created with the systemic target engagement information from a Phase I study (D9180C00001) that evaluated single doses of tozorakimab up to 300 mg SC or IV in healthy adult participants with a history of mild atopy and multiple doses up to 300 mg Q2W in patients with mild COPD. This PK/PD model was used to predict IL33 suppression in lung at trough assumed similar systemic exposure in patients and healthy subjects. However, emerging Phase II data in moderate to severe asthma patients (Study D9181C0001) and moderate to severe atopic dermatitis (Study D9182C0001) has shown 30% to 35% lower trough concentrations in

patients compared with healthy volunteers in Phase I (D9180C00001). The reduction in PK concentrations predicts a slightly below 80% IL33 lung inhibition for 300 mg Q8W but remains above 90% for tozorakimab 300 mg Q4W. For this reason, the sample size of the tozorakimab 300 mg Q8W arm has been reduced in the predecessor studies, and as of the protocol amendment (Version 2.0) participants receiving 300 mg Q8W regimen in the predecessor studies will be re-allocated to tozorakimab 300 mg Q4W in PROSPERO study.

4.4 End of Study Definition

For the purpose of Clinical Trial Transparency the definition of the end of the study differs under FDA and EU regulatory requirements:

- EU requirements define study completion as the last visit of the last subject for any protocol-related activity.
- FDA requirements defines 2 completion dates:
 - Primary Completion Date – the date that the final participant is examined or receives an intervention for the purposes of final collection of data for the primary outcome measure, whether the clinical study concluded according to the pre-specified protocol or was terminated. In the case of clinical studies with more than one primary outcome measure with different completion dates, this term refers to the date on which data collection is completed for all of the primary outcomes.
 - Study Completion Date – the date the final participant is examined or receives an intervention for purposes of final collection of data for the primary and secondary outcome measures and AEs (eg, last participant's last visit), whether the clinical study concludes according to the pre-specified protocol or is terminated.

An individual participant will be considered to have completed the study if the participant was followed through their last protocol-specified visit/assessment regardless of the number of doses of IP that was received.

Participants who prematurely discontinue IP should complete procedures described for the E/D visit and further phone follow-up of severe exacerbation and death information, if applicable (refer to Section 7.1.1 for details).

Participants will not be considered to have completed the study if consent was withdrawn or the participant was lost to follow-up (see Section 7.2 and Section 7.3, respectively).

The primary reporting of the study may take place once both predecessor studies have been unblinded and reported without waiting for all participants to complete treatment in this study. To ensure consistency with reporting PROSPERO and its predecessor studies, some participants may not complete the full 28-week treatment period. As a result, some participants may conclude the study prior to finishing the full treatment duration. Participants

will be made aware of this during the consent process. The participants who do not complete the full treatment period will be asked to attend the E/D visit, scheduled 12 weeks after their final IP dose, to complete their participation.

The number of severe exacerbations observed in the study will be monitored to estimate statistical power before the decision to unblind is confirmed. The study may also be unblinded to report early at the interim analysis due to overwhelming efficacy. In either scenario, the study team in direct contact with sites or with influence over management of participants or recording of endpoints in PROSPERO will remain blinded and be separate from the team involved in reporting of the predecessor studies. Details will be included in the Study Integrity Plan.

4.4.1 Study Stopping Criteria

Discontinuation of specific sites or of the study as a whole are handled in accordance with [Appendix A](#).

An external IDMC will be set up to provide systematic and unbiased assessment of safety for the predecessor studies and PROSPERO. The IDMC will oversee the safety of the study participants and will make appropriate recommendations regarding further study conduct based on the available data. The IDMC may recommend modification or termination of the study based on reviewed data, including regular unblinded safety review and at a planned interim analysis where decision rules relating to early reporting of the study for efficacy will be considered. The details of IDMC process and decision criteria will be detailed in the IDMC charter. If predefined criteria for stopping the predecessor studies for futility is met, and the predecessor studies are stopped for futility, this study will also be stopped.

If the Sponsor determines that temporary suspension or termination of the study is required, the Sponsor will discuss the reasons for taking such action with all participating Investigators (or heads of the medical institutions, where applicable). When feasible, the Sponsor will provide advance notice to all participating Investigators (or heads of the medical institutions, where applicable) of the impending action.

If the study is suspended or terminated for safety reasons, the Sponsor will promptly inform all Investigators, heads of the medical institutions (where applicable), and/or institutions conducting the study. The Sponsor will also promptly inform the relevant regulatory authorities of the suspension/termination along with the reasons for such action. Where required by applicable regulations, the Investigator or head of the medical institution must inform the IRB/IEC promptly and provide the reason(s) for the suspension/termination. If the study is suspended for safety reasons and it is deemed appropriate by the Sponsor to resume the study, approval from the relevant regulatory authorities (and IRBs/IECs when applicable) will be obtained prior to resuming the study. The Investigator shall also promptly inform the participant of the study suspension/termination and should assure appropriate participant therapy and/or follow-up.

5 STUDY POPULATION

Prospective approval of protocol deviations to recruitment and enrolment criteria, also known as protocol waivers or exemptions, is not permitted.

5.1 Inclusion Criteria

Participants are eligible to be included in the study only if all of the following criteria apply:

- 1 Participants who have completed the treatment period and have not been prematurely discontinued from IP in the predecessor studies.
- 2 Participants who received their last dose of IP in the predecessor studies within the previous 12 weeks and were not withdrawn from the predecessor study.

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- 3 FOCBP must have a negative urine pregnancy test at Visit 1.
 - (a) For a definition of FOCBP, refer to [Appendix E](#).
- 4 Participants who are willing to continue using contraceptive methods as agreed to for the predecessor OBERON or TITANIA studies. See [Table E1](#) in [Appendix E](#) for acceptable methods of contraception.

Contraceptive use by men or women should be consistent with local regulations regarding the methods of contraception for those participating in clinical studies.

(a) Male participants:

Non-sterile male participants who are sexually active with a FOCBP partner must agree to use a male condom while engaging in sexual activity from enrolment throughout the study duration and until 14 weeks after last dose of IP. For a definition of non-sterile male, refer to [Appendix E](#). In countries where spermicide is available, it is strongly recommended. It is also strongly recommended for the female partner of a male participant to use a highly effective method of contraception throughout this period.

Non-sterilised male patients should also refrain from biologically fathering a child or donating sperm during the same period.

(b) Female participants:

FOCBP who are sexually active with a non-sterilised male partner must agree to use one highly effective method of birth control, as defined below, from enrolment throughout the study and until at least 14 weeks after last dose of IP. Cessation of contraception after this point should be discussed with a responsible physician. Periodic abstinence (calendar, symptothermal, post-ovulation methods), withdrawal (coitus interruptus), spermicides only, and lactational amenorrhoea method are not

acceptable methods of contraception. Female condom and male condom should not be used together.

A highly effective method of contraception is defined as one that can achieve a failure rate of less than 1% per year when used consistently and correctly. Highly effective birth control methods can be found in [Appendix E](#). It is highly recommended for the male partner of a FOCBP participant to use a male condom while engaging in sexual activity throughout this period.

Note that there are no contraception requirements for female participants who are not of childbearing potential. However, all female participants should refrain from egg cell donation and breastfeeding throughout the study.

Informed Consent

- 5 Capable of giving signed informed consent as described in Appendix A, which includes compliance with the requirements and restrictions listed in the ICF and in the protocol.

5.2 Exclusion Criteria

Participants are excluded from the study if any of the following criteria apply.

Diagnostic Assessments/Medical Conditions

- 1 Any clinically significant disorder or abnormal findings (clinical, laboratory, instrumental, etc) or major physical and/or cognitive impairment, which, in the opinion of the Investigator, may put the participant at risk because of his/her participation in the study or impact the interpretation of the study results, or otherwise makes the participation of the participant inappropriate.
- 2 Participant meeting criteria for IP discontinuation (refer to Section [7.1.1](#)) as judged by the Investigator or the Sponsor.

Prior/Concomitant Clinical Study Experience

- 3 Concurrent enrolment in other interventional clinical study or treatment with another IP, with the exception of the OBERON and TITANIA predecessor studies.
- 4 Known history of:
 - (a) Severe allergic reaction to any monoclonal and polyclonal antibody.
 - (b) Allergy or reaction to any component of the IP formulation.
- 5 Chronic use (or expected need for chronic use during the study) of immunosuppressive medications (including, but not limited to, systemic corticosteroids), marketed or investigational biologic, or another prohibited medication that, in the opinion of the

Investigator or the Sponsor, may put the participant's safety at risk and/or impact the interpretation of the study results.

Other Exclusions

- 6 Involvement in the planning and/or conduct of the study (applies to both staff employed by the Sponsor and/or staff at the study site).
- 7 Participants who are not able to comply with the study requirements, procedures, and restrictions, as judged by the Investigator or the Sponsor.

5.3 Lifestyle Considerations

5.3.1 Contraception

Male participants who are sexually active with a FOCBP and participants that are FOCBP must use highly effective contraceptive methods from enrolment throughout the study and until 14 weeks after last dose of IP, as stated in inclusion criterion 4 (Section 5.1). See [Table E1](#) in [Appendix E](#) for acceptable methods of contraception.

There are no contraception requirements for female participants who are not of childbearing potential. However, all female participants should refrain from breastfeeding throughout the study.

5.3.2 Blood and Plasma Donation

Participants must abstain from donating blood and plasma from the time of informed consent, and for 14 weeks (5.5 half-lives) after last dose of IP.

5.3.3 Meals and Dietary Restrictions

Participants should avoid eating a large meal for at least 2 hours prior to all scheduled lung function assessments at the site.

5.3.4 Tobacco, E-cigarettes, and Vaping

Participants enrolled in the “former smokers” cohort should refrain from smoking throughout the entire duration of the study.

Participants who are current smokers (currently smoking tobacco) will be allowed to stop smoking during the study if desired.

Smoking and nicotine use status changes during the course of the study will be captured on the eCRF.

Use of e-cigarettes, or vaping of any products (eg, nicotine, THC) is discouraged during the study. Use of heated tobacco products (eg, IQOS) is discouraged in the former smoker cohort.

Participants should avoid smoking for at least one hour prior to all scheduled lung function assessments and blood collection at the site.

5.3.5 Activity

Participants should avoid engaging in strenuous exertion for at least 30 minutes prior to all scheduled lung function assessments and blood collection at the site.

5.4 Enrolment Failures

Enrolment failures are defined as participants who consent to participate in the clinical study but are not subsequently randomised. A minimal set of enrolment failure information is required to ensure transparent reporting of enrolment failure participants to meet the CONSORT publishing requirements and to respond to queries from regulatory authorities. Minimal information includes demography, enrolment failure details, eligibility criteria, and any SAE. These participants should have the reason for enrolment failure recorded in the eCRF.

All participants that are enrolment failures must continue with the follow-up period requirements as described in the protocol for the predecessor study.

6 STUDY INTERVENTIONS

Study intervention is defined as any investigational intervention(s), marketed product(s), placebo, intended to be administered, or medical device(s) utilised by a study participant according to the CSP. Study intervention in this CSP includes tozorakimab and placebo and will be referred to in the CSP as IP.

Tozorakimab and placebo will be supplied as solutions for injection in an APFS. Tozorakimab will be supplied as a 300 mg APFS containing 150 mg/mL tozorakimab, 20 mM L-histidine/L-histidine hydrochloride, 220 mM L-arginine hydrochloride, and 0.03% w/v polysorbate 80 with a pH of 5.5. Placebo will be supplied as an APFS containing 20 mM L-histidine/L-histidine hydrochloride, 220 mM L-arginine hydrochloride, and 0.03% w/v polysorbate 80 with a pH of 5.5. The label-claim volume for tozorakimab and placebo is 2 mL.

6.1 Study Interventions Administered

6.1.1 Study Interventions

Table 4 Study Interventions

Arm Name	Tozorakimab 300 mg Q4W	Tozorakimab 300 mg Q8W	Placebo
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Table 4 Study Interventions

Intervention Name	Tozorakimab	Tozorakimab	Placebo
Type	Biologic, combination product	Biologic, combination product	Placebo, combination product
Use	Active	Active	Placebo
Unit Dose Strength	150 mg/mL	150 mg/mL	Not applicable
Dosage Level	300 mg Q4W	300 mg Q8W (with alternating matching placebo between active tozorakimab administrations)	Matching volume to tozorakimab Q4W
Route of Administration	SC injection	SC injection	SC injection
IMP and NIMP	IMP	IMP	IMP
Sourcing	Provided centrally by the Sponsor	Provided centrally by the Sponsor	Provided centrally by the Sponsor
Packaging and Labelling	Study intervention in APFS with 2 mL label claim volume will be provided in kits. Each APFS and kit will be labelled and labels will be prepared in accordance with GMP and local regulatory requirements. The kits will include a unique number that is printed on all labels within the kit (ie, the outer carton label and the label of the APFS within the carton).	Study intervention in APFS with 2 mL label claim volume will be provided in kits. Each APFS and kit will be labelled and labels will be prepared in accordance with GMP and local regulatory requirements. The kits will include a unique number that is printed on all labels within the kit (ie, the outer carton label and the label of the APFS within the carton).	Study intervention in APFS with 2 mL label claim volume will be provided in kits. Each APFS and kit will be labelled and labels will be prepared in accordance with GMP and local regulatory requirements. The kits will include a unique number that is printed on all labels within the kit (ie, the outer carton label and the label of the APFS within the carton).

Note: The treatment period will last for 28 weeks or 52 weeks, respectively (see Section 4.1.2 for details).

APFS, accessorised pre-filled syringe; GMP, good manufacturing practice; IMP, investigational medicinal product; NIMP, non-investigational medical product; Q4W, every 4 weeks; Q8W, every 8 weeks;

SC, subcutaneous.

6.1.2 Investigational Product Administration Rescheduling

Every effort should be taken to keep IP administration within the visit window.

If a participant presents with a condition that contraindicates dosing, IP dose should be withheld and administered as soon as possible when the contraindicating condition resolves.

The IP should not be administered, and the dosing is to be rescheduled in the presence of any AE, laboratory abnormality, or another diagnostic finding that, in the opinion of the Investigator or the Sponsor, contraindicates dosing or could result in complications.

It is recommended that the Sponsor study physician/delegate is contacted with any questions.

If IP cannot be administered at a scheduled treatment visit (eg, due to conditions listed above), it can be postponed as necessary and administered as soon as possible, with the exception of Visit 1, in which the first dosing in PROSPERO should take place. When IP dosing needs to be postponed, it is recommended that all scheduled treatment visit procedures (except for IP administration) are still performed within the visit window.

Rescheduled IP dose can be then administered at an unscheduled visit. Physical exam, vital signs assessment, AE assessment, changes in concomitant medication, and pregnancy test (if applicable) are the minimum procedures at the time of dose administration visit; it may also include remaining visit procedures (not performed at the scheduled visit) and additional assessments as deemed necessary by the Investigator.

If the entire scheduled visit cannot be conducted within the window (eg, the participant is unable to attend the study site), then all visit procedures are rescheduled along with the IP dose.

If a dose is significantly delayed it is recommended to keep at least 2 weeks interval before the next dose. If a postponed dose overlaps with the next treatment visit window, the postponed dose will be skipped, and the next dose of IP given at the regularly scheduled visit. The visit schedule will be always calculated from Visit 1 of this study.

If the participant misses 2 consecutive or non-consecutive doses of IP, it is strongly recommended that the site discuss with the Sponsor study physician to review the participant's adherence to treatment and measures to ensure compliance. If the participant misses 3 or more consecutive doses of IP (>12 weeks elapses since the last dose of IP), the site must discuss with the study physician whether continuation or discontinuation of IP is in their best interests.

Discontinuation procedures are described in Section 7.1. All participants, whether they remain on IP or not, will be encouraged to stay in the study and be followed until the study completion (Section 7.1.1).

6.2 Preparation, Handling, Storage, and Accountability of Interventions

- 1 The Investigator or designee must confirm appropriate temperature conditions have been maintained during transit for all IP received and any discrepancies are reported and resolved before use of the IP.

- 2 Only participants enrolled in the study may receive IP and only authorised staff (with the exception of self-administration described in Section 6.2.2.3 and circumstances described in [Appendix H](#)) may dispense or administer IP. All IP must be stored in a secure, environmentally controlled, and monitored (manual or automated) area in accordance with the labelled storage conditions with access limited to the Investigator and authorised site staff.

For at-home self-administration, the participant will be responsible for refrigerating the product at home. The temperature is not monitored during storage by the participant; standard refrigeration is sufficient to maintain appropriate temperatures while the IP is in the participant's possession. The participant will be responsible for returning used APFS devices to the study centre.

- 3 The Investigator, institution, or the head of the medical institution (where applicable) is responsible for IP accountability, reconciliation, and record maintenance (ie, receipt, reconciliation, and final disposition records).
- 4 Further guidance and information for the final disposition of unused IPs are provided in the IP Manual.

6.2.1 Storage of Study Interventions: Tozorakimab and Placebo

IP (tozorakimab and tozorakimab placebo) APFS are stored at 2°C to 8°C (36°F to 46°F) and must not be frozen. The Investigator, or an approved representative (eg, pharmacist), will ensure that all study intervention is stored in a secured area, in refrigerated temperatures (2°C to 8°C; 36°F to 46°F) and in accordance with applicable regulatory requirements. A temperature log will be used to record the temperature of the storage area. Temperature excursions outside the permissible range listed in the clinical supply packaging are to be reported to the site monitor upon detection. A calibrated temperature-monitoring device will be used to record the temperature conditions in the drug storage facility. Storage conditions stated in the IB may be superseded by the label storage.

Study intervention must be kept in original packaging until time of use to prevent prolonged light exposure.

6.2.2 Tozorakimab and Placebo SC Injection Preparation and Administration

6.2.2.1 Preparation

Tozorakimab is a sterile, preservative-free, clear to opalescent, colourless to brown-yellow solution, practically free from visible particles.

Each APFS selected for administration should be inspected. If there are any defects noted with the IP, the Investigator and site monitor should be notified immediately. Refer to the Product Complaint section of the IP Manual and to Section 8.3.13.3.

Tozorakimab and placebo should be kept at room temperature for at least 60 minutes inside the original packaging prior to administration. The dose of tozorakimab or placebo must be administered at room temperature within 8 hours after removal of the APFS from refrigeration.

Tozorakimab and placebo do not contain preservatives; any unused portion of the APFS must be discarded immediately after use.

6.2.2.2 Administration

A dose of 300 mg tozorakimab or placebo will be administered using one APFS injection according to standard procedures for SC injections.

The first day of dosing in PROSPERO is considered Visit 1 (Week 0).

6.2.2.2.1 Before IP Administration

All applicable visit procedures, including sample collection, should be completed prior to IP administration. If the participant cannot tolerate a particular visit procedure during the visit (eg, spirometry), IP can be administered, and the procedure should be rescheduled for a different date, preferably within the visit window. No procedures should be performed after the dosing on the day of IP administration. Pharmacokinetic and ADA samples must always be collected before IP administration.

Before Each IP Administration

- FOCBP should have the urine pregnancy test performed before IP administration. The IP will be administered only when the result of the test is negative (Section 8.2.6).
- Investigator/authorised delegate will evaluate the participant's condition for potential contraindications for dosing.
- Investigator/authorised delegate will evaluate the injection site as per standards of medical care.

6.2.2.2.2 IP Administration

The IP will be administered SC by the Investigator/or an appropriate designee trained in study drug administration to the abdomen, upper arm, or thigh. The personnel administering the dose will wipe the skin surface of the administration site with alcohol and allow the skin surface to air dry. The skin will be pinched to isolate the SC tissue from the muscle. The needle will be fully inserted at a 45-degree angle into the SC tissue. The IP will be slowly

injected (at least 10 seconds duration is recommended). The area should not be massaged after injection.

It is recommended that the site of injection be rotated so that the participant receives IP injections in different anatomical regions from one visit to the next. In cases when rotation of the injection site is not feasible and/or the participant prefers not to rotate injection sites, the reason for not rotating the injection site should be documented in the source documents.

Each injection site must be documented on the eCRF and in the source documents at each IP visit.

6.2.2.3 Optional Self-Administration of IP

To reduce participant burden after Visit 4, participants will have an option for self-administration of IP using the APFS by the participant or their caregiver. The Investigator must evaluate the participant and/or their caregiver for suitability of performing the IP self-administration and provide them with appropriate training during onsite visits. All necessary supplies and instructions for administration and documentation of IP administration will be provided.

Starting from Visit 5, some visits do not include extensive onsite assessments and can optionally be performed as remote visits through telemedicine contact (Section 1.3) for participants who opt for IP self-administration. IP kit and urine dipsticks for pregnancy testing (for FOCBP) for the remote visits may be dispensed starting from Visit 4 (a single kit per time for the next remote visit), given that a mandatory training of optional IP self-administration is completed.

If the IP is self-administered by the participant/caregiver, IP should be administered on a day of a scheduled visit after all visit assessments are completed.

The Investigator or qualified designee will conduct a telemedicine contact on the day of the visit prior to IP self-administration to confirm that the participant does not have any contraindications for IP dosing. FOCBP will be asked about any concerns for pregnancy and will be advised to perform a urine pregnancy test.

If the urine pregnancy test result is positive, the participant should contact the site and should not self-administer IP. Site personnel will provide the participant with further instructions.

The participant must be contacted by the Investigator or qualified designee after the IP is self-administered, to evaluate for any potential acute reactions, in line with local clinical practice. For optional at-home self or caregiver administration, it is strongly recommended that the Investigator, or a designee, remains on the telemedicine call during the IP administration. Any AEs reported by the participant should be recorded in source documents and entered in the

eCRF. Furthermore, any device constituent deficiency reported by the participant should be recorded and handled as outlined in Section 8.3.12.

Refer to the Study Instructions for Self-Administration of IP by the Participant and/or their Caregiver for step-by-step guidance including Investigator assessment/training of participant and/or caregiver, IP dispensing and drug accountability.

The option of self-administration of IP will only be available in countries where allowed according to local regulations.

6.2.2.4 Monitoring Dose Administration

After the administration of IP, the SC injection site will be assessed for any reactions as detailed in Section 8.3.6, and the participant should be observed for potential acute drug reactions in line with local clinical practice for mAbs. For optional at-home self or caregiver administration, it is strongly recommended that the Investigator, or a designee, remains on the telemedicine call during the IP administration.

As with any biologic product, allergic reactions to dose administration are possible. Therefore, appropriate drugs and medical equipment to treat acute anaphylactic reactions should be immediately available, and study personnel should be trained to recognise and treat anaphylaxis ([Appendix F](#) and [Appendix G](#)).

6.2.3 Accountability

The site's designated IP manager is required to maintain accurate IP accountability records. Upon completion of the study, copies of IP accountability records will be returned to the Sponsor. All unused IP will be disposed of following site procedures and upon authorisation by the Sponsor or will be returned to a Sponsor-authorised depot if a site is unable to dispose of unused IP. The outer carton should be retained onsite to assist with kit assignment checks, if permitted by site procedures.

6.3 Measures to Minimise Bias: Randomisation and Blinding

6.3.1 Methods for Assigning Treatment Groups

All participants assigned to the placebo arm in the OBERON or TITANIA predecessor studies will be assigned to randomised IP in PROSPERO using an RTSM as follows:

Initial Randomisation Scheme (CSP Version 1.0)

- Participants previously treated with tozorakimab 300 mg Q4W SC will stay on the same blinded treatment.
- Participants previously treated with tozorakimab 300 mg Q8W SC will stay on the same blinded treatment.

- Participants previously randomised to placebo will be re-randomised in 1:1:2 ratio to blinded tozorakimab 300 mg Q4W SC, tozorakimab 300 mg Q8W SC, or placebo.

Current Randomisation Scheme (CSP Version 2.0)

- Participants previously treated with tozorakimab 300 mg Q4W SC will stay on the same blinded treatment.
- Participants previously treated with tozorakimab 300 mg Q8W SC will switch to tozorakimab 300 mg Q4W SC at the point of entry to PROSPERO.
- Participants previously randomised to placebo will be re-randomised in 1:1 ratio to blinded tozorakimab 300 mg Q4W SC or placebo.
- Participants allocated or re-randomised to tozorakimab 300 mg Q8W under CSP Version 1.0 will continue on the same treatment until the study completion.

The randomisation scheme will be generated for all OBERON and TITANIA participants regardless of predecessor study completion, and not only in those participants that complete OBERON and TITANIA; this will ensure that the comparison of treatment groups in an integrated analysis will be valid from the point of randomisation in the predecessor studies and not only in a selected subgroup of participants able to continue with long term treatment.

Randomisation codes will be assigned strictly sequentially in each stratum as participants become eligible for randomisation. Randomisation will be stratified by predecessor study (OBERON versus TITANIA), smoking status (current versus former), maintenance inhaled therapy (dual versus triple), and region, with these strata determined as per randomisation to the predecessor study regardless of any subsequent changes.

Participants who fail to meet the inclusion/exclusion criteria should not, under any circumstances, be randomised, or receive IP. There can be no exceptions to this rule. If a participant withdraws from the study after enrolment, then his or her enrolment/randomisation code cannot be reused. Withdrawn participants will not be replaced.

6.3.2 Methods for Ensuring Blinding

This is a double-blind study in which the investigational site staff and participants remain blinded during the clinical study. To ensure study blinding, yellow blinding labels will be affixed onto both tozorakimab and placebo syringe barrels during APFS manufacture. All packaging and labelling of IP will be done in such way as to ensure blinding for all participants, Sponsor, and investigational site staff.

Investigators, blinded Sponsor, and investigational site staff and participants will remain blinded to each participant's assigned IP throughout the course of the study, but will know which version of the CSP the participant was recruited under, and therefore if randomisation to Q8W arm is possible.

Tozorakimab could reduce eosinophil counts in blood over time. As a precaution, eosinophil, basophil, and monocyte data, with the exception of HP values, from the haematology laboratory tests will not be communicated to blinded Sponsor or site personnel during the intervention and follow-up periods. There is no need to apply blinding to local laboratory tests taken as part of treatment for acute illness (eg, actual/suspected COPD exacerbation). Procedures to maintain the blind are described in the Study Integrity Plan.

A Sponsor site monitor will perform IP accountability. In the event that the treatment allocation for a participant becomes known to the Investigator or other study staff involved in the management of study participants or needs to be known to treat an individual participant for an AE, the Sponsor must be notified immediately by the Investigator and, if possible, before unblinding.

The following personnel will have access to the randomisation list during the study, prior to database lock:

- Those carrying out the packaging and labelling of IP.
- Those generating the randomisation list.
- The Sponsor supply chain department.
- Unblinded Sponsor monitor (if applicable).
- Bioanalytical PK and ADA laboratories responsible for analysing the samples.

The information in the randomisation list will be kept from other personnel involved in the conduct of the study in a secure location until the primary analysis of the study. No other member of the extended study team at the Sponsor, or any CRO handling data, will have access to the randomisation scheme during this time.

Although the study will initially be completely double-blind (ie, blind for participants, Investigators/site staff, and AstraZeneca /designated CRO), at the conclusion of each of the predecessor studies (OBERON or TITANIA), the treatment allocation within the predecessor studies will become known to the AstraZeneca staff and/or designated CRO involved in reporting of the predecessor studies. The PROSPERO study team in direct contact with sites or with influence over management of participants or recording of endpoints will remain blinded and be separate from this team and access to systems or documents containing treatment allocations in OBERON/TITANIA will be controlled at least until after the unblinding of PROSPERO to conduct the primary analysis. The blind will be maintained for the Investigator, investigational site staff, and for the participants at least until the primary reporting of the PROSPERO study is completed. Procedures to maintain this blind will be described in the Study Integrity Plan.

6.3.3 Methods for Unblinding

Information indicating the treatment randomisation for each randomised participant, will be available to the Investigator(s) from the RTSM. Routines for this will be described in the RTSM user manual that will be provided to each site. The randomisation code should not be broken except in medical emergencies when the appropriate management of the participant requires knowledge of the treatment randomisation. The Investigator will document and report the action to the Sponsor, without revealing the participant's treatment randomisation to the Sponsor staff. The Sponsor retains the right to break the code for SAEs that are unexpected and are suspected to be causally related to IP and that potentially require expedited reporting to regulatory authorities. Randomisation codes will not be broken for the planned analyses of data until all decisions on the evaluability of the data from each individual participant have been made and documented.

6.4 Investigational Product Compliance

When participants are dosed at an onsite visit, they will receive IP under medical supervision. The date and time of dose administered in the clinic will be recorded in the source documents and recorded in the eCRF. The outer carton should be kept onsite to assist with kit assignment checks, if permitted by site procedures.

Refer to Section [6.2.2.3](#) and the Study Instructions for Self-Administration of Investigational Product by the Participant and/or their Caregiver for procedures related to participant/caregiver self-administration.

6.5 Concomitant Therapy

The Investigator/authorised delegate will collect and record any medication taken from the time of enrolment until the final study visit. All background COPD medications (eg, ICS/LABA/LAMA, antibiotics) and treatment administered for COPD exacerbations, all other medications taken for any reason, and any concomitant medication(s), including herbal preparations, taken during the study will be recorded in source documentation and in eCRF, along with:

- Reason for use.
- Dates of administration including start and end dates.
- Dosage information including dose, frequency, and route of administration.

The medical monitor should be contacted if there are any questions regarding concomitant or prior therapy.

Participants are prohibited from receiving certain concomitant medications (Section [6.5.4](#)) unless clinically indicated as determined by a physician.

Medications or therapies that are not prohibited and neither compromise participant safety nor affect study data, as judged by the Investigator, will be permitted and recorded in the source documentation and in the appropriate sections of the eCRF.

Refer to [Appendix J](#) for auxiliary medicinal products as applicable to EU CTR.

6.5.1 Rescue Medicine

Short-acting β_2 -agonists (SABAs, eg, salbutamol, albuterol, terbutaline, levalbuterol), SAMA, SABA/SAMA combination or alternative rescue medication, as per local SoC, may be used during the study in the event of worsening of COPD symptoms. Note medication restrictions prior to spirometry visits (Section [6.5.3](#)).

6.5.2 Maintenance Therapy

All participants are required to be treated with stable optimised maintenance inhaled therapy (ICS/LABA/LAMA triple therapy, or dual therapy if triple is not considered appropriate) during the study. Both dual and triple therapy may be in the form of separate inhalers or fixed dose combination inhalers. Individual component changes or switches between devices are allowed as long as the participant remains on the same equivalent doses. If antibiotics or PDE4 inhibitors were added to the background therapy and regimen, the doses should be stable throughout the study. Xanthines are allowed as maintenance therapy provided that the dose is either equivalent to theophylline ≤ 400 mg per day or serum drug level within the therapeutic range (recommend blood test to confirm < 12 mg/dL). Changes to the participant's maintenance therapies are strongly discouraged during the study, unless judged medically necessary by the Investigator. The justification and rationale for treatment changes should be documented in the source notes.

Note: Meaning of “stable” or “unchanged” COPD maintenance therapy in this protocol allows for individual component changes or switches between devices provided that the participant remains on the same class therapies at equivalent doses (for examples of ICS equivalent doses, refer to [Appendix I](#)).

Recommended COPD inhaled maintenance therapy compliance during the treatment period is $\geq 70\%$, as defined by medication start/stop dates, and the participant response to “in the past month, how many days a week have you taken your maintenance medication as scheduled?” at each visit, recorded in the source documentation. It is recommended that if compliance levels fall below 70%, the site/Investigator should evaluate the reasons for low compliance and to provide re-education to the participant in regard to the importance of compliance in taking their COPD inhaled maintenance therapy. It is strongly recommended that when compliance levels fall below 50%, the site contacts the Sponsor study physician in order to discuss the potential safety implication for the participant and future decisions.

All participants should be vaccinated against pneumococcus and influenza in accordance with local guidelines, unless vaccination is contraindicated or rejected by the participant, or is not available. Pneumococcal and influenza vaccination can be administered at any time through the study (except $\pm$ 7 days of IP administration) at the discretion of the Investigator.

6.5.3 Medication Withhold Periods

Spirometry assessments will be performed at the study site at scheduled visits (Section 1.3). There are restrictions regarding the participant's COPD medications prior to the spirometry assessments.

Participants will be asked to withhold their usual maintenance inhaled therapy (ICS/LABA/LAMA triple therapy, or dual therapy if triple is not considered appropriate) on the mornings of the scheduled pre-BD spirometry visits, preferably for at least 12 hours prior to the spirometry assessment (24 hours prior to the assessment for once-daily medications). In addition, SABA and SAMA should not be used within 6 hours prior to the spirometry assessments. The participant's usual COPD medications may be administered following completion of the pre-BD spirometry. The IP will be administered at a scheduled visit after the spirometry assessments.

If the participant has taken their usual maintenance inhaled therapy (ICS/LABA/LAMA triple therapy, or dual therapy if triple is not considered appropriate) on the morning of the scheduled spirometry visit, the Investigator/authorised delegate should remind the participant of the importance of withholding their usual morning maintenance inhaled therapy and reschedule the spirometry assessment for another day, within the allowed window. If rescheduling is absolutely not feasible for the participant, spirometry may be performed with a notation in source documentation and TPV system indicating that spirometry was performed after the usual optimised maintenance inhaled therapies were administered.

If the participant has taken rescue SABA or SAMA within 6 hours of the planned spirometry visit, they should ideally remain at the site until such time that the 6 hours withholding time has been reached or return on another day within the visit window. If neither option is feasible for the participant, spirometry may be performed with a notation indicating that the lung volume measurements and/or spirometry were conducted within 6 hours of SABA/SAMA use.

6.5.4 Allowed, Restricted, and Prohibited Medications

As in the predecessor studies, the following medications are allowed, restricted, or prohibited throughout the study duration in this study (unless the time period is specifically indicated, Table 5).

Table 5 Allowed, Restricted, and Prohibited Concomitant Medications

Medication	Allowed/ Restricted/ Prohibited	Details
Maintenance COPD medication: • dual (ICS/LABA, LABA/LAMA, ICS/LAMA) • triple (ICS/LABA/LAMA)	Allowed and should be stable throughout the study	• Participants should have evidence of having been treated with stable doses of optimised maintenance inhaled therapy (ICS/LABA/LAMA triple therapy, or dual therapy if triple is not considered appropriate) for COPD during the study. • Dual or triple maintenance inhaled therapy components may be used in fixed-dose combination(s) or separate devices. SAMA taken at regular scheduled interval (every 6 to 8 hours or at a minimum frequency of 3 times daily) will be considered equivalent to maintenance treatment with LAMA. • Nebulised ICS (eg, budesonide) and/or LABA, or LAMA are accepted as part of maintenance inhaled therapy.
SABA (eg, salbutamol, albuterol, terbutaline, levalbuterol)	Allowed as rescue medication and for treatment of acute COPD exacerbation	Prophylactic use of SABA in the absence of symptoms is discouraged; however, if deemed necessary by the participant and Investigator (eg, prior to planned exercise), it can be used, and such use should be documented in the source documentation and in the eCRF.
SAMA (short acting anticholinergics) (eg, ipratropium bromide)	Allowed	• SAMA taken at regular scheduled interval (every 6 to 8 hours or at a minimum frequency of 3 times daily) will be considered equivalent to maintenance treatment with LAMA. • Rescue use is allowed. • It is important that dose regimen is accurately recorded in the source documentation and in eCRF (scheduled dose versus PRN).
Systemic (oral or injectable) corticosteroids	Restricted	Prohibited throughout the study duration until the end of the follow-up period, with an exception for: • Treatment of acute exacerbations of COPD; • Treatment of an AE where there is no alternative treatment available, for the duration of < 4 weeks. If longer treatment is required, this should be discussed with the AZ study physician to decide on the patient's disposition.
Topical and intraarticular corticosteroids	Allowed	NA

Table 5 Allowed, Restricted, and Prohibited Concomitant Medications

Medication	Allowed/ Restricted/ Prohibited	Details
Immunosuppressive therapy (other than corticosteroids; including but not limited to: methotrexate, troleandomycin, cyclosporine, azathioprine)	Restricted	- Allowed for treatment of adverse events for the duration of < 4 weeks where there is no alternative treatment available. If longer treatment is required, this should be discussed with the Sponsor study physician to decide on the participant's disposition. - Prohibited for any other reason throughout the study.
Allergen immunotherapy	Restricted	Restricted except for stable maintenance dose allergen-specific immunotherapy until the end of the follow-up period. It is recommended that participant does not receive allergen immunotherapy on the same day as IP administration.
Interferon gamma	Prohibited	Prohibited until the end of the follow-up period.
Non-biologic IPs	Prohibited	Prohibited until the end of the follow-up period.
Investigational biologics other than tozorakimab	Prohibited	Prohibited until the end of the follow-up period.
Marketed biologics	Restricted	Prohibited until the end of the follow-up period except: - Injected biologic products for osteoporosis are allowed if on stable dosing prior to the study and not given within 14 days prior or after IP dosing. - Intravitreal injected biologics for neovascular (wet) age-related macular degeneration are allowed if on stable dosing prior to study start and not given within 7 days prior or after IP dosing. - Monoclonal antibodies targeting SARS-CoV-2 viral component for treatment or prophylaxis if not given within 14 days prior or after IP dosing. Above listed exceptions require discussion and approval from the Study Physician prior to IP administration.
Immunoglobulin or blood products	Prohibited	Prohibited until the end of the follow-up period.
Live attenuated vaccines	Prohibited	Prohibited until the end of the follow-up period (note: vaccines with adenoviral vectors that are unable to replicate, eg, ChAdOx1, are not considered live or attenuated).
Vaccination against COVID-19 (either first or subsequent dose)	Restricted	Prohibited within 7 days before or after any dose of IP.
Inactivated vaccine (eg. pneumococcal, inactivated influenza)	Restricted	Prohibited within 7 days before or after any dose of IP.
Antitussive medications	Allowed	NA

Table 5 Allowed, Restricted, and Prohibited Concomitant Medications

Medication	Allowed/ Restricted/ Prohibited	Details
Mucolytic medications (eg, carbocysteine)	Allowed	NA
Roflumilast (Daxas®, Daliresp®) and other PDE4 inhibitors	Allowed only as stable maintenance therapy for COPD	Allowed provided the participant has been treated with stable PDE4 inhibitors in the predecessor study and continues the treatment throughout the study.
Chronic use of macrolide or other antibiotic therapy	Restricted	Allowed provided the participant has been on a stable dose/regimen in the predecessor study and continues the treatment throughout the study.
Xanthines	Restricted	- For theophylline and aminophylline the dose should be either equivalent to theophylline ≤ 400 mg per day or serum drug level should be confirmed to be within the therapeutic range (recommended blood test to confirm < 12 mg/dL). - Other xanthines are allowed provided that the dose does not exceed the maximum locally approved, and the recommended precaution measures are taken.
Marijuana	Prohibited	Use of smoked marijuana is disallowed during the treatment period.
Parenteral herbal or traditional medicines	Prohibited	Any parenteral herbal or traditional medicines are disallowed.
Oral herbal or traditional medicines	Restricted	Oral herbal or traditional medicines are discouraged, and the Sponsor study physician should be consulted if any toxicities are expected.

AE, adverse event(s); COPD, chronic obstructive pulmonary disease; COVID-19, coronavirus disease 2019; eCRF, electronic case report form; ICS, inhaled corticosteroid; IP, investigational product; LABA, long-acting beta agonist; LAMA, long-acting muscarinic antagonist; NA, not applicable; PDE4, phosphodiesterase-4; PRN, pro re nata (as required); SABA, short-acting beta-2 agonist; SAMA, short-acting muscarinic antagonist; SARS-CoV-2, severe acute respiratory syndrome coronavirus 2.

6.5.5 COVID-19 Vaccination

There are no clinical data available to assess the interaction (if any) of tozorakimab with any COVID-19 vaccine.

The Sponsor accepts that vaccination against COVID-19, when and where it is available and if participants are eligible according to applicable guidelines, is in the participants' best interest. Consequently, vaccination during the study is, in general, permitted and any delays in vaccination due to study participation should be minimised.

To ensure the interpretability of safety data, receipt of COVID-19 vaccine (either first or any subsequent dose) is prohibited within 7 days before or after any dose of IP.

For all study participants, on-study COVID-19 vaccination including data regarding the date, dose, brand name, ordinal dose number (eg, first, second, etc), and site of administration (if applicable) (eg, Pfizer-BioNTech COVID-19 Vaccine first dose left deltoid) should be recorded in the participant's medical records and as a concomitant medication in eCRF. The site of administration should be collected if a participant receives a COVID-19 vaccination during study participation and an injection site reaction is observed after the vaccination.

Any AEs suspected to be due to the vaccination should be captured in the AE form including the causality assessment related to COVID-19 vaccine.

6.6 Dose Modification – Not Applicable

This protocol does not allow for alteration from the currently outlined dosing schedule.

6.7 Intervention After the End of the Study

After completing the study, participants should be given locally available SoC at the discretion of the Investigator/treating physician.

7 DISCONTINUATION OF INVESTIGATIONAL PRODUCT AND PARTICIPANT DISCONTINUATION/WITHDRAWAL

7.1 Discontinuation of Investigational Product

Discontinuation from IP does NOT automatically lead to a complete withdrawal from the study. Participants discontinuing from IP are strongly encouraged to continue in the study until the end of treatment period as described in Section 7.1.1.

Before a decision to discontinue a participant from IP is instituted, the Investigator should carefully consider whether continuation on IP or discontinuation of IP will be in the best interest of the participant and whether the issue can be mitigated by postponing or skipping the dose. It is recommended that the Sponsor study physician be consulted before the IP discontinuation transaction takes place. In case of safety concerns, IP administration must be immediately withheld until the final decision is made (refer to Section 6.1.2).

Participants may be discontinued from IP in the following situations:

- 1 Participant decision.
- 2 Risk to participant as judged by the Investigator or the Sponsor.
- 3 Anaphylactic reaction to IP requiring epinephrine administration.
- 4 Pregnancy.

- 5 Eligibility requirement found not to be fulfilled that in the opinion of the Investigator or Sponsor may put the participant at risk. If a participant does not meet all the eligibility criteria but is randomised in error, the Investigator should inform the Sponsor study physician immediately to discuss potential safety concerns and the best interests of the participant, and decide on the participant's further disposition.
- 6 The time elapsed since last dose of IP exceeds 12 weeks.
- 7 Severe noncompliance with the CSP.
- 8 Lost to follow-up.
- 9 Confirmed Hy's Law event.

Refer to the SoA (Section 1.3) for data to be collected at the time of intervention discontinuation and follow-up and for any further evaluations that need to be completed.

The reason for premature discontinuation of IP should be documented in source documentation and recorded in the eCRF. Conditions that require administration of IP to be temporarily withheld are described in Section 6.1.2.

7.1.1 Procedures for Discontinuation of Investigational Product

The Investigator should contact the participant if a decision to stop IP is taken. A participant who decides to discontinue IP will always be asked about the reason(s) and the presence of any AEs. The date of last IP dose, the date of IP discontinuation decision, and the reason(s) for discontinuation should be documented in source documentation and recorded in the eCRF.

Discontinuation of IP will be registered in the RTSM.

All participants discontinued from IP will continue in the study follow-up until the study completion.

Participants who prematurely discontinue IP should return to the study site and complete the E/D visit procedures 12 weeks after the last dose of IP (Section 1.3), or as soon as feasible after 12 weeks passed from the last dose of IP. In cases when the participant is unable or not willing to return to the site for the E/D visits at 12 weeks after the last IP dose, E/D visit assessments can be performed earlier than this, at a time point that is feasible for the participant. It is strongly recommended that all scheduled study visits and procedures are continued until the E/D Visit, if possible. If the participant is unable to attend a scheduled clinic visit, they should be contacted via phone to collect the following information pertaining to the respective visit:

- Concomitant medication
- Smoking status

- Supplemental O₂ status
- AEs
- SAEs
- COPD exacerbation evaluation
- HRU

After the E/D visit, the participant will continue in the study follow-up until Week 28 or Week 52 (unless the E/D visit occurs after the scheduled Visit 8 or Visit 14, respectively). To reduce participant burden, Follow-up Visits for participants who have discontinued IP prematurely will be performed remotely every 12 weeks until Week 28 or Week 52, respectively, (± 1 week) after the E/D visit to collect death, severe COPD exacerbation, and SAE over the phone. Ongoing AEs from the E/D visit will continue to be followed. Such Follow-up Visits may or may not fall within a scheduled visit window. No other assessments, such as blood test, spirometry, or imaging will be required after the E/D visit for discontinued participants. Follow-up may also include contact with a relative or treating physician and collection of information from medical records.

Participants attending the E/D visit at the time or after scheduled Visit 8 or Visit 14 (Week 28 or Week 52) exit the study. At the time of primary reporting, all participants in the treatment period will be asked to continue to the E/D visit 12 weeks after the final dose of IP and exit the study. Participants will not attend scheduled study visits after the last dose of IP until the E/D Visit and there will be no phone follow-up after the E/D visit.

7.2 Participant Withdrawal from the Study

- A participant may withdraw from the study at any time at his/her own request, or may be withdrawn at any time at the discretion of the Investigator for safety, behavioural, compliance, or administrative reasons. This is expected to be uncommon.
- A participant who considers withdrawing from the study must be informed by the Investigator about modified follow-up options (eg, telephone contact, a contact with a relative or treating physician, or information from medical records).
- At the time of withdrawal from the study, if possible, an E/D visit should be conducted, as described in Section 1.3. Refer to Section 1.3 for data to be collected at the time of study withdrawal and follow-up and for any further evaluations that need to be completed. The participant will discontinue the IP and be withdrawn from the study at that time.
- If the participant withdraws consent for disclosure of future information, the Sponsor may retain and continue to use any data collected before such a withdrawal of consent.
- If a participant withdraws from the study, it should be confirmed if he/she still agrees for existing samples to be used in line with the original consent. If he/she requests withdrawal

of consent for use of samples, destruction of any samples taken and not tested should be carried out in line with what was stated in the informed consent and local regulation. The Investigator must document the decision on use of existing samples in the site study records and inform the Global Study Team.

- A participant who withdraws consent will always be asked about the reason(s) and the presence of any AEs. The Investigator will follow-up with a participant as medically indicated. The Sponsor or its delegate will request Investigators to collect information on participants' vital status (dead or alive; date of death when applicable) at the end of the study from publicly available sources, in accordance with local regulations. Knowledge of the vital status at study end in all participants is crucial for the integrity of the study.
- It is the Investigator's responsibility to fully evaluate the participant upon discontinuation or withdrawal to ensure that the participant receives appropriate treatment according to his/her clinical status/condition.

7.2.1 Withdrawal of Informed Consent for Donated Biological Samples

If a participant withdraws consent to the use of donated biological samples, the samples will be disposed of/destroyed, and the action documented. The collection of the biological samples is an integral part of the study where mandatory sampling conform with the clinical safety laboratory test requirements.

A participant who does not allow for further collection or analysis of the mandatory biologic samples critical for clinical safety laboratory tests will need to be discontinued from IP.

Additional actions to be taken in the event of withdrawal of consent for donated biological samples are described in [Appendix C 2](#).

7.3 Lost to Follow-up

A participant will be considered lost to follow-up if he or she repeatedly fails to return for scheduled visits and is unable to be contacted by the study site.

To prevent participants being lost to follow-up, their contact details, including next of kin or other emergency contacts, should be collected initially and updated regularly by the site staff or representative (if allowed by national regulation). The Investigator should educate the participant on the importance of contact with the Investigator throughout the study.

The following actions must be taken if a participant is identified as a potential lost to follow-up:

- The site must make an effort to contact the participant to reschedule the missed visit as soon as possible and counsel the participant on the importance of maintaining the assigned visit schedule.
- Repeated attempts must be made to contact the participant or next of kin/emergency contact via telephone, fax, email, certified letter, and/or through participant locator agencies (if allowed by national regulation).
- Repeated attempts must be made to locate and obtain pertinent medical information for the participant via available hospital records, participant's primary physician, and a publicly available death registry to obtain a current survival status (if allowed by national regulation).
- Where permissible by local law, the ICF will include language to grant the option to employ outside companies to assist in obtaining updated contact information or ascertainment of vital status of lost participants using publicly available sources.

Efforts to reach the participant should continue until the end of the study.

A participant will be classified as lost to follow-up only if he/she has failed to return for the required study visits and his/her vital status remains unknown at the end of the study despite all above listed efforts.

Discontinuation of specific sites or of the study as a whole are handled as part of [Appendix A](#).

8 STUDY ASSESSMENTS AND PROCEDURES

Study procedures and their timing are summarised in the SoA (Section [1.3](#)). Protocol waivers or exemptions are not allowed.

The Investigator will ensure that data are recorded on the eCRF. The EDC system will be used for data collection and query handling.

Before randomisation and first dosing in PROSPERO, all relevant data including, but not limited to, AEs, SAEs, CM, HRU, and COPD exacerbations will be collected as part of the predecessor study and should be recorded and up to date in the predecessor study.

The Investigator ensures the accuracy, completeness, and timeliness of the data recorded and of the provision of answers to data queries according to the Clinical Study Agreement. The Investigator will sign the completed eCRF. A copy of the completed eCRF will be archived at the study site.

Safety concerns should be discussed with the Sponsor immediately upon occurrence or awareness to determine if the participant should continue or discontinue IP.

Adherence to the study design requirements, including those specified in the SoA (Section 1.3), is essential and required for study conduct.

All enrolment evaluations must be completed and reviewed to confirm that potential participants meet all eligibility criteria prior to randomisation transaction. The Investigator will maintain a log to record details of all participants that have undergone enrolment procedures and to confirm eligibility or record reasons for enrolment failure, as applicable.

Procedures conducted as part of the participant's routine clinical management (eg, echocardiography, radiological assessments) and obtained before signing of the ICF may be utilised for enrolment purposes provided the procedures met the protocol-specified criteria and were performed within the timeframe defined in the SoA.

Repeat or unscheduled samples may be taken for safety reasons or for technical issues with the samples.

8.1 Efficacy Assessments

8.1.1 Assessment of COPD Exacerbations

For the purpose of the protocol, a COPD exacerbation will be defined as a worsening in the participant's usual COPD symptoms that is beyond normal day-to-day variation, is acute in onset, lasts 2 or more days (or less if the worsening is so rapid and profound that the treating physician judges that intensification of treatment cannot be delayed), and may warrant a change in regular medication and leads to any of the following:

- Use of systemic corticosteroids for at least 3 days; a single depot injectable (IM) dose of corticosteroids will be considered equivalent to a 3-day course of systemic corticosteroids.
- Use of antibiotics.
- ER or ED visit < 24 hours requiring intensive treatment.
- An inpatient hospitalisation due to COPD (defined as an inpatient admission $\geq$ 24 hours in the hospital, an observation area, the ED, or other equivalent healthcare facility depending on the country and healthcare system).
- Results in death.

An exacerbation will be considered **moderate** if it requires treatment with systemic corticosteroids and/or antibiotics or resulted in an ER/ED visit < 24 hours requiring intensive treatment, and does not result in hospitalisation or death. An exacerbation will be considered **severe** if it results in hospitalisation or death due to COPD.

Participants will be encouraged to contact the study site and/or their primary care physician in case of COPD symptom worsening. The participants will be asked to inform the Investigator of any COPD-related events treated outside of the study site.

The Investigator should record all pertinent findings and symptoms associated with the exacerbation event (eg, dyspnoea, sputum volume, sputum purulence, cough, wheezing, and other COPD-related symptoms and/or findings) and their duration in source documents and in the eCRF COPD exacerbation module.

A vast majority of COPD exacerbations are associated with worsening of the symptoms as described above. Clinical presentations may, however, vary among participants. If it is not possible to capture the symptoms as described above (eg, participant intubated upon arrival to the ED), the Investigator must document the justification for defining the event as an exacerbation and record it in the source documentation and in the eCRF.

Death due to COPD will always be captured as severe COPD exacerbation irrespective of associated symptom worsening.

During a COPD exacerbation, it is important for participants to be treated as deemed appropriate by the treating HCP. However, all supplemental medication used to treat the COPD exacerbation should be discontinued as soon as it is considered safe by the Investigator and participants should return to their pre-exacerbation medication regimen as soon as practical.

It is recommended that medications to treat a COPD exacerbation are not used for more than 14 days. Recent data suggest that treatment with systemic corticosteroids for shorter periods of time results in similar outcomes with less systemic steroid exposure ([Leuppi et al 2013](#)).

The start of an exacerbation is defined as the start date of systemic corticosteroids or antibiotic treatment or hospital admission, whichever occurs earlier, and the end date is defined as the last day of systemic corticosteroids or antibiotic treatment or hospital discharge, whichever occurs later. A single depot injectable dose of corticosteroids will be considered equivalent to a 3-day course of systemic corticosteroids. The corresponding stop date for this treatment will consequently be determined as the date of administration plus 2 days. A minimum 7-day interval is required between the end of treatment (last dose of systemic corticosteroid or antibiotics or the last day of hospitalisation) of one exacerbation to the treatment start of the next exacerbation.

The participant may remain in the study after an exacerbation and continue to receive IP if the Investigator judges that it is medically appropriate for the participant to do so.

Study site evaluations for a COPD worsening may occur as a part of an ordinary onsite visit, via telephone contact, or as an unscheduled visit, if deemed necessary by the Investigator. A copy of the medical record should be obtained for exacerbations evaluated and treated at non-study sites (eg, by the primary care HCP or at an ED/hospital) and details entered into the eCRF exacerbation module in a timely fashion. Changes in concomitant medication due to an exacerbation must be recorded in the source documentation and in the appropriate module of the eCRF.

The investigator site is responsible for documenting the contacts (or other actions) following participant contact in the eCRF.

8.1.2 Assessment of Medication/Therapy Use

At study visits, the participant will be asked a single question, “in the past month, how many days a week have you taken your maintenance medication as scheduled?” The response must be recorded in the source documentation and in the appropriate module of the eCRF.

8.1.3 Spirometry (Pre- and Post-bronchodilator)

General Requirements

All spirometry is performed pre-dose.

Lung function (FEV₁ and FVC) will be measured by spirometry using equipment provided by a central vendor. Spirometry will be performed by the Investigator or authorised delegate according to American Thoracic Society (ATS)/European Respiratory Society (ERS) guidelines ([Graham et al 2019](#)).

The vendor providing central spirometry is responsible for assuring that the spirometer meets ATS/ERS recommendations and that the site personnel who will be performing the testing are properly certified. Spirometry calibration and data quality checks (including monitoring of unexpectedly high variability of results) will be detailed in a separate spirometry procedures manual and in the monitoring plan. Participants who exhibit unexpectedly high variability of FEV₁ will be evaluated by the Investigator for aetiology of variability, necessary follow-up actions, and assessment of continued participant eligibility. The Investigator will discuss unexpectedly high variability with the Sponsor study physician.

Participants should observe the relevant medication withhold periods and other restrictions beforehand (Section 5.3.3, Section 5.3.4, Section 5.3.5, and Section 6.5.3). If any of the restrictions are not met and the session cannot be sufficiently delayed on the day, the onsite visit and dosing should be rescheduled within the allowed visit window. All lung function measurements should take place prior to dosing, if dosing is also scheduled at the same visit.

Usual infection control measures should be followed including use of filters, per the instructions of the central vendor. Any additional infection control measures in accordance with COVID-19 related local and national guidelines should also be followed.

Time of Day for Scheduled Onsite Visit Spirometry

Spirometry testing should be performed at specified visits according to the SoA (Section 1.3). It is recommended that all post-enrolment spirometry assessments should be performed within ± 1.5 hours of the time that the enrolment spirometry in the predecessor study was performed, if possible. For example, if the enrolment spirometry in the predecessor study was started at 8:00 AM, then all subsequent spirometry testing in PROSPERO is recommended to be initiated between 6:30 AM and 9:30 AM.

Spirometry Technique

Detailed procedure for performing spirometry will be described in an instruction manual. Details regarding assessment of the quality of spirometry and the Best Test Review process will also be detailed in the manual and Monitoring Plan.

Spirometry References

The Global Lung Function Initiative equations will be used to determine the predicted normal value (PNV) and are pre-programmed into the spirometer (Quanjer et al 2012).

Forced expiratory volume in one second expressed as percent of the PNV, will be calculated as follows:

$$FEV_1\% \text{ of PNV} = (FEV_1 \text{ measured} / FEV_1 \text{ PNV}) \times 100$$

FEF_{25-75%} will be calculated using a similar method to FEV₁.

Post-BD Spirometry

Post-BD spirometry will be performed at specified visits as detailed in the SoA (Section 1.3). If a patient cannot tolerate post-BD spirometry, or their condition otherwise contraindicated a post-BD spirometry assessment, it can be skipped.

Endpoint maximal BD will be induced using albuterol (90 µg metered dose) or salbutamol (100 µg metered dose) with or without a spacer device up to a maximum of 4 inhalations, which are recommended to be taken within 30 minutes $\pm$ 15 minutes of the final pre-BD spirometry measurement. Post-BD spirometry will be performed 15 to 30 minutes later. If a participant cannot tolerate 4 puffs of albuterol or salbutamol, a lower number of inhalations may be considered at the clinical judgement of the Investigator.

Order of Administration for the COPD Maintenance Inhaled Therapy and IP Relative to the Scheduled Pre- and Post-BD Spirometry

The participant's usual COPD morning maintenance inhaled therapy and rescue medicine must not be given until spirometry assessments are complete. Dosing of IP should also be performed after spirometry is complete.

If the participant cannot tolerate a spirometry assessment during the visit, it can be rescheduled for a different date, preferably within the visit window. There is no need to withhold IP in this case; the IP can be administered during the visit as scheduled.

Record Keeping

A signed and dated copy of the pre- and post-BD printout must be kept at study centre for source data verification. The printout must be marked with the study code, enrolment code, date and time of measurement, and visit number. If a printout cannot be printed, the mean value of the measurements will be recorded in the participant's charts.

8.1.4 Quantitative Computed Tomography Assessment

If feasible, a low radiation dose inspiratory and expiratory CT scans should be performed according to a standardised protocol at the end of treatment period as detailed in the SoA (Section 1.3), for participants with a qualified baseline CT in the predecessor study (as confirmed by the imaging vendor).

For participants who discontinue IP early, it is highly recommended that a CT is performed at the E/D visit instead of at the end of treatment period. However, it should not be performed if less than 24 weeks since PROSPERO randomisation.

CT scanning is optional.

The analysis of the CT images performed by the imaging vendor will enable quantification of mucous plugging, emphysema, and airway wall thickening. The follow-up scan performed in this study will be used to determine the long-term effect of treatment on change from baseline in the CT parameters.

Ideally, new CT scans should be performed in a standardised way that will facilitate a more detailed assessment by the imaging vendor of:

- Mucous plugging
- Airway wall and lumen dimensions
- Emphysema
- Air trapping
- Other quantitative assessments of airway morphometry and lung density

A radiology manual, as well as further training by the imaging vendor, will describe the process of obtaining the CT scans and how to submit these to the imaging vendor. An optimised CT protocol will also be provided to the sites that includes all required scan parameters (same as in the predecessor study).

All CT scans should, if possible be submitted to the central imaging vendor for central review.

8.2 Safety Assessments

Planned time points for all safety assessments are provided in the SoA (Section 1.3).

8.2.1 Physical Examinations, Weight, and Supplemental Oxygen Status

Complete and abbreviated physical examinations will be performed by a licensed HCP (eg, physician, physician's assistant, or licensed nurse practitioner). A complete physical examination will include, but will not be limited to, assessment of general appearance, head, ears, eyes, nose, throat, neck, skin, as well as cardiovascular, respiratory, abdominal, and nervous systems.

An abbreviated physical examination will include, at a minimum, assessments of cardiovascular, respiratory, skin and nervous systems.

Before randomisation and first dosing in PROSPERO, each clinically significant abnormal finding from physical examinations will be recorded as an AE in the predecessor study, and each treatment-emergent abnormality, including injection site reactions identified by assessing SC injection sites, will be recorded as an AE.

Physical examinations, weight, and supplemental O₂ status will be recorded at timepoints specified in the SoA (Section 1.3).

8.2.2 Smoking Status

Smoking status will be assessed at visit by collecting the participant's response to a single yes/no question from study personnel: 'What is your smoking status as of today, do you currently smoke?'.

Use of e-cigarettes or vaping of any products (eg, nicotine, THC) is discouraged during the study. Use of heated tobacco products (eg, IQOS) is discouraged in the former smoker cohort; it should be captured in source documentation and eCRF.

8.2.3 Vital Signs

Vital signs will be assessed pre-dose as described in the SoA (Section 1.3). It is recommended that vital signs are assessed before any interventional study procedures (blood test collection, spirometry, IP administration), and prior to administration of COPD maintenance inhaled

therapy, if possible. The participant should be in a resting position for at least 5 minutes prior to the collection of vital signs, as follows:

- Body temperature (measured in Celsius). Oral measurement is preferred, but other methods are acceptable as per local standards of care.
- Heart (pulse) rate
- Blood pressure (systolic and diastolic)
- Respiratory rate
- Oxygen saturation (SpO₂) measured by pulse oximetry

8.2.4 Electrocardiogram

Single 12-lead ECGs will be performed at timelines as specified in the SoA (Section 1.3). The ECG assessment is recommended to be performed before other interventions (eg, blood sampling, spirometry, and administration of the COPD maintenance medications), but must be before administration of IP. The participant should be in a resting supine position for at least 10 minutes prior to the collection of ECGs.

Electrocardiograms will be recorded using an ECG machine provided by a central vendor that automatically calculates the heart rate and measures PR, QRS, and QT intervals; only this equipment should be used for assessment throughout the participant's participation in the study. Electrocardiogram device software will be used to assess ECG parameters. The ECG results will be transmitted to a central reader. The Investigator or authorised delegate will be responsible for the overall interpretation and determination of clinical significance of any potential ECG findings. Any findings will be recorded in the source documentation and eCRF.

Electrocardiogram variables will be collected, as follows:

- Heart (pulse) rate
- RR interval
- QRS interval
- PR interval
- QT interval

8.2.5 Echocardiography

All participants who were eligible for echocardiography at baseline, and/or received echocardiography performed in a predecessor study or during this study will have a follow-up echocardiography at the end of the treatment period or E/D visit (see Section 1.3).

During the study, additional echocardiography assessments must be performed as soon as practical if new onset or worsening of heart failure is suspected, and may also be performed for other reasons as clinically indicated.

Echocardiography will be performed locally using local equipment. It is strongly recommended that any repeated assessments throughout the study for a given participant are performed on the same equipment by the same echocardiography specialist. LVEF and left ventricular volume will be assessed, and local echocardiography results should include clinical interpretation by the qualified specialist.

The Investigator/delegate will be responsible for the overall interpretation and determination of clinical significance of any potential echocardiography findings and their recording in source notes and eCRF.

8.2.6 Clinical Safety Laboratory Assessments

The nominal timing of blood draws is described in the SoA (Section 1.3). A Laboratory Manual that specifies the procedures for collection, processing, storage, and shipment of samples, as well as laboratory contact information, specific to this clinical research study will be provided to the sites. Clinical laboratory safety tests will be performed in a central clinical laboratory. Urine pregnancy tests, if applicable, will be performed at the site using a licensed test (dipstick). In case of a telemedicine visit, a urine pregnancy test will be performed elsewhere, as appropriate.

Clinically significant abnormal laboratory results should be repeated as soon as possible (preferably within 24 to 48 hours). NT-proBNP should be additionally performed when new onset heart failure is suspected, or as soon as possible thereafter. If clinically indicated, other laboratory tests, including troponin, may be performed locally per local SoC.

The following laboratory variables will be measured (Table 6). For haematology, refer to blinding procedures for eosinophil data in Section 6.3.2.

Table 6 Laboratory Safety Variables

Haematology/Haemostasis - Blood - Haematocrit - Hb - HbA1c - Leukocyte count - Leukocyte differential count (absolute count) ^a - Platelet count - RBC Urinalysis - Microscopy including white blood cell/RBC and casts - Urine - Appearance and colour - Bilirubin - Blood - Glucose - Ketones - Leukocytes - pH - Protein - Specific gravity Endocrinology - Serum - FSH ^b - HCG ^c	Clinical Chemistry - Plasma - Fibrinogen - Serum - Albumin - ALP - ALT - AST - Bilirubin, total - Bicarbonate - Blood urea nitrogen - Creatinine - GGT - Glucose - High sensitivity CRP - NT-ProBNP - Potassium - Sodium - Total Protein - Tryptase (only to be taken in event of suspected anaphylaxis; Appendix G) - Uric acid Pregnancy Dipstick Test - Pregnancy test (urine HCG) ^d
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Note: For serum chemistry, tests for AST, ALT, ALP, and S-bilirubin must be conducted concurrently and assessed concurrently. In case a participant shows an AST **or** ALT $\geq 3 \times$ ULN together with TBL $\geq 2 \times$ ULN refer to [Appendix D](#) 'Actions required in cases of increases in liver biochemistry and evaluation of Hy's Law (HL)', for further instructions.

^a Eosinophil, basophil, and monocyte counts to be redacted.

^b If needed to confirm post-menopausal status in female participants aged < 50 years.

^c Serum HCG test will be performed at Visit 1 for FOCBP. If a urine stick test is positive, then a serum HCG test is required for confirmation.

^d FOCBP only.

ALP, alkaline phosphatase; ALT, alanine aminotransferase; AST, aspartate aminotransferase; CRP, C-reactive protein; FOCBP, female(s) of child-bearing potential; FSH, follicle-stimulating hormone; GGT, gamma-glutamyl transferase; Hb, haemoglobin; HbA1c, haemoglobin A1c; HCG, human chorionic gonadotropin; NT-ProBNP, N-terminal pro-brain natriuretic peptide; RBC, red blood cell(s); TBL, total bilirubin; ULN, upper limit of normal.

Pregnancy Test

The following tests are applicable only to FOCBP participants and will be conducted as per the schedule provided in the SoA (Section 1.3):

- FSH: The test will be performed, for female participants, to confirm postmenopausal status in females < 50 years of age who have been amenorrhoeic for ≥ 12 months.
- Serum beta-HCG: The test will be performed, for FOCBP (analysed at the central laboratory).
- Urine HCG: The dipstick test will be performed for FOCBP. At each treatment visit, the test will be performed before IP administration. In case of a positive dipstick test, IP must be withheld and a confirmatory serum beta-HCG test must be performed.

Other Safety Tests

The following tests may be conducted locally if clinically indicated and as per local SoC:

- Serology: Hepatitis B (HBsAg and anti-HBc) and hepatitis C antibodies
- p24 antigen, HIV-1 and HIV-2 antibodies
- TB test: If applicable, TB test will be performed according to the local SoC as determined by local guidelines.

8.3 Adverse Events and Serious Adverse Events

The Investigator is responsible for ensuring that all staff involved in the study are familiar with the content of this section.

The definitions of an AE or SAE can be found in [Appendix B](#).

Adverse events will be reported by the participant (or, when appropriate, by a caregiver, surrogate, or the participant's legally authorised representative).

The Investigator and any designees are responsible for detecting, documenting, and recording events that meet the definition of an AE.

8.3.1 Time Period and Frequency for Collecting AE and SAE Information

Adverse events will be collected from randomisation (V1) throughout the treatment period and including the follow-up period.

SAEs will be collected from signing the ICF throughout the duration of the study in the predecessor and/or PROSPERO study eCRF.

Before randomisation and first dosing in PROSPERO, AEs and SAEs will be collected as part of the predecessor study and should be recorded and up to date in the predecessor study eCRF.

In case of AEs and SAEs not resolved in predecessor studies at time of randomisation to PROSPERO, they will be also recorded in the PROSPERO study eCRF.

If the Investigator becomes aware of an SAE with a suspected causal relationship to the IP that occurs after the end of the clinical study in a participant treated by him or her, the Investigator shall, without undue delay, report the SAE to the Sponsor.

8.3.2 Follow-up of AEs and SAEs

Any AEs that are unresolved at the participant's last AE assessment in the study are followed up by the Investigator for as long as medically indicated, but without further recording in the eCRF. The Sponsor retains the right to request additional information for any participant with ongoing AE(s)/SAE(s) at the end of the study, if judged necessary.

Adverse Event Variables

The following variables will be collected for each AE:

- Adverse event (verbatim). Note: for COVID-related events, the Investigator should add 'COVID' as part of the verbatim term.
- The date and time when the AE started and stopped
- Intensity
- Whether the AE is serious or not
- Investigator causality rating against the IP(s) (yes or no)
- Action taken with regard to IP(s)
- Adverse event caused participant's withdrawal from study (yes or no)
- Outcome

In addition, the following variables will be collected for SAEs:

- Date AE met criteria for SAE
- Date Investigator became aware of SAE
- Adverse event is serious due to
- Date of hospitalisation
- Date of discharge
- Probable cause of death
- Date of death
- Autopsy performed
- Causality assessment in relation to study procedure(s)
- Causality assessment to other medication

8.3.3 Causality Collection

The Investigator should assess causal relationship between IP and each AE, and answer ‘yes’ or ‘no’ to the question ‘Do you consider that there is a reasonable possibility that the event may have been caused by the IP?’

For SAEs, causal relationship should also be assessed for other medication and study procedures. Note that for SAEs that could be associated with any study procedure the causal relationship is implied as ‘yes’.

A guide to the interpretation of the causality question is found in [Appendix B](#) to the CSP.

8.3.4 Adverse Events Based on Signs and Symptoms

All AEs spontaneously reported by the participant or reported in response to the open question from the study site staff: ‘Have you had any health problems since the previous visit/you were last asked?’ or revealed by observation will be collected and recorded in the source documentation and eCRF. When collecting AEs, the recording of diagnoses is preferred (when possible) to recording a list of signs and symptoms. However, if a diagnosis is known and there are other signs or symptoms that are not generally part of the diagnosis, the diagnosis and each sign or symptom will be recorded separately.

8.3.5 Adverse Events Based on Examinations and Tests

The results from the CSP-mandated laboratory tests and vital signs will be summarised in the CSR.

Deterioration as compared to baseline in protocol-mandated laboratory values, vital signs, or spirometry values should therefore only be reported as AEs if they fulfil any of the SAE criteria, are the reason for discontinuation of treatment with the IP, or are considered to be clinically relevant as judged by the Investigator (which may include, but is not limited to, consideration as to whether treatment or non-planned visits were required or other action was taken with the IP [eg, IP interruption]).

If deterioration in a laboratory value/vital sign is associated with clinical signs and symptoms, the sign or symptom will be reported as an AE and the associated laboratory result/vital sign will be considered as additional information. Wherever possible the reporting Investigator uses the clinical, rather than the laboratory term (eg, anaemia versus low haemoglobin value). In the absence of clinical signs or symptoms, clinically relevant deteriorations in non-mandated parameters should be reported as AE(s).

Deterioration of a laboratory value, which is unequivocally due to disease progression, should not be reported as an AE/SAE.

Any new or aggravated clinically relevant abnormal medical finding at a physical examination as compared with the baseline assessment will be reported as an AE unless unequivocally related to the disease under study.

8.3.6 Adverse Events of Special Interest

Adverse events of special interest are events of scientific and medical interest specific to understanding of tozorakimab and may require close monitoring and rapid communication by the Investigator to the Sponsor. An AESI may be serious or non-serious.

The following AESIs will be particularly monitored in this study:

- Serious hypersensitivity (including Type 1 to 4 hypersensitivity reactions)
- Injection site reactions.
- New onset or progression of heart failure.
- Serious infections: Any event that is defined by the seriousness criteria and falls into the Infections and Infestations System Organ Class.
- Opportunistic infections: Any event, both serious or non-serious, caused by microorganisms that are normally non-pathogenic or rarely pathogenic in individuals with normal immune function or cause an infection of a type or severity not seen in the normal host. Examples of common opportunistic infections include, but are not limited to: Cryptococcosis, cytomegalovirus infection, Pneumocystis jirovecii pneumonia, and tuberculosis.
- Gastrointestinal AEs.

8.3.7 Hy's Law

Cases where a participant shows elevations in liver biochemistry may require further evaluation and occurrences of $AST \text{ or } ALT \geq 3 \times ULN$ together with $TBL \geq 2 \times ULN$ may need to be reported as SAEs. Please refer to [Appendix D](#) for further instruction on cases of increases in liver biochemistry and evaluation of HL.

8.3.8 Disease Under Study

Disease progression can be considered as worsening of participant's condition attributable to the disease for which the IP is being studied (increase in the COPD symptoms and/or increase in the severity of the disease). Symptoms of disease under study are those which might be expected to occur as a direct result of COPD (eg, dyspnoea, mucus production, cough etc.). Events which are unequivocally due to disease under study and/or its progression should not be reported as an AE during the study unless they meet SAE criteria or lead to discontinuation of the IP.

Worsening of COPD that meets the protocol definition for COPD exacerbation (Section [8.1.1](#)) will be captured in the source notes and on a specific eCRF module. A COPD exacerbation

should not be reported as an AE unless it meets the definition of an SAE or leads to discontinuation of the IP.

8.3.9 Reporting of Serious Adverse Events

All SAEs must be reported, whether or not considered causally related to the IP, or to the study procedure(s). All SAEs will be recorded in source documentation and eCRF.

If any SAE occurs during the course of the study, Investigators or other site personnel will inform the appropriate Sponsor representatives within one day, ie, immediately but **no later than 24 hours** of when he or she becomes aware of it.

The designated Sponsor representative will work with the Investigator to ensure that all the necessary information is provided to the Sponsor Patient Safety data entry site **within one calendar day** of initial receipt for fatal and life-threatening events **and within 5 calendar days** of initial receipt for all other SAEs.

For fatal or life-threatening AEs where important or relevant information is missing, active follow-up will be undertaken immediately. Investigators or other site personnel will inform the Sponsor representatives of any follow-up information on a previously reported SAE within one calendar day, ie, immediately but **no later than 24 hours** of when he or she becomes aware of it.

Once the Investigators or other site personnel indicate an AE is serious in the EDC system, an automated email alert is sent to the designated Sponsor representative.

If the EDC system is not available, then the Investigator or other study site staff reports an SAE to the appropriate Sponsor representative by telephone.

The Sponsor representative will advise the Investigator/study site staff how to proceed.

The reference document for definition of expectedness/listedness is the IB for the Sponsor drug.

8.3.10 Pregnancy

All pregnancies and outcomes of pregnancy should be reported to the Sponsor, except for if the pregnancy is discovered before the study participant has received any IP.

8.3.10.1 Maternal Exposure

If a participant becomes pregnant during the course of the study, IP should be discontinued immediately.

Pregnancy itself is not regarded as an AE unless there is a suspicion that the IP under study may have interfered with the effectiveness of a contraceptive medication. Congenital anomalies/birth defects and spontaneous miscarriages should be reported and handled as SAEs. Elective abortions without complications should not be handled as AEs. The outcome of all pregnancies (spontaneous miscarriage, elective termination, ectopic pregnancy, normal birth, or congenital anomaly/birth defect) should be followed up and documented even if the participant was discontinued from the study.

If any pregnancy occurs in the course of the study or within the 14-week post-last dose of IP follow-up period, then the Investigator or other site personnel must inform the appropriate Sponsor representatives within **one day** (ie, immediately but **no later than 24 hours** of when he or she becomes aware of it).

The designated Sponsor representative works with the Investigator to ensure that all relevant information is provided to the Sponsor Patient Safety data entry site **within 1 or 5 calendar days** for SAEs (Section 8.3.9) and **within 30 days** for all other pregnancies.

The same timelines apply when outcome information is available.

The Pregnancy Report (PREGREP) module in the eCRF is used to report the pregnancy and the paper-based PREGOUT form is used to report the outcome of the pregnancy.

8.3.10.2 Paternal Exposure

Non-sterilised male participants must use a male condom from enrolment through 14 weeks following the last dose of IP. It is strongly recommended for the female partner of a male participant to also use a highly effective method of contraception throughout this period. Non-sterilised male participants should also refrain from biologically fathering a child or donating sperm during the same period. Pregnancy of the participant's partners is not considered to be an AE. However, the outcome of all pregnancies (spontaneous miscarriage, elective termination, ectopic pregnancy, normal birth, or congenital anomaly), occurring from the date of the first dose until a specified time period after the last dose and as indicated by previous studies (preclinical and clinical) should, if possible, be followed up and documented in the Pregnancy Report Form. Consent from the partner must be obtained before the Pregnancy Report Form is completed.

8.3.11 Medication Error, Drug Abuse, and Drug Misuse

8.3.11.1 Timelines

If an event of medication error, drug abuse, or drug misuse occurs during the course of the study, then the Investigator or other site personnel informs the appropriate Sponsor representatives within **one calendar day**, ie, immediately but **no later than 24 hours** of when they become aware of it.

The designated Sponsor representative works with the Investigator to ensure that all relevant information is completed within **one** (initial fatal/life-threatening or follow-up fatal/life-threatening) or **5** (other serious initial and follow-up) **calendar days** if there is an SAE associated with the event of medication error, drug abuse, or misuse (see Section 8.3.9) and **within 30 days** for all other events.

8.3.11.2 Medication Error

For the purposes of this clinical study a medication error is an **unintended** failure or mistake in the treatment process for an IMP or AstraZeneca NIMP that either causes harm to the participant or has the potential to cause harm to the participant.

The full definition and examples of a medication error can be found in Appendix B 4.

8.3.11.3 Drug Abuse

Drug abuse is the persistent or sporadic **intentional**, non-therapeutic excessive use of IMP or AstraZeneca NIMP for a perceived reward or desired non-therapeutic effect.

The full definition and examples of drug abuse can be found in Appendix B 4.

8.3.11.4 Drug Misuse

Drug misuse is the **intentional** and inappropriate use (by a study participant) of IMP or AstraZeneca NIMP for medicinal purposes outside of the authorised product information, or for unauthorised IMPs or AstraZeneca NIMPs, outside of the intended use as specified in the protocol and includes deliberate administration of the product by the wrong route.

The full definition and examples of drug misuse can be found in Appendix B 4.

8.3.12 Reporting of Overdose

For this study, any dose of tozorakimab greater than planned dose in the study will be considered an overdose.

- An overdose with associated AEs is recorded as the AE diagnosis/symptoms on the relevant AE modules in the eCRF and on the Overdose eCRF module.
- An overdose without associated symptoms is only reported on the Overdose eCRF module.

If an overdose on a Sponsor IP occurs in the course of the study, the Investigator or other site personnel inform appropriate Sponsor representatives immediately, but **no later than 24 hours** of when he or she becomes aware of it.

The designated Sponsor representative works with the Investigator to ensure that all relevant information is provided to the Sponsor Patient Safety data entry site **within one or 5 calendar**

days for overdoses associated with an SAE (see Section 8.3.9) and **within 30 days** for all other overdoses.

8.3.13 Device Constituent and Third Party Medical Device Deficiencies

Combination products with a device constituent are being provided for use in this study. In order to fulfil regulatory reporting obligations worldwide, the Investigator is responsible for the detection and documentation of events meeting the definitions of a deficiency that occur during the study with the device constituent of tozorakimab combination product.

The definition of a deficiency in the device constituent of a combination product is an inadequacy of a device with respect to its identity, quality, durability, reliability, safety, or performance. Device constituent deficiencies include malfunctions, use errors, and inadequate information supplied by the manufacturer.

Device constituent deficiencies from this study will be collected and monitored to ensure the safety of participants and to improve the safety and performance of the device.

Device constituent deficiencies will not be presented in the CSR, but where required by local regulations, deficiencies will be summarised in the relevant period report.

The Product Complaint Intake Form will be used to collect the deficiency. Additionally, where an SAE has occurred in addition to a malfunction, the Sponsor Clinical Study Medical Device/Device Constituent Report Form will be used.

In addition, third-party spirometers and ECG machines provided by a central vendor are being provided for use in this study. Instructions for medical device use are provided in device-specific Instructions For Use. All deficiencies relating to these third-party medical devices, will be reported to the manufacturer, who will be responsible for fulfilling their regulatory obligations. The site will directly report deficiencies using the manufacturer's contact details in the Instructions for Use.

8.3.13.1 Time Period for Detecting Device Constituent Deficiencies

- Any incident or malfunction related to the IPs will be detected, documented, and reported during all periods of the study in which the IP is used.
- If the Investigator learns of any deficiency at any time after a participant has been discharged from the study, and such incident is considered reasonably related to a device constituent provided for the study, the Investigator will promptly notify the Sponsor.
- New or updated information will be recorded on the originally completed form with all changes signed and dated by the Investigator.

8.3.13.2 Follow-up of a Device Deficiency

- Follow-up applies to all participants, including those who discontinue study intervention.
- The Investigator is responsible for ensuring that follow-up includes any supplemental investigations as indicated to elucidate the nature and/or causality of the deficiency.
- New or updated information will be recorded on the originally completed form with all changes signed and dated by the Investigator.

8.3.13.3 Prompt Reporting of a Device Constituent Deficiency to the Sponsor

- All device deficiencies will be reported to the Sponsor within 24 hours after the Investigator determines that the event meets the protocol definition of a device constituent deficiency.
- The Sponsor Clinical Study Medical Device/Device Constituent Report Form and/or Product Complaint Intake Form will be sent to the Sponsor by email.
- Where an SAE has occurred in addition to the malfunction, the SAE will be recorded in the eCRF as detailed in Section [8.3.9](#).

The Sponsor will be the contact for the receipt of device deficiency reports.

8.3.14 Independent Data Monitoring Committee

An IDMC will be established to oversee the safety of the study participants, ensuring that the studies are being conducted with the highest scientific and ethical standards. The IDMC will periodically review unblinded safety summary tables and listings to evaluate participant safety and make appropriate recommendations, including at the time of interim analysis (refer also to Section [9.5](#) and Section [9.6](#)). The IDMC will function independently of all other individuals associated with the conduct of the studies, including the Sponsor. The committee will operate in accordance with an IDMC Charter.

8.3.15 Independent Adjudication Committee

An Independent Adjudication Committee will be constituted to provide an independent, external, systematic, and unbiased assessment of heart failure events (worsening and new onset), MACE, and all AEs with fatal outcome.

8.4 Human Biological Samples

Instructions for the collection and handling of biological samples will be provided in the study-specific laboratory manual. Samples should be stored in a secure storage space with adequate measures to protect confidentiality. Research on biological samples, including any possible future research, is subject to national regulations and will only be conducted in a particular country if approved in that country. For further details on Handling of Human Biological Samples, refer to [Appendix C](#).

Samples will be stored for a maximum of 15 years from the date of the issue of the CSR in line with consent and local requirements, after which they will be destroyed/repatriated.

- Pharmacokinetic samples will be disposed of after the Bioanalytical Report finalisation or 6 months after issuance of the draft Bioanalytical Report (whichever is earlier).
 - Pharmacokinetic samples may be disposed of or anonymised by pooling. Additional analyses may be conducted on the anonymised, pooled PK samples to further evaluate and validate the analytical method. Any results from such analyses may be reported separately from the CSR.
- Remaining ADA sample aliquots will be retained by the Sponsor or its designee for a maximum of 15 years following issue of the CSR. Additional use includes, but is not limited to, further characterisation of any ADAs, confirmation and/or requalification of the assay, as well as additional assay development work. The results from future analysis will not be reported in the CSR.

8.4.1 Pharmacokinetics

Serum samples will be collected for measurement of serum concentrations of IP as specified in the SoA (Section 1.3).

Samples may be collected at additional time points during the study if warranted and agreed upon between the Investigator and the Sponsor (eg, for safety reasons). The timing of sampling may be altered during the course of the study based on newly available data (eg, to obtain data closer to the time of peak or trough matrix concentrations) to ensure appropriate monitoring.

Serum samples will be used to analyse the PK of tozorakimab. Samples collected for analyses of tozorakimab serum concentration may also be used to evaluate safety or efficacy aspects related to concerns arising during or after the study.

Samples will be collected, labelled, stored, and shipped as detailed in the Laboratory Manual.

8.4.1.1 Determination of Drug Concentration

Samples for determination of IP concentration in serum will be assayed by bioanalytical test sites operated by or on behalf of the Sponsor, using an appropriately validated bioanalytical method. Full details of the analytical method used will be described in a separate bioanalytical report.

Samples from participants who received placebo may be analysed at minimum selected time points to confirm no dosing with tozorakimab took place.

IP concentration information that may unblind the study will not be reported to investigative sites or blinded personnel until the study has been unblinded.

Incurred sample reproducibility analysis, if any, will be performed alongside the bioanalysis of the test samples. The results from the evaluation, if performed, will be reported in a separate Bioanalytical Report.

8.4.2 Immunogenicity Assessments

Blood samples for determination of ADA in serum will be assayed by bioanalytical test sites operated by or on behalf of the Sponsor, using an appropriately validated bioanalytical method. Tiered analyses will be performed to include screening, confirmatory, and titre assay components. All ADA positive samples from the study will be tested for nAb using a validated bioanalytical method. Full details of the methods used will be described in a separate report.

ADA samples may also be further tested for characterisation of ADA response, including possible assessment of neutralising antibody.

Samples will be collected, labelled, stored, and shipped as detailed in the Laboratory Manual.

8.4.3 Pharmacodynamics

There are no established PD measures for tozorakimab and no PD analyses are intended to be performed in this study.

8.5 Human Biological Sample Biomarkers – Not Applicable

8.6 Optional Genomics Initiative Sample – Not Applicable

8.7 Healthcare Resource Utilisation

Healthcare resource utilisation data will be collected by the Investigator/authorised delegate in accordance with the schedule provided in the SoA (Section 1.3) and recorded in source notes as well as the appropriate eCRF module. All visits will collect HRU information with a 4-week recall period.

Overall and COPD-related HRU data will be collected:

- Annualised rate of hospitalisations.
- Annualised rate of ER/ED visits.
- Annualised rate of hospitalisations and ER/ED visits combined.
- Number of days in hospital, measured by total number of days in hospital, general ward, intensive care unit, and critical care unit.
- Rate of hospital readmissions.

Note that cases of hospitalisation also must be reported as an SAE (Section 8.3.9).

9 STATISTICAL CONSIDERATIONS

9.1 Statistical Hypotheses

This study acts as an LTE following an initial 52-week treatment period in predecessor studies OBERON and TITANIA. Analyses will include data from participants enrolled in the predecessor studies whenever applicable in order to minimise the impact of bias caused by dropout and in order to allow inferences to be made from the time of first randomisation and dosing. The primary reporting of the study may take place once both predecessor studies have been unblinded and reported without waiting for all participants to complete the treatment period in this study. The number of severe exacerbations observed in the study by this point will be monitored to estimate statistical power before the decision to unblind is confirmed.

The primary efficacy endpoint is the annualised rate of severe COPD exacerbations over the treatment period incorporating both the predecessor studies and PROSPERO. Here the treatment period will vary between participants (28 or 52 weeks) dependent upon the CSP version under which they were recruited. The primary endpoint will be assessed first in the Primary Population of former smokers, followed by the Overall Population of former or current smokers.

The tozorakimab 300 mg Q8W arm is used in the predecessor studies only for characterisation of the dose-response curve, and not for confirmatory hypothesis testing and, as of CSP Version 2.0, no new participants will enter PROSPERO receiving this regimen. As such, confirmatory hypothesis tests will only be conducted on the comparison of tozorakimab 300 mg Q4W to placebo.

The null hypothesis for tozorakimab is that the annualised severe COPD exacerbation rate on tozorakimab 300 mg Q4W is equal to the severe COPD exacerbation rate on placebo, or that the rate ratio equals 1. The alternative hypothesis is that the severe COPD exacerbation rate on tozorakimab is not equal to the exacerbation rate on placebo, ie:

H_0 : Rate ratio of tozorakimab 300mg Q4W SC versus placebo = 1

H_1 : Rate ratio of tozorakimab 300mg Q4W SC versus placebo $\neq$ 1

Hypothesis testing for the primary analysis, annualised rate of severe COPD exacerbations, will be performed at the 2-sided 5% significance level. If the p-value is less than 0.05 and the treatment effect favours tozorakimab, H_0 will be rejected.

A multiple testing procedure will be applied to control type I error rate at the 2-sided significance level of 5% for the primary and key secondary endpoints; details are provided in

Section 9.1.2. Significance of hypothesis tests will be determined according to the significance levels defined by this testing procedure.

9.1.1 Analysis Methods for Estimands

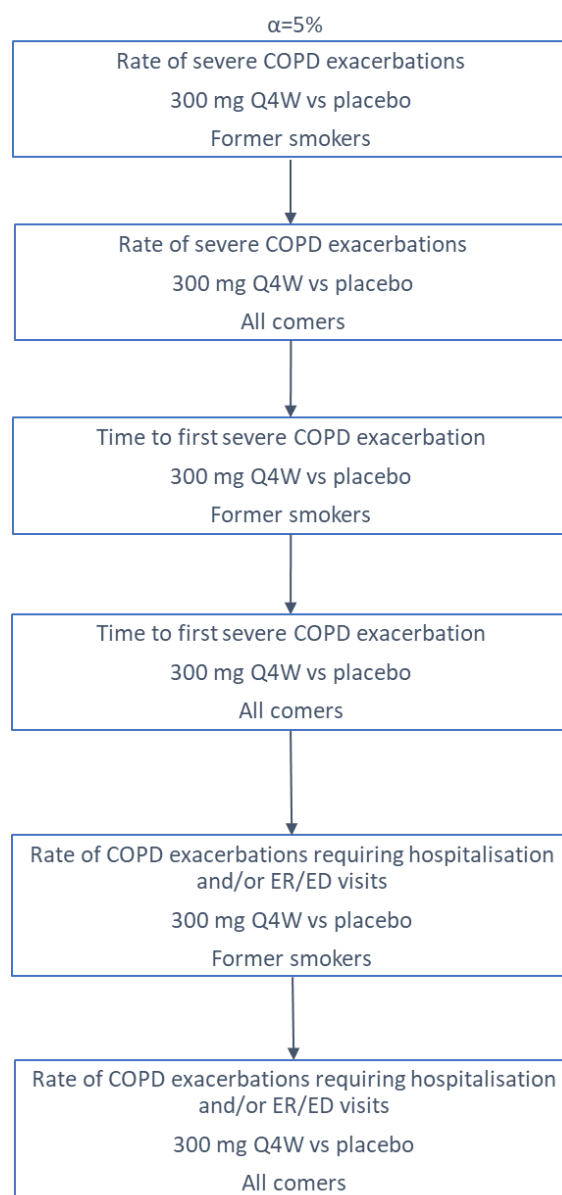
Analyses of the treatment policy estimand will be conducted in the FAS, in which all observed data in participants who were dosed will be used regardless of whether participants remain on randomised treatment, aside from data after planned re-allocation of IP. This will be the primary estimand of interest.

Analysis of the while-on-treatment estimand will be conducted using the on-treatment period applied to the FAS, where only data obtained prior to participants experiencing intercurrent events of discontinuation from randomised treatment and/or initiation of prohibited biologic medication will be used. For those discontinuing randomised treatment, this assumes that efficacy observed on treatment is reflective of what would have occurred had the participant not discontinued treatment. The while-on-treatment estimand will be provided as a supporting analysis for the primary and key secondary endpoints and time to death and will be the main estimand for the remaining secondary endpoints and exploratory endpoints for which assessments are only conducted up until the E/D visit.

9.1.2 Type I Error Control

To account for multiplicity in testing for the primary endpoint (annualised rate of severe COPD exacerbations) and key secondary endpoint (time to first severe COPD exacerbation) for tozorakimab in both the Primary Population of former smokers and the Overall Population of former or current smokers, a testing strategy will be used to strongly control the overall type I error rate at the 2-sided 5% level. The testing strategy is depicted in [Figure 3](#).

Figure 3 Type I Error Control for Comparison of 300 mg Q4W Versus Placebo



COPD, chronic obstructive pulmonary disease; ED, emergency department; ER, emergency room; Q4W, every 4 weeks.

For the primary endpoint in the Primary Population of former smokers, hypothesis testing will first be conducted for the comparison of tozorakimab 300 mg SC Q4W versus placebo at the 2-sided 5% significance level.

If the null hypothesis for this comparison is rejected, with a treatment effect in favour of tozorakimab 300 mg SC Q4W and $p\text{-value} < 0.05$, testing will proceed to the key secondary endpoints for the tozorakimab 300 mg SC Q4W dosing regimen as outlined below, with each

of these tests conducted at a 5% 2-sided significance level. Otherwise, no null hypotheses will be rejected for any other endpoint in either population.

Hierarchical testing will continue at the given significance level until a null hypothesis cannot be rejected; at this time, further testing will stop, and no subsequent null hypotheses in the testing hierarchy will be rejected for that dose.

These hypothesis tests for key secondary endpoints will proceed in each study as follows:

- 1 The primary endpoint for the annualised rate of severe COPD exacerbations will be tested at the 5% 2-sided significance level in the overall All Comer Population (former or current smokers).
- 2 If this null hypothesis is rejected, then the key secondary endpoint of time to first severe COPD exacerbation will be tested in the population of former smokers.
- 3 If this null hypothesis is rejected, then the key secondary endpoint of time to first severe COPD exacerbation will be tested in the Overall Population of former or current smokers (all-comers).
- 4 If this null hypothesis is rejected, then the key secondary endpoint of annualised rate of COPD exacerbation requiring hospitalisation and/or ER/ED visits will be tested in the population of former smokers.
- 5 If this null hypothesis is rejected, then the key secondary endpoint of annualised rate of COPD exacerbation requiring hospitalisation and/or ER/ED visits will be tested in the Overall Population of former or current smokers (all-comers).

9.2 Sample Size Determination

The number of participants entering PROSPERO is determined by the number of participants who complete the treatment period in the predecessor studies (OBERON and TITANIA) and have not been prematurely discontinued from IP. As such the sample size entering PROSPERO is anticipated to be up to 2120 participants, although this is dependent upon the number of participants recruited to the tozorakimab Q8W arm and the proportion of participants who complete the predecessor study and consent to enter this extension so the actual number may be either higher or lower.

However, as the analyses will be conducted using an integrated analysis incorporating the predecessor studies all participants randomised in OBERON and TITANIA will be utilised and therefore the total number of participants utilised in the primary comparison of tozorakimab 300 mg Q4W to placebo in former smokers will be approximately 1272 (636 participants in each arm) or in the overall population of former or current smokers approximately 1696 (848 participants per arm).

In the Primary Population of former smokers, assuming an average duration of follow-up of 1.7 years, a severe COPD exacerbation rate of 0.25 events per participant per year, negative

binomial dispersion of 2.75 and a rate ratio of 0.7 (a 30% relative reduction in rates), this sample size is expected to provide approximately 77% power to demonstrate a statistically significant difference at a 2-sided 5% significance level. The power to demonstrate statistical significance assuming a 35% reduction in rates would be approximately 90% under similar assumptions. The critical value to demonstrate significance would be a relative reduction of approximately 22.5%.

9.3 Populations for Analyses

All participants randomised and dosed in the predecessor studies OBERON and TITANIA will be incorporated into analyses to support this CSR, even if they did not enter the PROSPERO study. This allows valid randomised comparisons to be made from the point of randomisation and first dose onwards. The populations are defined as indicated in [Table 7](#) and treatment groups to be compared within these populations in [Table 8](#).

Table 7 Populations for Analysis

Population/ Analysis Set	Description
Full Analysis Set (FAS)	All participants randomised and receiving at least one dose of IP in OBERON or TITANIA, irrespective of their continued participation into PROSPERO. Participants will be analysed according to their randomised treatment, irrespective of whether or not they have prematurely discontinued, according to the ITT principle. Participants who withdraw consent to participate in the study, who did not enter PROSPERO, or who were re-randomised from placebo to tozorakimab treatment in PROSPERO will be included up to the date of their study termination or re-randomisation.
Safety Analysis Set (SAF)	All participants who have received at least one dose of IP in OBERON or TITANIA, irrespective of their continued participation into PROSPERO. Erroneously treated participants (eg, those randomised to Treatment A but actually given Treatment B) are accounted for in the treatment group of the treatment they actually received. A participant who has on one or several occasions received active IP is classified as active.
PK Analysis Set	Participants who received at least one dose of tozorakimab and had at least one detectable serum concentration measurement post first dose of tozorakimab.

IP, investigational product; ITT, intent to treat; PK, pharmacokinetic(s).

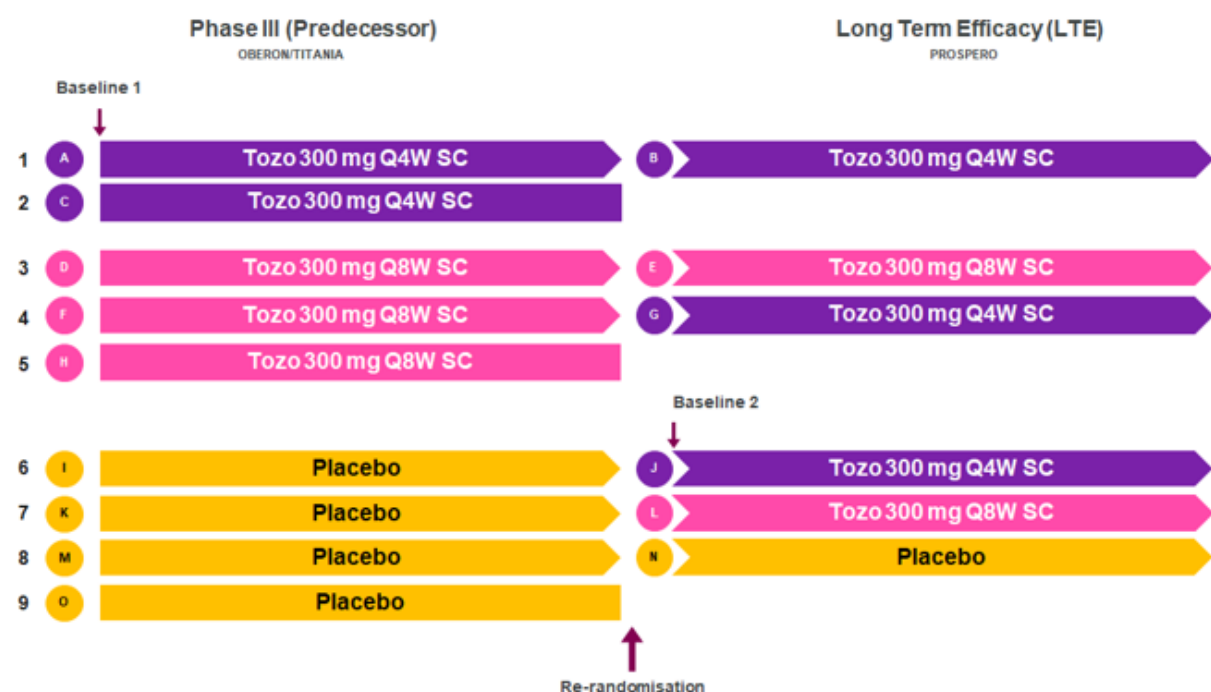
In addition to the populations defined above, subsets of each population (SAF-LTE or FAS-LTE) will be defined, which includes only participants who enrolled and who have received at least 1 dose of IP in the PROSPERO LTE study. As these are based on completers of the predecessor study, they may be subject to selection bias, where less severe participants may be more likely to complete the predecessor study and enrol into the LTE study. These populations will be used for supportive analyses only for making comparisons of the first year of data in the predecessor studies to the second year of data in PROSPERO in a similar set of participants. The number of participants discontinuing IP and patterns of discontinuations will be considered when interpreting the data. Reporting will be performed over particular periods

of follow-up, either on-study or on-treatment (according to the planned 28- or 52-week treatment duration for the given participant and allowing for inclusion of the washout follow-up period up to the E/D visit). These periods will be precisely defined in the SAP. The full on-study period can only be used for the severe exacerbation, hospitalisation and death data collected in the post discontinuation telephone calls as described in Section 7.1.1.

9.3.1 Treatment Groups for Analysis

There are 9 possible combinations of treatments that participants could receive across the predecessor and LTE studies, as shown in Figure 4, where C, H, and O indicate participants who were not dosed in PROSPERO.

Figure 4 Treatment Groups for Analysis



Tozo, tozorakimab; Q4W, every 4 weeks; Q8W, every 8 weeks; SC, subcutaneous(ly).

From these treatment assignments, various possible treatment groups can be formed for comparison, as described in Table 8. When applied to the Safety Analysis Set, reporting will be according to treatments actually received.

Table 8 Treatment Groups for Analysis

Treatment Group	Description
Randomised treatment groups, for primary interpretation from first dose in predecessor studies:	
Randomised tozorakimab 300 mg SC Q4W	Study participants randomised to tozorakimab 300 mg SC Q4W in predecessor studies, using data from start of the predecessor study to the end of the PROSPERO study. A, B, and C in Figure 4
Randomised tozorakimab 300 mg SC Q8W	Study participants randomised to tozorakimab 300 mg SC Q8W in predecessor studies, using data from start of the predecessor study to the end of the PROSPERO study or until re-allocation to tozorakimab 300 mg Q4W. D, E, F, and H in Figure 4
Randomised placebo	Study participants randomised to placebo in predecessor studies, using data from start of the predecessor study up until the switch to tozorakimab for participants re-randomised to tozorakimab or to the end of the PROSPERO study for participants re-randomised to placebo in PROSPERO study. I, K, M, N, and O in Figure 4
Continuation treatment groups, for comparing observations in the predecessor studies to the LTE study amongst the same set of participants	
Tozorakimab 300 mg SC Q4W predecessor + tozorakimab 300 mg SC Q4W LTE	Study participants randomised and dosed with tozorakimab 300 mg SC Q4W in predecessor studies, who were also dosed with Q4W in PROSPERO, using data from start of the predecessor study to the end of the PROSPERO study. A and B in Figure 4
Tozorakimab 300 mg SC Q8W predecessor + tozorakimab 300 mg SC Q8W LTE	Study participants randomised and dosed with tozorakimab 300 mg SC Q8W in predecessor studies, who were also dosed with Q8W in PROSPERO, using data from start of the predecessor study to the end of PROSPERO study. D and E in Figure 4
Tozorakimab 300 mg SC Q8W predecessor + tozorakimab 300 mg SC Q4W LTE	Study participants randomised and dosed with tozorakimab 300 mg SC Q8W in predecessor studies and re-allocated to and dosed with tozorakimab SC Q4W in PROSPERO study. F and G in Figure 4
Placebo predecessor + placebo LTE	All participants randomised to placebo in the predecessor studies, and who were re-randomised to and dosed with placebo in PROSPERO. Will include all data from first dose in predecessor studies until end of the PROSPERO study. M and N in Figure 4

Table 8 Treatment Groups for Analysis

Treatment Group	Description
Placebo predecessor + tozorakimab 300 mg SC Q4W LTE	All participants originally randomised to placebo in the predecessor studies and re-randomised to and dosed with tozorakimab 300 mg SC Q4W in PROSPERO. When presented by visit or study period (predecessor versus PROSPERO) will include all data from first dose/randomisation in the predecessor studies to end of the PROSPERO study. I and J in Figure 4 Otherwise, if not split by time only data from re-randomisation/first dose in PROSPERO will be included. J only in Figure 4
Placebo predecessor + tozorakimab 300 mg SC Q8W LTE	All participants originally randomised to placebo in the predecessor studies and re-randomised to and dosed with tozorakimab 300 mg SC Q8W in PROSPERO. When presented by visit or study period (predecessor versus PROSPERO) will include all data from first dose/randomisation in the predecessor studies to end of the PROSPERO study allocated. K and L in Figure 4 Otherwise, if not split by time only data from re-randomisation/first dose in PROSPERO will be included. L only in Figure 4
All exposed treatment groups, considering all participants receiving active treatment	
All tozorakimab 300 mg SC Q4W	All participants randomised to tozorakimab 300 mg SC Q4W in the predecessor studies and participants randomised to tozorakimab 300 mg SC Q4W in the LTE study as their first dosing regimen of tozorakimab. It will include all data from time of first dose/randomisation to tozorakimab 300 mg SC Q4W. A, B, C, and J in Figure 4
All tozorakimab	All participants randomised to any dose regimen of tozorakimab in the predecessor studies or the LTE study. It will include all data from time of first dose/randomisation to tozorakimab. A, B, C, D, E, F, G, H, J, and L in Figure 4
All placebo only	All participants randomised to placebo in the predecessor studies and re-randomised to placebo in the LTE study. It will include all data from time of first dose/randomisation to placebo. M, N, and O in Figure 4

LTE, long-term extension; Q4W, every 4 weeks; Q8W, every 8 weeks; SC, subcutaneous(ly).

The FAS using the randomised treatment groups from the start of the predecessor studies will be the primary method used for analysis of efficacy endpoints using the time to event- or exposure-adjusted rates. This allows a randomised comparison of all IP-dosed participants according to their randomised treatment from their first dose of IP in the predecessor studies onwards for as long as randomisation would have allowed them to stay on the same treatment so as to address the treatment policy estimand.

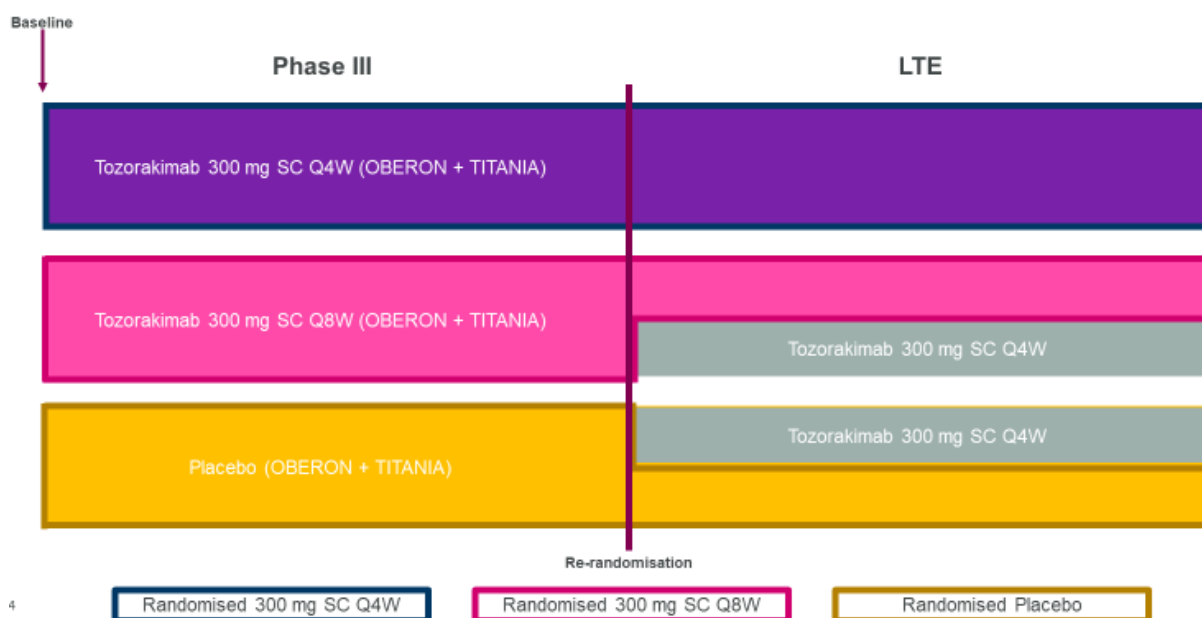
For participants switching from the 300 mg Q8W treatment arm to tozorakimab 300 mg Q4W upon entry to PROSPERO only data up until the point of switching dose regimen will be used for comparisons using randomised treatment groups.

For the All exposed treatment groups, a single All tozorakimab group will be produced as well as a treatment group for tozorakimab 300 mg Q4W.

Supportive summaries will be produced using on-treatment periods to address the while on treatment estimand.

The FAS is illustrated in [Figure 5](#).

Figure 5 Full Analysis Set: Randomised Tozorakimab 300 mg SC Q4W, Randomised Tozorakimab 300 mg SC Q8W, and Randomised Placebo (Main Comparison)



LTE, long-term extension; Q4W, every 4 weeks; Q8W, every 8 weeks; SC, subcutaneous(ly); Tozo: tozorakimab.

Similar treatment groups and periods based upon the Safety Analysis Set will be used for summaries of exposure-adjusted AE rates from the time of first dose IP in the predecessor studies. Safety summaries will also be produced using the all-exposed treatment groups.

9.4 Statistical Analyses

The SAP will be finalised prior to the first IDMC meeting, BSSR, or interim analysis, (whichever occurs first) involving data from this study and will describe the participant populations to be included in the analyses, and procedures for accounting for missing, unused, and spurious data along with a more technical and detailed description of the statistical analyses described in this section. Any deviations from this plan will be reported in the CSR.

This section is a summary of the planned statistical analyses of the most important endpoints including primary and key secondary endpoints.

Analyses will be performed by Sponsor or its representatives.

9.4.1 General Considerations

The primary endpoint is the annualised rate of severe COPD exacerbations. The primary analysis will be conducted in the former smoker population of the FAS population pooling data from the predecessor studies and PROSPERO study, comparing the Randomised tozorakimab 300 mg SC Q4W, and Randomised placebo treatment groups as described in Section 9.3. Participants randomised to placebo in the predecessor studies but re-randomised to tozorakimab in the PROSPERO study will only have data from the predecessor studies used in the primary analysis, with data for the remaining period imputed as if the participants had remained on placebo, given that this data is missing by design and determined by randomisation. This will provide an estimate of the treatment policy estimand in the scenario where all participants had remained on their initially randomised treatment for the duration of the study.

A supportive on-treatment analysis of severe exacerbation endpoints will also be produced.

9.4.2 Efficacy

9.4.2.1 Primary Endpoint

The primary endpoint is the annualised rate of severe COPD exacerbations. This will be assessed for tozorakimab versus placebo primarily in the population of former smokers. Analyses will be conducted in the FAS, using all available data, where all observed data will be used regardless of treatment or study discontinuation, background treatment switching, or change in smoking status, aside from data after planned re-allocation of IP. The use of this population will provide an estimate of the treatment policy estimand.

The rate of severe COPD exacerbations in the tozorakimab 300 mg Q4W group will be compared to the exacerbation rate in the placebo group using a negative binomial model. The response variable in the model will be the number of severe COPD exacerbations over the duration of the treatment period after applying dropout based multiple imputation. The model will include treatment, predecessor study, region, maintenance inhaled therapy, and baseline COPD exacerbation history as categorical covariates, and baseline post-BD percent predicted FEV₁ and log blood eosinophil count as continuous covariates. The log of the follow-up time will be used as an offset variable in the model with this taken or imputed up to the end of the planned treatment period.

The estimated treatment effect (ie, the rate ratio of tozorakimab versus placebo), corresponding 95% CI, and 2-sided p-value for the rate ratio will be presented. In addition, the exacerbation rate in each treatment group will be presented.

Any COPD exacerbation that occurs within 7 days of the last dose of systemic corticosteroids or antibiotics or the last day of hospitalisation for a prior COPD exacerbation will be combined and counted as a single exacerbation event and as the maximum severity between the individual events. Further details on combining exacerbation events will be provided in the SAP. Similar criteria will be applied to all exacerbation endpoints.

Participants who remain in PROSPERO at the time of data cutoff will have data utilised up to this point. For participants lost to follow-up, or prematurely withdrawn from the study, data from the date of last contact to the date of data cutoff or end of planned treatment period (28 or 52 weeks, respectively) will be imputed. Similarly, participants re-randomised from placebo to active therapy will have data imputed from the end of the predecessor study to the date of data cutoff or end of planned treatment period. Only data prior to participants being switched off the 300 mg Q8W arm will be utilised.

Handling of Missing Data

In the primary analysis multiple imputation will be carried out where event counts post study withdrawal will be imputed conditional upon the observed number of events prior to the withdrawal ([Gottlow et al 2015](#), [Wan et al 2015](#)). The missing data for participants who withdraw early will be imputed based on dropout reason-based multiple imputation (DRMI) where event counts after withdrawals unrelated to treatment are imputed under missing at random assumptions and those related to treatment are imputed under missing not at random assumptions. Event counts post withdrawal for participants in the tozorakimab treatment arm who dropped out for a treatment-related reason are imputed conditional upon the negative binomial rate parameter for the placebo control arm given baseline covariates. For all other participants the rate conditional upon their randomised treatment will be used. Reasons for dropout will be categorised based upon blinded data. Imputation will be applied to each of multiple bootstrap datasets before conducting the primary analysis model and estimates based upon each dataset will then be combined using Rubin's rules ([Fleming 2011](#), [Ratitch et al 2013](#)).

9.4.2.2 Secondary Endpoints

Analysis for all secondary endpoints will be conducted in the FAS in the Primary Population of former smokers and in the Overall Population of former or current smokers.

Annualised Rate of Severe COPD Exacerbations

The annualised rate of severe COPD exacerbations will be analysed as a key secondary endpoint in the All Comer Population of former or current smokers, using similar methods as in the primary analysis (Section 9.4.2.1), with the addition of a covariate for smoking status.

Time to First Severe COPD Exacerbation

A key secondary endpoint is the time to first severe COPD exacerbation over the treatment period. Analyses will be conducted in the former smoker population, followed by the Overall Population of former or current smokers.

The time to first severe COPD exacerbation is defined as:

$$\text{Start date of first severe COPD exacerbation} - \text{Date of randomisation} + 1$$

The hazard ratio of severe COPD exacerbations in tozorakimab 300 mg Q4W group relative to the placebo group will be derived using a Cox-proportional hazards model. The response variable in the model will be the time to first severe COPD exacerbation. The model will include covariates of treatment group, predecessor study, region, maintenance inhaled therapy, and baseline COPD exacerbation history as categorical covariates, and baseline post-BD percent predicted FEV₁ and log blood eosinophil count as continuous covariates. Smoking status will be included as a covariate in analyses in the Overall Population of former or current smokers.

Estimated adjusted hazard ratios relative to the placebo comparator will be displayed along with the associated 2-sided 95% CI and p-values.

Participants who remain in PROSPERO at the time of data cutoff and do not experience a severe COPD exacerbation will be censored at the date of data cutoff. For participants who did not experience a COPD exacerbation before being lost to follow-up, or prematurely withdrawn from the study, data from the date of last contact to the date of data cutoff or end of planned treatment period (28 or 52 weeks, respectively) will be imputed before censoring is applied. Similarly, participants re-randomised from placebo to active therapy will have data imputed from the end of the predecessor study to the date of data cutoff or end of planned treatment period. Participants switched off the 300 mg Q8W arm will be censored at the point of switching dose regimen in the primary efficacy analyses.

Annualised Rate of COPD Exacerbation Requiring Hospitalisation and/or ER/ED Visits

The annualised rate of COPD exacerbations requiring hospitalisation and/or ER/ED visits will be analysed as a key secondary endpoint in the former smoker population, followed by the Overall Population of former or current smokers, using similar methods as in the primary analysis.

Time to First Moderate to Severe COPD Exacerbation

The time to first moderate to severe COPD exacerbation will be analysed as a secondary efficacy endpoint in former smokers. Participants are not asked to fill in a daily symptom diary in this study and so once participants discontinue randomised treatment the identification of moderate COPD exacerbations may not be systematic. Therefore, only the while-on-treatment estimand will be addressed for endpoints relating to moderate and severe COPD exacerbations using the on-treatment period.

A Cox proportional hazard model will be fitted with similar covariates and methods to the while-on-treatment supportive analysis of the primary endpoint as described in Section 9.4.2.1. Hazard ratios, 95% CI, and p-values will be reported as well as the proportion of participants with an event. The time to first moderate or severe COPD exacerbation among former or current smokers will be analysed in a similar manner.

Annualised Rate of Moderate-to-Severe COPD Exacerbations

The secondary endpoint of annualised rate of moderate to severe COPD exacerbations during the on-treatment period will be assessed for each dose of tozorakimab versus placebo in former smokers in a similar manner to while-on-treatment supportive analysis of the annualised rate of severe COPD exacerbations as described above. A similar analysis in former or current smokers will be conducted. Smoking status will be included as a covariate in analyses in the all-comer population (ie, former or current smokers).

Time to Death (All-cause Mortality and Respiratory-related Mortality)

The incidence of death, all cause and respiratory related, will be summarised in the FAS. If there are sufficient number of events, a Cox proportional hazards model may be used to compare time to event for these mortality endpoints between treatment groups. The number and percentage of participants who die from any cause will be presented by treatment group.

Pharmacokinetics

Serum concentrations of tozorakimab will be summarised in the PK analysis set using descriptive statistics at each visit applying the continuation treatment groups comparing the data from participants who were first dosed with tozorakimab in predecessor studies separately from those participants who newly received active treatment in PROSPERO. Details of the PK evaluation or any additional PK or PK/PD analyses will be outlined in the SAP.

Immunogenicity

Immunogenicity information will be analysed for tozorakimab and placebo treatments using the information for both predecessor and PROSPERO studies.

The ADA and corresponding titre will be summarised at each visit by treatment group. Additionally, the ADA status across studies will also be summarised using continuation treatment groups and ADA categories. The potential effects of ADA status and ADA titre on PK, efficacy, and safety will be evaluated.

9.4.2.3 Exploratory Endpoints

Refer to the SAP for details on analyses of exploratory endpoints.

Change from Baseline in Pre-dose/Pre-BD FEV₁

Change from baseline in pre-dose/pre-BD FEV₁ will be analysed between tozorakimab and placebo using a repeated measures linear model in the on-treatment period of the FAS. The dependent variable will be the change from baseline in pre-dose/pre-BD FEV₁ total score at post-baseline visits up to the Follow-up Visit. Treatment, visit, treatment-by-visit interaction, predecessor study, region, maintenance inhaled therapy, and the number of exacerbations in the previous year will be fitted as categorical covariates and baseline FEV₁ and post-BD percent FEV₁ predicted, and log baseline blood eosinophil count as will be fitted as continuous covariates. An unstructured variance-covariance matrix will be used to model correlation per participant. If this model fails to converge, an alternative structure will be employed as pre-specified in the SAP.

Contrasts will be used to produce treatment effect estimates at each visit. This will be reported alongside the 2-sided 95% CI and p-value.

Rate of Decline in Trough and Post-BD FEV₁

The rate of decline in trough and post-BD FEV₁ will be analysed between tozorakimab and placebo using the on-treatment period of the FAS. Pre-dose trough FEV₁ will be analysed with a linear mixed model with random participant slopes of FEV₁ versus time and random participant intercepts. The time is calculated as weeks between the date of the first dose of study treatment and the date of the visit:

$$(Date\ of\ Visit - Date\ of\ First\ Dose + 1) / 7$$

Slopes from Week 4 of the predecessor studies will be considered for the analysis. An unstructured correlation matrix will be employed for these 2 random effects. The model will fit a distinct fixed mean for each treatment and a distinct fixed slope (of FEV₁ versus time) for each treatment. The model will include the interaction between treatment (as a categorical covariate) and time (as a continuous covariate). Other covariates in the model will be percentage baseline FEV₁ and baseline blood eosinophil count as continuous covariates and treatment, predecessor study, region, maintenance inhaled therapy, and the number of exacerbations in the previous year as categorical covariates. The rate of decline (the negative of the slope) will be estimated and compared between treatments for the treatment policy estimand. Similar models will be fitted for the former smoker and Overall Population of

former or current smokers, with smoking status included as a fixed covariate and as an interaction with time.

Change from Baseline in CT Mucous Score and Imaging Parameters

Analyses for all CT endpoints will be reported outside of the CSR.

Change from baseline in the CT mucus score will be summarised descriptively by treatment group and imaging parameters will be analysed using the methods to be described in SAP.

COPD-Related Healthcare Resource Utilisation

All HRU endpoints will be summarised descriptively such as with annualised rates or percentage by treatment group.

9.4.3 Safety Endpoints

Adverse Events

Safety summaries will be produced using alternative ways of handling participants re-randomised from placebo to active treatment such that participants only appear in at most one treatment arm within each summary table. Using the Safety Analysis Set, summaries will be produced using the treatment actually received in the predecessor study and excluding data from placebo participants re-randomised to active therapy. Summaries will be produced while-on-treatment for all AE types, including data up to the E/D visit. Summaries will also be produced while-on-study for SAEs and deaths. In addition, summaries will be produced using the All Exposed treatment groups applied to the Safety Analysis Set from the time of first dose of tozorakimab for any participants exposed, compared to participants never exposed to tozorakimab.

AEs during the Treatment Period will be summarised by the number of participants experiencing an event and using exposure-adjusted event rates. Adverse events will be tabulated at the level of the Medical Dictionary for Regulatory Activities (MedDRA) System Organ Class and Preferred Term. The version of MedDRA current at the time of database lock will be used in the tabulations and listings.

Adverse events will be presented for each treatment group by System Organ Class and Preferred Term covering number and percentage of participants reporting at least one event and number of events where appropriate.

An overview of AEs will be presented for each treatment group, the number, exposure adjusted rate, and percentage of participants with any AE, AEs with outcome of death, serious AEs, and AEs leading to discontinuation of IP.

Separate AE tables will be provided taking into consideration relationship as assessed by the Investigator, intensity, seriousness, death, and events leading to discontinuation of IP.

An additional table will present number, exposure adjusted rate, and percentage of participants with the most common AEs (frequency of > 3% or other appropriate threshold as defined in the SAP).

In accordance with the requirements of the FDA, a separate table will present non-serious AEs occurring in more than 5% of patients in any treatment group.

Key patient information will be presented for participants with AEs with outcome of death, serious AEs, and AEs leading to discontinuation of IP.

An AE listing for the safety analysis set will cover details for each individual AE.

Full details of AE analyses will be provided in the SAP.

The following events are considered treatment emergent:

- Adverse events with an onset date on or after first dose of IP
- Worsening of pre-existing events on or after first dose of IP

Vital Signs

Summary statistics (mean, median, SD, and range) of change from baseline will be tabulated by vital sign parameter and treatment for each scheduled assessment time. For vital signs, baseline will be defined as the average of the values prior to dosing on the day of randomisation, or prior to first dose for the All Exposure treatment groups. In addition, potentially clinically significant values will be identified and summarised.

Clinical Laboratory Values

Summary statistics (mean, median, SD, and range) of change from baseline values will be tabulated for each treatment and each assessment time. For clinical laboratory measurements, baseline will be defined as the last available value prior to randomisation, or prior to first dose for the All Exposure treatment groups. Potentially clinically significant values will be identified and summarised.

ECGs

Summary statistics (mean, median, SD, and range) for absolute values and change from baseline will be tabulated by ECG parameter and treatment for each scheduled assessment time. For ECG parameters, baseline values will be defined as the last value obtained prior to

randomisation, or prior to first dose for the All Exposure treatment groups. In addition, potentially clinically significant values will be identified and summarised.

9.4.4 Other Analyses

In addition to the analyses based upon smoking status (current versus former), subgroup analyses may be carried out by other stratification factors. Details will be outlined in the SAP.

9.5 Interim Analysis

An IDMC will oversee a futility analysis in the predecessor studies to assess whether the studies can be stopped for futility and may oversee an interim analysis in this study and the predecessor studies based upon the primary endpoints to assess whether the studies can be reported early for overwhelming efficacy. The IDMC will also periodically review unblinded safety summary tables and listings to evaluate participant safety and make appropriate recommendations, including at the time of interim analysis (Section 9.6). The committee will operate in accordance with an IDMC charter.

Implications of Interim Analysis for Futility in OBERON and TITANIA

If pre-defined stopping criteria in OBERON and TITANIA futility analysis is met, and the predecessor studies are stopped for futility, PROSPERO study will also be stopped.

Interim Analysis for Overwhelming Efficacy

An interim analysis for efficacy may also be conducted and an alpha adjustment will be made only if in the Primary Population of former smokers analysis, the following point is reached:

- At least 80% information fraction for the rate of moderate or severe exacerbations is reached in each predecessor study, and
- At least 70% information fraction for the rate of severe exacerbations are observed in a pooled analysis of this study and the 2 predecessor studies, and
- This is reached by at least 6 months prior to the point where every participant has been treated for one year in the predecessor studies.

Decision guidance will be provided to the IDMC. The interim analysis could lead to early reporting of PROSPERO if the interim analysis for efficacy is met. The study would still be continuing with sites and patients remaining blinded to collect further longer-term safety information.

The significance level for the interim analysis will be set to 0.005, one-sided, regardless of expected information fraction available. The significance level for the final primary analysis will then be adjusted based on the fraction of information available at the interim relative to final analysis such that the Type I error for the primary analysis is controlled at 0.025 one-sided, or equivalently 2-sided 0.05 for the tozorakimab Q4W comparison with

placebo. The information fraction is defined by the number of severe exacerbations. If the study does not stop for efficacy at the interim analysis, then the adjusted significance levels to be used at final reporting will be calculated using the bivariate normal method described by [Stone 2010](#) based upon the observed information. Further details will be provided in the IDMC charter and in the SAP and Interim Analysis Plan.

9.6 Independent Data Monitoring Committee

An IDMC will be established to oversee the safety of the study participants and make appropriate recommendations regarding further study conduct based on the available data, including regular review of safety data and at planned interim analyses where the opportunity to report on the study early for overwhelming efficacy may be considered.

The IDMC will also oversee a futility analysis of the predecessor studies which may impact PROSPERO.

Details of the composition of the IDMC, the data to be reviewed, and the frequency of the meetings can be found in the IDMC charter.

10 SUPPORTING DOCUMENTATION AND OPERATIONAL CONSIDERATIONS

Appendix A Regulatory, Ethical, and Study Oversight Considerations

A 1 Regulatory and Ethical Considerations

This study will be conducted in accordance with the protocol and with the following:

- Consensus ethical principles derived from international guidelines including the Declaration of Helsinki as amended at 64th WMA General Assembly, Fortaleza, Brazil, October 2013 and Council for International Organisations of Medical Sciences (CIOMS) International Ethical Guidelines.
- Applicable ICH GCP Guidelines.
- Applicable laws and regulations.
- The protocol, revised protocol, ICF, IB, and other relevant documents (eg, advertisements) must be submitted to an IRB/IEC by the Investigator and reviewed and approved by the IRB/IEC before the study is initiated.
- Any revised protocol will require IRB/IEC and applicable Regulatory Authority approval before implementation of changes made to the study design, except for changes necessary to eliminate an immediate hazard to study participants.
- The Sponsor will be responsible for obtaining the required authorisations to conduct the study from the concerned Regulatory Authority. This responsibility may be delegated to a CRO, but the accountability remains with the Sponsor.
- The Investigator will be responsible for providing oversight of the conduct of the study at the site and adherence to requirements of 21 CFR 312.120, ICH guidelines, the IRB/IEC, European Regulation 536/2014 for clinical studies (if applicable), European Medical Device Regulation 2017/745 for clinical device research (if applicable), and all other applicable local regulations.

Regulatory Reporting Requirements for SAEs

- Prompt notification by the Investigator to the Sponsor of an SAE is essential so that legal obligations and ethical responsibilities towards the safety of participants and the safety of a study intervention under clinical investigation are met.
- The Sponsor has a legal responsibility to notify both the local regulatory authority and other regulatory agencies about the safety of a study intervention under clinical investigation. The Sponsor will comply with country-specific regulatory requirements relating to safety reporting to the regulatory authority, IRB/IEC, and Investigators.
- In the European Union, the Sponsor will comply with safety reporting requirements and procedures as described in the European Clinical Trials Regulation (EU) No 536/2014. All Suspected Unexpected Serious Adverse Reactions (SUSARs) to investigational medicinal product will be reported to the EudraVigilance database within the required regulatory timelines
- For all studies except those utilising medical devices, Investigator safety reports must be prepared for SUSARs according to local regulatory requirements and Sponsor policy and forwarded to Investigators as necessary.

- Adherence to European Medical Device Regulation 2017/745 for clinical device research (if applicable), and all other applicable local regulations.
- An Investigator who receives an investigator safety report describing an SAE or other specific safety information (eg, summary or listing of SAEs) from the Sponsor will review and then file it along with the IB and will notify the IRB/IEC, if appropriate according to local requirements.

Regulatory Reporting Requirements for Serious Breaches, Including Personal Data Breaches

- Prompt notification by the Investigator to the Sponsor of any (potential) serious breach of the protocol or regulations (including if a data breach compromises the integrity, confidentiality, or availability of the personal data of participants) is essential so that legal and ethical obligations are met.
 - A “serious breach” means a breach likely to affect to a significant degree the safety and rights of a participant or the reliability and robustness of the data generated in the clinical study.
- If any (potential) serious breach occurs in the course of the study, Investigators or other site personnel will inform the appropriate Sponsor representatives immediately after he or she becomes aware of it.
- In certain regions/countries, the Sponsor has a legal responsibility to notify both the local regulatory authority and other regulatory agencies about such breaches.
- The Sponsor will comply with country-specific regulatory requirements relating to serious breach reporting to the regulatory authority (including data protection authorities and, if applicable, affected participants in case of a personal data breach), IRB/IEC, and Investigators. If EU CTR 536/2014 applies, the Sponsor is required to enter details of serious breaches into the European Medicines Agency CTIS. It is important to note that redacted versions of serious breach reports will be available to the public via CTIS.
- The Investigator should have a process in place to ensure that:
 - The site staff or service providers delegated by the Investigator/institution are able to identify the occurrence of a (potential) serious breach., including personal data breaches.
 - A (potential) serious breach is promptly reported to the Sponsor or delegated party, through the contacts (email address or telephone number) provided by the Sponsor.
- The Sponsor and the site have taken all necessary steps to avoid personal data breaches and have undertaken measures to prevent such breaches from occurring in the first place and to mitigate the impact of occurred data breaches (eg, applying encryption, maintaining and keeping systems and IT security measures up-to-date, regular reviews and testing, regular training of employees, and developed security policies and standards).
- Both the Sponsor and the site have developed an internal data breach reporting and investigation process and internal protocols with guidance on how to respond swiftly and diligently to the occurrence of a personal data breach.

- In compliance with applicable laws, the Data Controller for the processing activity where the personal data breach occurred (the Sponsor or respectively the site), will notify the data protection authorities without undue delay within the legal terms provided for such notification and within the prescribed form and content.
- If the personal data breach needs to be notified to participants, the notification to participants is done by the site for the data breaches that occurred within the processing activities for which the site is the Data Controller. For data breaches that occurred within the processing activities of the Sponsor as the Data Controller, the notification is done in collaboration with the site and is performed by the site and/or Principal Investigator, acting on behalf of the Sponsor, so that the Sponsor has no access to the identifying personal information of the participants. The site and/or Principal Investigator shall conduct the notification by contacting the participants using the information that they gave for communication purposes in clinical research.
- If a personal data breach occurs in a processor's systems, engaged by the Sponsor, the processor under contractual obligations with the Sponsor promptly and in due course after discovering the breach notifies the Sponsor and provides full cooperation with the investigation. In these cases, to the extent the Sponsor is the Data Controller for the processing activity where the breach occurred, it will be responsible for the notification to data protection authorities and, if applicable, to participants. If the personal data breach needs to be notified to the participants, the notification to participants is done in collaboration with the site and is performed by the site and/or Principal Investigator, acting on behalf of the Sponsor, so that the Sponsor has no access to the identifying personal information of the participants.
- Where a personal data breach is suffered by the Study Monitor, the latter will provide the Sponsor with all of the information needed for notification of the breach, without disclosing data that allows the Sponsor directly or indirectly to identify the participants. The notification will be done by the Sponsor solely with the information provided by the Study Monitor and in no event with access to information that could entail a risk of re-identification of the participants. If the data breach must be notified to the data participants, the notification will be done directly by the Study Monitor in collaboration with the site and/or Principal Investigator, acting on behalf of the Sponsor, so that the Sponsor has no access to the identifying personal information of the participants. The contract between the Sponsor and the Study Monitor shall expressly specify these conditions.
- The contract between the site and the Sponsor for performing the clinical research includes the provisions and rules regarding who is responsible for coordinating and directing the actions in relation to the breaches and performing the mandatory notifications to authorities and participants, where applicable.

A 2 Financial Disclosure

Investigators and sub-Investigators will provide the Sponsor with sufficient, accurate financial information as requested to allow the Sponsor to submit complete and accurate financial certification or disclosure statements to the appropriate regulatory authorities. Investigators are responsible for providing information on financial interests during the course of the study and for 1 year after completion of the study.

A 3 Informed Consent Process

- The Investigator or his/her representative will explain the nature of the study to the participant and answer all questions regarding the study.
- Participants must be informed that their participation is voluntary and they are free to refuse to participate and may withdraw their consent at any time and for any reason during the study. Participants will be required to sign a statement of informed consent that meets the requirements of 21 CFR 50, local regulations, ICH guidelines, HIPAA requirements, where applicable, and the IRB/IEC or study centre.
- The medical record must include a statement that written informed consent was obtained before the participant was enrolled in the study and the date the written consent was obtained. The authorised person obtaining the informed consent must also sign the ICF.
- Participants must be re-consented to the most current version of the ICF(s) during their participation in the study.
- A copy of the ICF(s) must be provided to the participant.

Participants who re-undergo enrolment procedures are required to sign a new ICF.

The ICF will contain a separate section that addresses and documents the collection and use of any mandatory human biological samples. The Investigator or authorised designee will explain to each participant the objectives of the analysis to be performed on the samples.

A 4 Data Protection

- Participants will be assigned a unique identifier by Trusted Third Party contracted by the Sponsor or by the Principal Investigator. Any participant records or datasets that are transferred to the Sponsor will contain the identifier only; participant names or any information which would make the participant identifiable will not be transferred.
- The participant must be informed that his/her personal study-related and coded (pseudonymised) data will be used by the Sponsor in accordance with local data protection law. The level of disclosure and use of their data must also be explained to the participant in the informed consent.
- The participant must be informed that his/her medical records may be examined by Clinical Quality Assurance auditors or other authorised personnel appointed by the Sponsor, by appropriate IRB/IEC members, and by inspectors from regulatory authorities.
- General Data Protection Regulation defines pseudonymisation as the processing of personal data in such a way that the personal data can no longer be attributed to a specific individual without the use of additional information, provided that such additional information is kept separately and protected by technical and organisational measures to

ensure that the personal data are not attributed to an identified or identifiable natural person.

- Appropriate safeguards will be implemented to protect coded data during and after the study that include:
 - Access to the coded data will be limited to specific individuals subject to confidentiality obligations (including the obligation to not attempt to re-identify individuals/decode the clinical data).
 - The coded data will be protected with security measures to avoid data alteration, loss, and unauthorised accesses and further de-identification techniques may be applied.
 - A data protection impact assessment, where required, will apply to identify and mitigate privacy risks, if any, associated with each scientific research.
 - The coded data will not be shared for direct marketing purposes or other purposes that are not legal duties or are not considered scientific research according to the applicable data protection legislation. In particular, it will not be used to make decisions about future services available to the participant, such as insurance.
- In addition to having the participants' data and biosamples coded, the data is also protected by high-standard technical security means, such as strong access control and encryption.
- Participants are also protected legally by the following means if the level of disclosure of the coded data includes sharing of the latter with other third parties as the participants will be explained in the ICF:
 - Within the Sponsor group, the participants' coded data are protected by Binding Corporate Rules. More information about the Sponsor's BCR are provided here: www.astrazenecabindingcorporaterules.com.
 - In all other cases, the participants' coded data is protected by contractual arrangements, Codes of Conduct, or certifications that set the rules for personal information protection to those available in European countries or other alternatives set forth in the law, as well as any supplementary technical and organisational measures that may result out of conducted transfer impact assessments.

A 5 Committees Structure

For details on the IDMC, refer to the IDMC charter. For details on the Independent Adjudication Committee, refer to the charter.

A 6 Dissemination of Clinical Study Data

Any results both technical and lay summaries for this trial will be submitted to EU CTIS within a year from the global End of Trial Date in all participating countries, due to scientific reasons, as otherwise statistical analysis is not relevant.

A description of this clinical study will be available on <http://astrazenecagrouptrials.pharmacm.com> and <http://www.clinicaltrials.gov> as will the summary of the study results when they are available. The clinical study and/or summary of study results may also be available on other websites according to the regulations of the countries in which the study is conducted.

A 7 Data Quality Assurance

All participant data relating to the study will be recorded on eCRF unless transmitted to the Sponsor or designee electronically (eg, laboratory data). The Investigator is responsible for verifying that data entries are accurate and correct by electronically signing the eCRF.

The Investigator must maintain accurate documentation (source data) that supports the information entered in the eCRF.

The Investigator must permit study-related monitoring, audits, IRB/IEC review, and regulatory agency inspections and provide direct access to source data documents.

Quality tolerance limits will be predefined in the IQRMP to identify systematic issues that can impact participant safety and/or reliability of study results. These predefined parameters will be monitored during the study, and important deviations from the quality tolerance limits and remedial actions taken will be summarised in the CSR.

Monitoring details describing strategy, including definition of study-critical data items and processes (eg, risk-based initiatives in operations and quality such as Risk Management and Mitigation Strategies and Analytical Risk-based Monitoring), methods, responsibilities and requirements, including handling of noncompliance issues and monitoring techniques (central, remote, or onsite monitoring) are included in the monitoring plan.

AstraZeneca or designee is responsible for medical oversight throughout the conduct of the study which includes clinical reviews of study data in accordance with the currently approved protocol. Monitoring details describing clinical reviews of study data from a medical perspective are included in more detail in the monitoring plan.

The Sponsor or designee is responsible for the data management of this study including quality checking of the data.

The Sponsor assumes accountability for actions delegated to other individuals (eg, Contract Research Organisations).

Study monitors will perform ongoing source data verification as per the Monitoring Plan to confirm that data entered into the eCRF by authorised site personnel are accurate, complete, and verifiable from source documents; that the safety and rights of participants are being protected; and that the study is being conducted in accordance with the currently approved protocol and any other study agreements, ICH GCP, and all applicable regulatory requirements.

Records and documents, including signed ICFs, pertaining to the conduct of this study must be retained by the Investigator for a minimum of 25 years after study archiving or as required by local regulations, according to the AstraZeneca GRAD schedule. No records may be destroyed during the retention period without the written approval of the Sponsor. No records may be transferred to another location or party without written notification to the Sponsor.

A 8 Source Documents

Source documents provide evidence for the existence of the participant and substantiate the integrity of the data collected. Source documents are filed at the Investigator's site.

Data entered in the eCRF that are transcribed from source documents must be consistent with the source documents or the discrepancies must be explained. The Investigator may need to request previous medical records or transfer records, depending on the study. Also, current medical records must be available.

Definition of what constitutes source data can be found in the monitoring plan.

A 9 Study and Site Start and Closure

The study start date is the date on which the clinical study will be open for recruitment of participants.

The first act of recruitment is the first site open and will be the study start date.

The Sponsor designee reserves the right to close the study site or terminate the study at any time for any reason at the sole discretion of the Sponsor. Study sites will be closed upon study completion. A study site is considered closed when all required documents and study supplies have been collected and a study-site closure visit has been performed.

The Investigator may initiate study-site closure at any time, provided there is reasonable cause and sufficient notice is given in advance of the intended termination.

Reasons for the early closure of a study site by the Sponsor or Investigator may include but are not limited to:

- Failure of the Investigator to comply with the protocol, the requirements of the IRB/IEC or local health authorities, the Sponsor's procedures, or GCP guidelines
- Inadequate recruitment of participants by the Investigator
- Discontinuation of further IP development

If the study is prematurely terminated or suspended, the Sponsor shall promptly inform the Investigators, the IECs/IRBs, the regulatory authorities, and any contract research organisation(s) used in the study of the reason for termination or suspension, as specified by the applicable regulatory requirements. The Investigator shall promptly inform the participant and should assure appropriate participant therapy and/or follow-up.

Participants from terminated sites will have the opportunity to be transferred to another site to continue the study.

A 10 Publication Policy

The results of this study may be published or presented at scientific meetings. If this is foreseen, the Investigator agrees to submit all manuscripts or abstracts to the Sponsor before submission. This allows the Sponsor to protect proprietary information and to provide comments.

The Sponsor will comply with the requirements for publication of study results. In accordance with standard editorial and ethical practice, the Sponsor will generally support publication of multicentre studies only in their entirety and not as individual site data. In this case, a coordinating Investigator will be designated by mutual agreement.

Authorship will be determined by mutual agreement and in line with International Committee of Medical Journal Editors authorship requirements.

Appendix B Adverse Events: Definitions and Procedures for Recording, Evaluating, Follow-up, and Reporting

B 1 Definition of Adverse Events

An AE is the development of any untoward medical occurrence in a patient or clinical study participant administered a medicinal product and which does not necessarily have a causal relationship with this treatment. An AE can therefore be any unfavourable and unintended sign (eg, an abnormal laboratory finding), symptom (for example nausea, chest pain), or disease temporally associated with the use of a medicinal product, whether or not considered related to the medicinal product.

The term AE is used to include both serious and non-serious AEs and can include a deterioration of a pre-existing medical occurrence.

B 2 Definition of Serious Adverse Events

An SAE is an AE occurring during any study phase (ie, enrolment, treatment, washout, follow-up), that fulfils one or more of the following criteria:

- Results in death
- Is immediately life-threatening
- Requires in-participant hospitalisation or prolongation of existing hospitalisation
- Results in persistent or significant disability or incapacity
- Is a congenital anomaly or birth defect
- Is an important medical event that may jeopardise the participant or may require medical treatment to prevent one of the outcomes listed above.

AEs for **malignant tumours** reported during a study should generally be assessed as **SAEs**. If no other seriousness criteria apply, the ‘Important Medical Event’ criterion should be used. In certain situations, however, medical judgement on an individual event basis should be applied to clarify that the malignant tumour event should be assessed and reported as a **non-serious AE**. For example, if the tumour is included as medical history and progression occurs during the study, but the progression does not change treatment and/or prognosis of the malignant tumour, the AE may not fulfil the attributes for being assessed as serious, although reporting of the progression of the malignant tumour as an AE is valid and should occur. Also, some types of malignant tumours, which do not spread remotely after a routine treatment that does not require hospitalisation, may be assessed as non-serious; examples in adults include Stage 1 basal cell carcinoma and Stage 1A1 cervical cancer removed via cone biopsy.

Life-threatening

‘Life-threatening’ means that the participant was at immediate risk of death from the AE as it occurred or it is suspected that use or continued use of the product would result in the

participant's death. 'Life-threatening' does not mean that if an AE occurred in a more severe form it might have caused death (eg, hepatitis that resolved without hepatic failure).

Hospitalisation

Outpatient treatment in an ER is not in itself an SAE, although the reasons for it may be (eg, bronchospasm, laryngeal oedema). Hospital admissions and/or surgical operations planned before or during a study are not considered AEs if the illness or disease existed before the participant was enrolled in the study provided that it did not deteriorate in an unexpected way during the study.

Important Medical Event or Medical Treatment

Medical and scientific judgement should be exercised in deciding whether a case is serious in situations where important medical events may not be immediately life threatening or result in death, hospitalisation, disability, or incapacity but may jeopardise the participant or may require medical treatment to prevent one or more outcomes listed in the definition of serious. These should usually be considered as serious.

Simply stopping the suspect drug does not mean that it is an important medical event; medical judgement must be used. The following are examples of an Important Medical Event:

- Angioedema not severe enough to require intubation but requiring IV hydrocortisone treatment
- Hepatotoxicity caused by paracetamol (acetaminophen) overdose requiring treatment with N-acetylcysteine
- Intensive treatment in an ER or at home for allergic bronchospasm
- Blood dyscrasias (eg, neutropenia or anaemia requiring blood transfusion, etc.) or convulsions that do not result in hospitalisation
- Development of drug dependency or drug abuse

Intensity rating scale:

- Mild (awareness of sign or symptom, but easily tolerated)
- Moderate (discomfort sufficient to cause interference with normal activities)
- Severe (incapacitating, with inability to perform normal activities)

It is important to distinguish between serious and severe AEs. Severity is a measure of intensity whereas seriousness is defined by the criteria in Appendix B 2. An AE of severe intensity need not necessarily be considered serious. For example, nausea that persists for several hours may be considered severe nausea, but not an SAE unless it meets the criteria shown in Appendix B 2. On the other hand, a stroke that results in only a limited degree of disability may be considered a mild stroke but would be an SAE when it satisfies the criteria shown in Appendix B 2.

B 3 A Guide to Interpreting the Causality Question

When making an assessment of causality consider the following factors when deciding if there is a ‘reasonable possibility’ that an AE may have been caused by the drug.

- Time Course. Exposure to suspect drug. Has the participant actually received the suspect drug? Did the AE occur in a reasonable temporal relationship to the administration of the suspect drug?
- Consistency with known drug profile. Was the AE consistent with the previous knowledge of the suspect drug (pharmacology and toxicology) or drugs of the same pharmacological class? Or could the AE be anticipated from its pharmacological properties?
- De-challenge experience. Did the AE resolve or improve on stopping or reducing the dose of the suspect drug?
- No alternative cause. The AE cannot be reasonably explained by another aetiology such as the underlying disease, other drugs, other host or environmental factors.
- Re-challenge experience. Did the AE reoccur if the suspected drug was reintroduced after having been stopped? The Sponsor would not normally recommend or support a re-challenge.
- Laboratory tests. A specific laboratory investigation (if performed) has confirmed the relationship.

In difficult cases, other factors could be considered such as:

- Is this a recognised feature of overdose of the drug?
- Is there a known mechanism?

Causality of ‘related’ is made if following a review of the relevant data, there is evidence for a ‘reasonable possibility’ of a causal relationship for the individual case. The expression ‘reasonable possibility’ of a causal relationship is meant to convey, in general, that there are facts (evidence) or arguments to suggest a causal relationship.

The causality assessment is performed based on the available data including enough information to make an informed judgement. With no available facts or arguments to suggest a causal relationship, the event(s) will be assessed as ‘not related’.

Causal relationship in cases where the disease under study has deteriorated due to lack of effect should be classified as no reasonable possibility.

B 4 Medication Error, Drug Abuse, and Drug Misuse

Medication Error

For the purposes of this clinical study a medication error is an unintended failure or mistake in the treatment process for an IMP or Sponsor NIMP that either causes harm to the participant or has the potential to cause harm to the participant.

A medication error is not lack of efficacy of the drug, but rather a human- or process-related failure while the drug is in control of the study site staff or participant.

Medication error includes situations where an error:

- Occurred
- **Was identified and** intercepted before the participant received the drug
- Did not occur, but circumstances were recognised that could have led to an error

Examples of events to be reported in clinical studies as medication errors:

- Drug name confusion
- Dispensing error, eg, medication prepared incorrectly, even if it was not actually given to the participant
- Drug not administered as indicated, eg, wrong route or wrong site of administration
- Drug not taken as indicated, eg, tablet dissolved in water when it should be taken as a solid tablet
- Drug not stored as instructed, eg, kept in the refrigerator when it should be at room temperature
- Wrong participant received the medication (excluding IRT/RTSM errors)
- Wrong drug administered to participant (excluding IRT/RTSM errors)

Examples of events that **do not** require reporting as medication errors in clinical studies:

- Errors related to or resulting from IRT/RTSM – including those which led to one of the above listed events that would otherwise have been a medication error
- Participant accidentally missed drug dose(s), eg, forgot to take medication
- Accidental overdose (will be captured as an overdose)
- Participant failed to return unused medication or empty packaging

Medication errors are not regarded as AEs but AEs may occur as a consequence of the medication error.

Drug Abuse

For the purposes of this study, drug abuse is defined as the persistent or sporadic intentional, non-therapeutic, excessive use of IMP or Sponsor NIMP for a perceived reward or desired non-therapeutic effect.

Any events of drug abuse, with or without associated AEs, are to be captured and forwarded to the DES using the Drug Abuse Report Form. This form should be used both if the drug abuse happened in a study participant or if the drug abuse involves a person not enrolled in the study (such as a relative of the study participant).

Examples of drug abuse include, but are not limited to:

- The drug is used with the intent of receiving a perceived reward (by the study participant or a person not enrolled in the study).
- The drug in the form of a tablet is crushed and injected or snorted with the intent of getting high.

Drug Misuse

Drug misuse is the intentional and inappropriate use (by a study participant) of IMP or Sponsor NIMP for medicinal purposes outside of the authorised product information, or for unauthorised IMPs or Sponsor NIMPs, outside the intended use as specified in the protocol and includes deliberate administration of the product by the wrong route.

Events of drug misuse, with or without associated AEs, are to be captured and forwarded to the DES using the Drug Misuse Report Form. This form should be used both if the drug misuse happened in a study participant or if the drug misuse regards a person not enrolled in the study (such as a relative or study participant).

Examples of drug misuse include but are not limited to:

- The drug is used with the intention to cause an effect in another person.
- The drug is sold to other people for recreational purposes.
- The drug is used to facilitate assault in another person.
- The drug is deliberately administered by the wrong route.
- The drug is split in half because it is easier to swallow, when it is stated in the protocol that it must be swallowed whole.
- Only half the dose is taken because the study participant feels that he/she is feeling better when not taking the whole dose.
- Someone who is not enrolled in the study intentionally takes the drug.

Appendix C Handling of Human Biological Samples

C 1 Chain of Custody

A full chain of custody is maintained for all samples throughout their lifecycle.

The Investigator at each centre maintains full traceability of collected biological samples from the participants while in storage at the centre until shipment or disposal (when appropriate) and records relevant to processing information related to the samples while at site.

The sample receiver keeps full traceability of the samples while in storage and during use until used or disposed of or until further shipment and keeps record of receipt of arrival and onward shipment or disposal.

The Sponsor or delegated representatives will keep oversight of the entire life cycle through internal procedures, monitoring of study sites, auditing or process checks, and contractual requirements of external laboratory providers.

Samples retained for further use will be stored in the Sponsor-assigned biobanks or other sample archive facilities and will be tracked by the appropriate Sponsor team for the remainder of the sample life cycle.

If required, the Sponsor will ensure that remaining biological samples are returned to the site according to local regulations or at the end of the retention period, whichever is sooner.

C 2 Withdrawal of Informed Consent for Donated Biological Samples

If a participant withdraws consent to the use of donated biological samples, the samples will be disposed of/destroyed/repatriated, and the action documented. If samples are already analysed, the Sponsor is not obliged to destroy the results of this research.

Following withdrawal of consent for biological samples, further study participation should be considered in relation to the withdrawal processes outlined in the informed consent.

The investigator:

- Ensures participant's withdrawal of informed consent to the use of donated samples is highlighted immediately to the Sponsor or delegate.
- Ensures that relevant human biological samples from that participant, if stored at the study site, are immediately identified, disposed of as appropriate, and the action documented.
- Ensures that the participant and the Sponsor are informed about the sample disposal.

The Sponsor ensures the organisation(s) holding the samples is/are informed about the withdrawn consent immediately and that samples are disposed of or repatriated as appropriate, and the action is documented and study site is notified.

C 3 International Airline Transportation Association 6.2-Guidance Document

LABELLING AND SHIPMENT OF BIOHAZARD SAMPLES

International Airline Transportation Association (IATA)

(<https://www.iata.org/whatwedo/cargo/dgr/Pages/download.aspx>) classifies infectious substances into 3 categories: Category A, Category B, or Exempt

Category A Infectious Substances are infectious substances in a form that, when exposure occurs, is capable of causing permanent disability, life-threatening or fatal disease in otherwise healthy humans or animals.

Category A Pathogens are, eg, Ebola, Lassa fever virus. Infectious substances meeting these criteria which cause disease in humans or both in humans and animals must be assigned to UN 2814. Infectious substances which cause disease only in animals must be assigned to UN 2900.

Category B Infectious Substances are infectious substances that do not meet the criteria for inclusion in Category A. Category B pathogens are, eg, hepatitis A, C, D, and E viruses. They are assigned the following UN number and proper shipping name:

- UN 3373 – Biological Substance, Category B are to be packed in accordance with UN 3373 and IATA 650

Exempt – Substances which do not contain infectious substances or substances which are unlikely to cause disease in humans or animals are not subject to these regulations unless they meet the criteria for inclusion in another class.

Clinical study samples will fall into Category B or exempt under IATA regulations.

Clinical study samples will routinely be packed and transported at ambient temperature in IATA 650 compliant packaging

(<https://www.iata.org/contentassets/b08040a138dc4442a4f066e6fb99f32a/dge/-62enpi650.pdf>).

Biological samples transported in dry ice require additional dangerous goods specification for the dry-ice content.

Appendix D Actions Required in Cases of Increases in Liver Biochemistry and Evaluation of Hy's Law

D 1 Introduction

This appendix describes the process to be followed in order to identify and appropriately report potential Hy's Law cases and Hy's Law cases. It is not intended to be a comprehensive guide to the management of elevated liver biochemistries. Specific guidance on managing liver anomalies can be found in Section 8.3.7 of the CSP.

During the study, the Investigator will remain vigilant for increases in liver biochemistry. The Investigator is responsible for determining whether a participant meets potential Hy's Law criteria at any point during the study.

All sources of laboratory data are appropriate for the determination of potential Hy's Law and HL events; this includes samples taken at scheduled study visits and other visits including central and all local laboratory evaluations even if collected outside of the study visits; for example, potential Hy's Law criteria could be met by an elevated ALT from a central laboratory **and/or** elevated TBL from a local laboratory.

The Investigator will also review AE data (for example, for AEs that may indicate elevations in liver biochemistry) for possible potential Hy's Law events.

The Investigator participates, together with Sponsor clinical project representatives, in review and assessment of cases meeting potential Hy's Law criteria to agree whether Hy's Law criteria are met. Hy's Law criteria are met if there is no alternative explanation for the elevations in liver biochemistry other than DILI caused by the IP.

The Investigator is responsible for recording data pertaining to PHL/HL cases and for reporting SAEs and AEs according to the outcome of the review and assessment in line with standard safety reporting processes.

D 2 Definitions

Potential Hy's Law

AST or ALT $\geq 3 \times$ ULN **together with** TBL $\geq 2 \times$ ULN at any point during the study following the start of study medication irrespective of an increase in ALP.

Hy's Law

AST or ALT $\geq 3 \times$ ULN **together with** TBL $\geq 2 \times$ ULN, where no other reason, other than the IP, can be found to explain the combination of increases (eg, elevated ALP indicating cholestasis, viral hepatitis, another drug).

For potential Hy's Law and Hy's Law, the elevation in transaminases must precede or be coincident with (ie, on the same day) the elevation in TBL, but there is no specified timeframe within which the elevations in transaminases and TBL must occur.

D 3 Identification of Potential Hy's Law Cases

In order to identify cases of potential Hy's Law it is important to perform a comprehensive review of laboratory data for any participant who meets any of the following identification criteria in isolation or in combination:

- $ALT \geq 3 \times ULN$
- $AST \geq 3 \times ULN$
- $TBL \geq 2 \times ULN$

Central Laboratories Being Used:

When a participant meets any of the potential Hy's Law identification criteria, in isolation or in combination, the central laboratory will immediately send an alert to the Investigator (also sent to the Sponsor representative).

The Investigator will also remain vigilant for any local laboratory reports where the potential Hy's Law identification criteria are met; where this is the case the Investigator will:

- Notify the Sponsor representative
- Request a repeat of the test (new blood draw) by the central laboratory without delay
- Complete the appropriate unscheduled laboratory eCRF module(s) with the original local laboratory test result

When the identification criteria are met from central or local laboratory results the Investigator will without delay:

- Determine whether the participant meets potential Hy's Law identification criteria (refer to Section D 2 for definition) by reviewing laboratory reports from all previous visits (including both central and local laboratory results)

Local Laboratories Being Used:

The Investigator will without delay review each new laboratory report and if the identification criteria are met will:

- Notify the Sponsor representative
- Determine whether the participant meets potential Hy's Law identification criteria (refer to Section D 2 for definition) by reviewing laboratory reports from all previous visits
- Promptly enter the laboratory data into the laboratory eCRF

D 4 Follow-up

D 4.1 Potential Hy's Law Criteria not met

If the participant does not meet potential Hy's Law identification criteria the Investigator will:

- Inform the Sponsor representative that the participant has not met PHL criteria.
- Perform follow-up on subsequent laboratory results according to the guidance provided in the CSP.

D 4.2 Potential Hy's Law Criteria met

If the participant does meet potential Hy's Law criteria the Investigator will:

- Notify the Sponsor representative who will then inform the central Study Team
- Within one day of potential Hy's Law criteria being met, the Investigator will report the case as an SAE of potential Hy's Law; serious criteria 'Important medical event' and causality assessment 'yes/related' according to CSP process for SAE reporting.
- For participants that met potential Hy's Law criteria prior to starting IP, the Investigator is not required to submit a potentially Hy's Law SAE unless there is a significant change ^a in the participant's condition
- The study physician contacts the Investigator, to provide guidance, discuss, and agree on an approach for the study participants' follow-up (including any further laboratory testing) and the continuous review of data
- Subsequent to this contact the Investigator will:
 - Monitor the participant until liver biochemistry parameters and appropriate clinical symptoms and signs return to normal or baseline levels, or as long as medically indicated. Complete follow-up SAE Form as required.
 - Investigate the aetiology of the event and perform diagnostic investigations as discussed with the study physician. This includes deciding which tests available in the Hy's law lab kit should be used.
 - Complete the three Liver eCRF Modules as information becomes available

^a A '**significant**' change in the participant's condition refers to a clinically relevant change in any of the individual liver biochemistry parameters (ALT, AST, or TBL) in isolation or in combination, or a clinically relevant change in associated symptoms. The determination of whether there has been a significant change will be at the discretion of the Investigator; this may be in consultation with the study physician if there is any uncertainty.

D 5 Review and Assessment of Potential Hy's Law Cases

The instructions in this section should be followed for all cases where potential Hy's Law criteria are met.

As soon as possible after the biochemistry abnormality is initially detected, the study physician contacts the Investigator in order to review available data and agree on whether there is an alternative explanation for meeting potential Hy's Law criteria other than DILI caused by the IP, to ensure timely analysis and reporting to health authorities within 15 calendar days from date PHL criteria was met. The Sponsor Global Clinical Lead or equivalent and Global Safety Physician will also be involved in this review together with other subject matter experts as appropriate.

According to the outcome of the review and assessment, the Investigator will follow the instructions below.

Where there is an agreed alternative explanation for the ALT or AST and TBL elevations, a determination of whether the alternative explanation is an AE will be made and subsequently whether the AE meets the criteria for a SAE:

- If the alternative explanation is **not** an AE, record the alternative explanation on the appropriate eCRF
- If the alternative explanation is an AE/SAE: update the previously submitted Potential Hy's Law SAE and AE eCRFs accordingly with the new information (reassessing event term; causality and seriousness criteria) following the Sponsor standard processes.

If it is agreed that there is **no** explanation that would explain the ALT or AST and TBL elevations other than the IP:

- Send updated SAE (report term 'Hy's Law') according to Sponsor standard processes.
 - The 'Medically Important' serious criterion should be used if no other serious criteria apply
 - As there is no alternative explanation for the Hy's Law case, a causality assessment of 'related' should be assigned.

If, there is an unavoidable delay, of over 15 calendar days in obtaining the information necessary to assess whether or not the case meets the criteria for Hy's Law identification, then it is assumed that there is no alternative explanation until such time as an informed decision can be made:

Provide any further update to the previously submitted SAE of potential Hy's Law, (report term now 'Hy's Law case') ensuring causality assessment is related to IP and seriousness criteria is medically important, according to CSP process for SAE reporting.

Continue follow-up and review according to agreed plan. Once the necessary supplementary information is obtained, repeat the review and assessment to determine whether Hy's Law

criteria are still met. Update the previously submitted potential Hy's Law SAE report following CSP process for SAE reporting, according to the outcome of the review and amending the reported term if an alternative explanation for the liver biochemistry elevations is determined.

D 6 Laboratory Tests

Hy's Law Lab Kit for Central Laboratories

Additional standard chemistry and coagulation tests	GGT LDH Prothrombin time INR
Viral hepatitis	IgM anti-HAV HbsAg IgM and IgG anti-HBc HBV DNA ^a IgG anti-HCV HCV RNA ^b IgM anti-HEV HEV RNA
Other viral infections	IgM & IgG anti-CMV IgG anti-HSV, and HSV 1/2 IgM testing or HSV PCR ^c IgM & IgG anti-EBV
Alcoholic hepatitis	Carbohydrate deficient transferrin (CD-transferrin)
Autoimmune hepatitis	Antinuclear antibody (ANA) Anti-Liver/Kidney Microsomal Ab (Anti-LKM) Anti-Smooth Muscle Ab (ASMA)
Metabolic diseases	Alpha-1-antitrypsin Ceruloplasmin Iron Ferritin Transferrin Transferrin saturation

^a HBV DNA is only recommended when IgG anti-HBc is positive.

^b HCV RNA is only recommended when IgG anti-HCV is positive or inconclusive.

^c HSV 1/2 IgM testing or HSV 1/2 PCR testing depending on region.

Appendix E Contraception Guidance

Females are considered to be of childbearing potential unless they meet either of the criteria, as follows:

- Surgically sterilised (including bilateral salpingectomy, bilateral oophorectomy, or hysterectomy), or
- Post-menopausal.

For females aged < 50 years, post-menopausal is defined as having both:

- A history of ≥ 12 months amenorrhoea, without an alternative cause, following cessation of exogenous sex-hormonal treatment and,
- A FSH level in the post-menopausal range.

For females aged ≥ 50 years, post-menopausal is defined as having a history of ≥ 12 months amenorrhoea, without an alternative cause, following cessation of exogenous sex-hormonal treatment.

A highly effective method of contraception is defined as one that results in a low failure rate (ie, less than 1% per year) when used consistently and correctly. The acceptable methods of contraception are described in [Table E1](#) below. Female patients must use one of the following non-hormonal or hormonal methods along with a condom.

Periodic abstinence, the rhythm method, and withdrawal are NOT acceptable methods of contraception.

Table E1 Highly Effective Methods of Contraception

Non-hormonal Methods	Hormonal Methods
Copper intrauterine device Bilateral tubal occlusion (caveat: failure rate >1%) Vasectomised partner ^a Sexual abstinence ^b	Combined (estrogen and progestogen containing hormonal contraception) Hormone containing intrauterine devices (IUD, eg, Mirena, Liletta) Oral (combined pill) Medroxyprogesterone injections (eg, Depo- Provera) Transdermal (patch) Intravaginal device (eg, ethinylestradiol and etonogestrel [Nuvaring]) Etonogestrel, subdermal implants (SDI)

^a With appropriate post-vasectomy medical testing of surgical success (ie, absence of sperm in ejaculate).

^b Sexual abstinence is considered a highly effective method only if defined as refraining from heterosexual intercourse during the entire period of the study and if it is the preferred and usual lifestyle of the participant.

The following constitute ineffective methods of contraception:

- Periodic abstinence (calendar, symptom-thermal, post-ovulation methods)
- Coitus interruptus (withdrawal method)

- Spermicides
- Lactational amenorrhea
- Other non-hormonal or non-copper-containing IUDs (or others not listed in the acceptable methods of contraception)
- Triphasic Combined Oral Contraceptives
- Progesterone only oral contraceptives (except Cerazette)
- Sponge, diaphragm, or male or female condoms without secondary methods
- Tubal occlusion without a secondary method

Male participants are considered non-sterile unless they have previously had bilateral orchidectomy, which should be confirmed by medical assessment and documented in the subject's medical records. Males who have undergone vasectomy, which should be documented in the subject's medical records, must have the lack of sperm in the ejaculate confirmed a minimum of 30 days after the procedure by their physician on 2 separate occasions separated by a minimum of 30 days.

Appendix F National Institute of Allergy and Infectious Diseases and Food Allergy and Anaphylaxis Network Guidance for Anaphylaxis Diagnosis

The National Institute of Allergy and Infectious Diseases and Food Allergy and Anaphylaxis Network define anaphylaxis as a serious allergic reaction that is rapid in onset and may cause death ([Sampson et al 2006](#)). They recognise 3 categories of anaphylaxis, with criteria designated to capture from 80% of cases (Category 1) to > 95% of all cases of anaphylaxis (for all 3 categories).

- 1 Acute onset of an illness (minutes to several hours) with involvement of the skin, mucosal tissue, or both (eg, generalised hives, pruritus or flushing, swollen lips-tongue-uvula) AND AT LEAST ONE OF THE FOLLOWING:
 - Respiratory compromise (eg, dyspnoea, wheeze-bronchospasm, stridor, hypoxaemia).
 - Reduced BP or associated symptoms of end-organ dysfunction (eg, hypotonia [collapse], syncope, incontinence).
- 2 Two or more of the following that occur rapidly after exposure to a likely allergen for that participant (minutes to several hours):
 - Involvement of the skin-mucosal tissue (eg, generalised hives, itch-flush, swollen lips-tongue-uvula).
 - Respiratory compromise (eg, dyspnoea, wheeze-bronchospasm, stridor, hypoxaemia).
 - Reduced BP or associated symptoms (eg, hypotonia [collapse], syncope, incontinence)
 - Persistent gastrointestinal symptoms (eg, crampy abdominal pain, vomiting).
- 3 Reduced BP after exposure to known allergen for that participant (minutes to several hours):
 - Adults: systolic BP of less than 90 mmHg or greater than 30% decrease from that person's baseline.

For the purpose of AE reporting, the above criteria should be used to guide retrospective judgment as to whether an event was true anaphylaxis. Guidance on the recognition of possible anaphylaxis at the time of the event is provided ([Appendix G](#)).

Appendix G Signs and Symptoms and Management of Acute Anaphylaxis

Appropriate drugs, such as epinephrine, antihistamines, corticosteroids, etc, and medical equipment to treat anaphylactic reactions must be immediately available at study sites, and study personnel should be trained to recognise and treat anaphylaxis. Local or national guidelines for the recognition and management of acute anaphylaxis should be followed where available. Where not available, guidance is provided below.

G 1 Signs and Symptoms of Acute Anaphylaxis

Anaphylaxis is an acute and potentially lethal multi-system allergic reaction in which some or all of the following signs and symptoms occur:

- Diffuse erythema.
- Pruritus.
- Urticaria and/or angioedema.
- Bronchospasm.
- Laryngeal oedema.
- Hypotension.
- Cardiac arrhythmias.
- Feeling of impending doom.
- Unconsciousness.
- Shock.

Other earlier or concomitant signs and symptoms can include:

- Itchy nose, eyes, pharynx, genitalia, palms, and soles.
- Rhinorrhoea.
- Change in voice.
- Metallic taste.
- Nausea, vomiting, diarrhoea, abdominal cramps, and bloating.
- Light-headedness.
- Headache.
- Uterine cramps.
- Generalised warmth.

G 2 Management of Acute Anaphylaxis

G 2.1 Immediate Intervention

- 1 Assessment of airway, breathing, circulation, and adequacy of mentation.
- 2 Administer epinephrine IM every 5 to 15 minutes, in appropriate doses, as necessary, depending on the presenting signs and symptoms of anaphylaxis, to control signs and symptoms and prevent progression to more severe symptoms such as respiratory distress, hypotension, shock, and unconsciousness.

G 2.2 Possibly Appropriate, Subsequent Measures Depending on Response to Epinephrine

- 1 Place participant in recumbent position and elevate lower extremities.
- 2 Establish and maintain airway.
- 3 Administer oxygen.
- 4 Establish venous access.
- 5 Normal saline IV for fluid replacement.

G 2.3 Specific Measures to Consider After Epinephrine Injections, Where Appropriate

- 1 Consider epinephrine infusion.
- 2 Consider H1 and H2 antihistamines.
- 3 Consider nebulised beta-2 agonist [eg, albuterol (salbutamol)] for bronchospasm resistant to epinephrine.
- 4 Consider systemic corticosteroids.
- 5 Consider vasopressor (eg, dopamine).
- 6 Consider glucagon for participant taking b-blocker.
- 7 Consider atropine for symptomatic bradycardia.
- 8 Consider transportation to an ED or an intensive care facility.
- 9 For cardiopulmonary arrest during anaphylaxis, high-dose epinephrine and prolonged resuscitation efforts are encouraged, if necessary.

If a suspected anaphylactic reaction occurs during or within a 24-hour period after administration of IP, blood samples for serum tryptase should be collected as soon as possible after the event, at 60 ± 30 minutes after the event, at discharge, and between 2 and 4 weeks post discharge. Immediate care of the participant and treatment of the reaction must take priority over collecting blood samples.

Adapted from [Kemp et al 2008](#).

Appendix H Changes Related to Mitigation of Study Disruptions Due to Cases of Civil Crisis, Natural Disaster, or Public Health Crisis

Changes below should be implemented only during study disruptions due to any of or a combination of civil crisis, natural disaster, or public health crisis (eg, during quarantines and resulting site closures, regional travel restrictions and considerations if site personnel or study participants become infected with SARS-CoV-2 or similar pandemic infection) during which participants may not wish to or may be unable to visit the study site for study visits. These changes should only be implemented if allowable by local/regional guidelines and following notification from the Sponsor, and instructions on how to perform these procedures will be provided at the time of implementation.

During civil crisis, natural disaster, or public health crisis, some study assessment and procedures may not be conducted due to international or local policies or guidelines, hospital or clinic restrictions, and other measures implemented to ensure participant's safety. If in doubt, please contact the Sponsor study physician.

H 1 Consent of Study Participants During Study Interruptions

During study interruptions, it may not be possible for the participants to complete study visits and assessments on site and alternative means for carrying out the visits and assessments may be necessary, eg, remote visits. Consent should be obtained for the alternative means of carrying out visits and assessments and should be obtained prior to performing the procedures described in Sections [H 2](#) to [H 6](#). Local and regional regulations and/or guidelines regarding re-consent of study participants should be checked and followed. Consent may be verbal if allowed by local and regional guidelines (note, in the case of verbal consent the ICF should be signed at the participant's next contact with the study site). Visiting the study sites for the sole purpose of obtaining consent should be avoided.

H 2 Re-enrolment Procedures for Participants to Reconfirm Study Eligibility

Additional re-enrolment procedures for enrolment failure due to study disruption can be performed in previously participants that previously underwent enrolment procedures. The Investigator should confirm this with the designated Sponsor study physician.

In addition, during study disruption there may be a delay between confirming eligibility of a participant and either enrolment into the study or commencing of dosing with IP. If this delay occurs, the participant will need to re-undergo enrolment procedures to reconfirm eligibility before commencing study procedures. This will provide another opportunity to re-perform enrolment procedures for an enrolment failure participant in addition to that detailed in Section [5.4](#). The procedures detailed in Section [5](#) must be undertaken to confirm eligibility using the same randomisation number as for the participant.

H 3 Home or Remote Visit to Replace Onsite Visit (Where Applicable)

A qualified HCP from the study site or TPV service will visit the participants' home or other remote location as per local Standard Operating Procedures, as applicable. Supplies will be provided for a safe and efficient visit. The qualified HCP will be expected to collect information per the CSP.

If applicable, safety procedures and sample collection may be performed according to the revised SoA in the Study Instruction for Mitigation Due to Civil Crisis, Natural Disaster, or Public Health Crisis.

H 4 Telemedicine Visit to Replace Onsite Visit (Where Applicable)

In this appendix, the term telemedicine visit refers to remote contact with the participants using telecommunications technology including phone calls, virtual or video visits, and mobile health devices.

During a civil crisis, natural disaster, or public health crisis, onsite visits may be replaced by a telemedicine visit if allowed by local/regional guidelines. Having a telemedicine contact with the participants will allow AEs, concomitant medication, medical and COPD history, COPD treatment, COPD exacerbations, smoking status, supplemental oxygen status, HRU, and other information at deemed necessary by the Investigator to be reported and documented.

H 5 At-home or Remote Location IP Administration Instructions

If an onsite visit is not possible, at-home or remote location administration of IP may be performed by a qualified HCP, provided this is acceptable within local regulation/guidance. The option of at-home or remote location IP administration ensures participants' safety in cases of a pandemic where participants may be at increased risk by traveling to the site/clinic. This will also minimise interruption of IP administration during other study disruptions (eg, site closures due to natural disaster).

Section [H 5.1](#) to be used only when applicable for the IP formulation being used, eg, with injectable biologics.

H 5.1 At-home or Remote Location IP Administration by a Qualified HCP or TPV Service

A qualified HCP from the study site or TPV service should administer the IP at the participant's home or other remote location according to the CSP. All necessary supplies and instructions for administration and documentation of IP administration will be provided. Additional information related to the visit can be obtained via a telemedicine or home visit.

H 6 Data Capture During Telemedicine or Home/Remote Visits

Data collected during telemedicine or home / remote visits will be captured by the qualified HCP from the study site or TPV service.

Appendix I Examples of Inhaled Corticosteroids Equivalent Doses

ICS Component	Conversion Factor (ie, Multiply Dose by X)	Dose Equivalent to Fluticasone Propionate 250 µg (µg)
Fluticasone propionate	1	250
Fluticasone propionate HFA (Flovent HFA)	1.14	220 (per actuation)
Fluticasone furoate ^a	2.5	100 (metered)/ 92 (delivered dose)
Budesonide	0.63/ 0.78	400 (metered) / 320 (delivered dose)
Budesonide (nebulised)	0.5	0.5mg (1 ml)
Beclomethasone	0.5	500
Beclomethasone (Trimbow)	1.25	200 (metered)/ 174 (delivered dose)
Beclomethasone extra-fine particles and HFA (Fostair, QVAR)	1.04/ 1.25	250 (metered)/ 200 (per actuation)
Ciclesonide	1.56	160
Mometasone	1.14	220 (metered) 200 (per actuation)
Flunisolide	0.25	1000
Triamcinolone	0.25	1000

^a If any combination has only one ICS dose regimen approved for treatment of COPD, that may be considered as an equivalent ICS dose to any other ICS therapy for this study.

Example conversion

Converting budesonide (delivered dose) to fluticasone propionate (in mcg): For a budesonide dose of 160 µg, use a conversion factor of 0.78.

160 µg of budesonide × 0.78 = 124.8 µg fluticasone propionate equivalent.

Appendix J Country-Specific Requirements

EU Markets

Section 6.5, Concomitant Therapy

As per Section 6.5.2 of the CSP all participants are required to be treated with stable optimised background maintenance inhaled therapy (ICS/LABA/LAMA triple therapy, or dual therapy if triple is not considered appropriate) during the study. Both dual and triple therapy may be in the form of separate inhalers (ATC codes R03AL, R03BB, R03BA) or fixed dose combination inhalers (ATC codes R03AL or R03AK).

Short-acting β 2-agonists (SABAs, [ATC codes R03AC]), SAMAs [ATC codes R03BB], SABA/SAMA [ATC code R03AL] combination or alternative symptom relief medications, as per local SoC, may be used during the study in the event of worsening of COPD symptoms and referred throughout the protocol as “rescue medications”. Day-to-day variation of symptoms requiring use of symptom relief (rescue) medications is expected in patients with COPD, and use of such medication does not necessarily reflect on the efficacy of the IMP.

Salbutamol (ATC code R03AC02) will also be used for post-BD spirometry.

Auxiliary medicinal products referred to here should be locally approved and administered per the prescribing information.

Section 8.2.6, Clinical Safety Laboratory Assessments

In markets where pregnancy testing has to be performed at monthly intervals during the same period as contraception is mandatory (in this study, until 14 weeks after the last dose of IP), two additional pregnancy tests in FOCBP will be performed at Week 56 and Week 62.

Participants can perform this test at home and confirm the results of the test via a phone call/virtual call with the site staff. Alternatively, participants may visit the site to perform this test, if needed. If a participant discontinues IP early, then a pregnancy test will be performed monthly until 14-weeks post-last IP administration. If a participant discontinues IP, then a pregnancy test must be completed monthly until 14-weeks post-last IP administration. The results of the pregnancy must be documented in the eCRF.

Results of the pregnancy test at Week 56 and Week 62 should be entered in the eCRF as an unscheduled visit.

China

With reference to [Appendix D 6](#), CD-transferrin and transferrin are not available in China. The Hy's Law Lab Kit for central laboratories will be amended by the local study team accordingly.

For China, ADA samples will be retained for a maximum of 1 year.

Czech Republic

Section 6.2.1.2, Monitoring Dose Administration

With reference to Section 6.2.2.4 of the CSP. After administration of IP, the participant should be observed for potential acute reactions in line with local clinical practice for mAbs. In markets where a specific defined time period for observation of potential acute reactions after IP dose is required, it is recommended that for the first 2 doses of study intervention, the participant is to remain at the site for ≥ 2 hours after study intervention administration to be observed by the qualified HCP. For subsequent doses of study intervention, participants are to remain at the site for ≥ 1 hour after study intervention to be observed by the qualified HCP. Self-administration of IP at home by the participant or their caregiver is not applicable in the Czech Republic.

Appendix D, Actions Required in Cases of Increases in Liver Biochemistry and Evaluation of Hy's Law

With reference to [Appendix D](#) Actions Required in Cases of Increases in Liver Biochemistry and Evaluation of Hy's Law. Further clarification in addition to [Appendix D](#) is provided to Investigators.

In the event that there is an increase in liver enzymes AST and/or ALT $\geq 3 \times$ ULN and TBL $\geq 2 \times$ ULN with no alternative explanation, and the Investigator is confident that Hy's Law criteria have been met and that the abnormalities are due to the IP, and if the PI believes it is in the best interest of the participant, then the IP should be immediately suspended until all liver tests are corrected to within normal limits. The Investigator should arrange any further appropriate investigations required and the Medical Monitor must be notified. This will be documented as a SAE in line with standard safety reporting processes.

If further use of the IP is commenced, with the agreement of the Investigator, participant, and Sponsor, and if there is a further increase in the liver enzymes after reintroduction, then the participant will be definitively excluded from further doses of the IP. Discontinuation from IP does not automatically lead to a complete withdrawal from the study. Participants discontinuing from IP are strongly encouraged to continue on-study for collection of safety and clinical data.

Germany

- a) As per Article 77 of the Regulation (EU) No. 536/2014 i.c.w. Sec 42 Par 6. AMG, the study will be terminated in Germany if approval by the National Authority has been revoked.
- b) In compliance to. Article 76 of the Regulation (EU) No. 536/2014 i.c.w. Sec 40a S. 1 No. 3 lit b) AMG. if the risk-benefit ratio changes on the basis of the IDMC recommendation, the maximum sum insured can be adjusted accordingly.

Appendix K Abbreviations

Abbreviation or special term	Explanation
ADA	antidrug antibody(ies)
AE	adverse event(s)
AESI	adverse event of special interest
ALP	alkaline phosphatase
ALT	alanine aminotransferase/transaminase
APFS	accessorised pre-filled syringe
AST	aspartate aminotransferase/transaminase
BCR	blinding corporate rules
BD	bronchodilator
CI	confidence interval
CONSORT	Consolidated Standards of Reporting Trials
COPD	chronic obstructive pulmonary disease
COVID-19	coronavirus disease 2019
CSP	clinical study protocol
CSR	clinical study report
CT	computed tomography
CTR	clinical trials regulation
CTIS	Clinical Trials Information System
DILI	drug induced liver injury
ECG	electrocardiogram
eCRF	electronic case report form
ED	emergency department
E/D	early discontinuation
EDC	electronic data capture
ER	emergency room
EU	European Union
FAS	Full Analysis Set
FEV ₁	forced expiratory volume in one second
FOCBP	female(s) of childbearing potential
FSH	follicle-stimulating hormone
FVC	forced vital capacity
GCP	good clinical practice
GRAD	Global Retention and Disposal
HBc	hepatitis B core

Abbreviation or special term	Explanation
HBsAg	hepatitis B surface antigen
HCG	human chorionic gonadotropin
HCP	healthcare professional
HIV	human immunodeficiency virus
HRU	healthcare resource utilisation
IB	Investigator's Brochure
ICF	informed consent form
ICH	International Council for Harmonisation
ICS	inhaled corticosteroid
IDMC	independent data monitoring committee
IEC	independent ethics committee
IL	interleukin
IM	intramuscular(ly)
IMP	investigational medicinal product
IP	investigational product
IQRMP	Integrated Quality and Risk Management Plan
IRB	institutional review board
IRT	interactive response technology
LABA	long-acting beta agonist
LAMA	long-acting muscarinic antagonist
LTE	long-term extension
LVEF	left ventricular ejection fraction
mAb	monoclonal antibody
MACE	major acute cardiovascular events
NIMP	non-investigational medicinal product
NT-proBNP	N-terminal pro-brain natriuretic peptide
NYHA	New York Heart Association
PD	pharmacodynamic(s)
PHL	potential Hy's Law
PK	pharmacokinetic(s)
Q2W	every 2 weeks
Q4W	every 4 weeks
Q8W	every 8 weeks
RTSM	randomisation and trial supply management
SABA	short-acting beta agonist

Abbreviation or special term	Explanation
SAE	serious adverse event(s)
SAF	Safety Analysis Set
SAMA	short-acting muscarinic antagonist(s)
SAP	statistical analysis plan
SARS-CoV-2	severe acute respiratory syndrome coronavirus 2
SC	subcutaneous(ly)
SoA	schedule of activities
SoC	standard of care
SUSAR	suspected unexpected serious adverse reaction(s)
TBL	total bilirubin
THC	tetrahydrocannabinol
TPV	third-party vendor
ULN	upper limit of normal
w/v	weight over volume

Appendix L Protocol Version History

The Summary of Changes Table for the current revision is located directly before the Table of Contents.

CSP Version 3.0 26 March 2024

This amendment is considered to be substantial based on the criteria set forth in Article 10(a) of Directive 2001/20/EC of the European Parliament and the Council of the EU, and in the EU CTR Article 2, 2 (13).

Overall Rationale for the Amendment:

The aim of this amendment is to change the primary endpoint from time to first severe COPD exacerbation to annualised rate of severe COPD exacerbations in the primary population (previously a key secondary endpoint). The key secondary endpoint is correspondingly changed as a reflection of this from annualised rate of severe COPD exacerbations to time to first severe COPD exacerbation.

Rate of exacerbations is a well-established and clinically relevant endpoint for COPD trials. Based upon a review of blinded aggregate data, greater rates of severe exacerbation events and higher dispersion have been detected throughout the predecessor studies OBERON and TITANIA and through this LTE PROSPERO than was predicted when the studies were initiated, while discontinuation rates have also been lower. In this situation there can be attenuation of treatment effects in a time to event analysis compared to a rate-based analysis and so a rate-based analysis is expected to provide greater statistical power.

Summary of Changes:

List of Substantial Modifications

Section Number and Name	Description of Change	Brief Rationale
1.1 Synopsis (Objectives and Endpoints; Primary Endpoint Analysis), 3 Objectives and Endpoints, 4.2.2 Rationale for Selection of Primary Endpoint, 9.1 Statistical Hypotheses, 9.1.2 Type I Error Control, and Figure 3 Type I Error Control for Comparison of 300 mg Q4W Versus Placebo, 9.3.1 Treatment Groups for Analysis, 9.4.1 General Considerations, 9.4.2.1 Primary Endpoint, 9.4.2.2 Secondary Endpoints	Revised primary endpoint to annualised rate of severe COPD exacerbations (previously time to first severe COPD exacerbation) in former smokers. Revised first key secondary endpoint to annualised rate of severe COPD exacerbations (previously time to first severe COPD exacerbation) in former or current smokers. Revised second key secondary endpoint to time to first severe COPD exacerbation (previously annualised rate of COPD exacerbations) in former smokers, with further analyses in former or current smokers. Revised corresponding statistical analysis details for the endpoints accordingly throughout Section 9.	See overall rationale for the amendment described above.
9.2 Sample Size Determination	Revised the sample size justification to apply to a rate-based endpoint.	Updated to correspond to the primary endpoint of annualised rate of severe COPD exacerbations.
1.1 Synopsis (Interim Analysis), 9.5 Interim Analysis	Revised text as follows to apply for a rate-based endpoint (bold indicates added text; strikethrough indicates deleted text): At least 70% of information fraction for the expected number of participants with rate of severe exacerbations are observed in a pooled analysis of this study and the 2 predecessor studies, where this expected count is based on the total sample size known after the approval of CSP Version 2.0 in all countries, and	Updated to correspond to the primary endpoint of annualised rate of severe COPD exacerbations, for which information fraction is no longer defined by the number of participants with events.

List of Non-substantial Modifications

Section Number and Name	Description of Change	Brief Rationale
8.4 Human Biological Samples, Appendix A3 Informed Consent Process	Removed text regarding future research and optional samples that are not applicable for the study.	Clarification.
9.4.2.2 Secondary Endpoints	Added “Smoking status will be included as a covariate in analyses in the all-comer population (ie, former or current smokers)” for the secondary endpoint annualised rate of moderate-to-severe COPD exacerbations.	Clarify analysis details for the all-comer population.
10 Appendix E Contraception Guidance	The number of months (12) was missing from the definition of post-menopausal due to a typographical error.	Correction of typographical error.
10 Appendix L Protocol Version History	Moved the summary of changes for CSP v2 27 July 2023 to appendix.	New appendix for previous versions of the protocol.
11 References	Added references Fleming 2011, Gottlow et al 2015, Ratitch et al 2013, and Wan et al 2015. Removed Lipkovich et al 2016 and Moscovici et al 2017.	Update references.

CSP Version 2.0 27 July 2023

This amendment is considered to be substantial based on the criteria set forth in Article 10(a) of Directive 2001/20/EC of the European Parliament and the Council of the EU, and in the EU CTR Article 2, 2 (13).

Overall Rationale for the Amendment:

The primary objective of this amendment is to account for an update in the predecessor studies to reduce the sample size of the tozorakimab 300 mg Q8W arm and to update the duration of the treatment period.

The rationale for this change follows recent data from Phase II studies in moderate to severe atopic dermatitis and asthma, in which lower trough PK concentrations (30% to 35%) were observed compared with the Phase I healthy volunteer study. With this updated exposure data, predicted IL-33 lung inhibition reduces from 94% to 92%, but remains above 90% for the 300 mg Q4W regimen; the 2% drop suggests this dose is on the flat portion of the dose-response curve. However, the predicted IL-33 inhibition for the 300 mg Q8W regimen reduces by approximately 5% to below 80%.

On this basis, together with the acceptable safety profile of tozorakimab across all dosing regimens, randomisation to the tozorakimab 300 mg Q8W arm will be stopped in the predecessor studies, OBERON and TITANIA. It follows that no new participants will be allocated to tozorakimab 300 mg Q8W in this study after the amendment to PROSPERO.

The treatment period in this study is reduced to 28 weeks since this is a sufficient treatment period, when pooled with the predecessor studies, to allow the collection of efficacy, long-term safety data, and the timely completion of the study. Those participants who are already enrolled under CSP Version 1.0 can continue to complete the study with the same treatment regimen as assigned and for a treatment period of 52 weeks after this amendment.

Participants will therefore be enrolled in the study prior to the amendment under CSP Version 1.0 or under this version (Version 2.0) as follows:

- CSP Version 1.0: Participants receiving tozorakimab in the predecessor study will have been allocated to the same active treatment regimen as in the predecessor study. Participants from the placebo arm of the predecessor study will have been re-randomised in PROSPERO in a 1:1:2 ratio to tozorakimab 300 mg Q4W, tozorakimab 300 mg Q8W and placebo, for a treatment period of 52 weeks.
- CSP Version 2.0: Participants on 300 mg Q4W in the predecessor studies will remain on 300 mg Q4W. Participants on 300 mg Q8W in the predecessor studies will be allocated to 300 mg Q4W in PROSPERO, and participants on placebo from the predecessor studies will be re-randomised 1:1 to tozorakimab 300 mg Q4W or placebo. The treatment period will be 28 weeks. Those participants who are already enrolled under CSP Version 1.0 can continue to complete the study with the same treatment regimen as assigned and for a treatment period of 52 weeks after this amendment.

It follows that under CSP Version 2.0 no new participants from the predecessor studies will be allocated to 300 mg Q8W.

The data collected from the tozorakimab 300 mg Q8W arm will only be used for characterisation of the dose-response curve, and not for confirmatory hypothesis testing. The overall sample size and multiple testing procedure have been amended accordingly.

Additional non-substantial updates were also made to: implement mandatory changes bringing the AstraZeneca global CSP template into alignment with industry standards; update per requirements for EU CTR; clarify procedures, statistical analyses, and CSP and appendices wording.

Summary of Changes:

List of Substantial Modifications

Section Number	Description of Change	Brief Rationale
Section 1 (1.1, 1.2, 1.3), Section 4 (4.1, 4.1.2, 4.2.3, 4.3), Section 6.3.1, Section 9 (9.1, 9.1.1, 9.2, 9.3.1, 9.4.2.1, 9.5)	No new participants from the predecessor studies will be allocated to 300 mg Q8W after the amendment and sample size of the Q8W arm is reduced as such. Study design, sample size, and statistical analysis (including confirmatory hypothesis testing and the multiple testing procedure) are updated to reflect this change. Update of the percentage of IL33:ST2 inhibition expected in lung at trough for both tozorakimab dose regimens based on emerging Phase II data.	The rationale for this change follows recent data from Phase II studies in moderate to severe atopic dermatitis (D9182C00001) and asthma, in which lower trough PK concentrations (30% to 35%) were observed compared with the Phase I healthy volunteer study (D9182C00001). This study follows the predecessor study, and no new participants will be allocated to tozorakimab 300 mg Q8W.
Section 1 (1.1, 1.2, 1.3, 1.3.1, 1.3.2), Section 4 (4.1, 4.1.2, 4.2.4, 4.4), Section 6.1.1, Section 7.1.1, Section 8.1.4, Section 9 (9.2, 9.3.1)	Reduced the treatment period to 28 weeks for subjects recruited under CSP Version 2.0 impacting study design schema, schedule of activities, rationale, treatment period and statistical analysis.	The duration is reduced to 28 weeks since this is a sufficient study duration, when pooled with the predecessor studies, to provide long term efficacy and safety data.

List of Non-substantial Modifications

Section Number and Name	Description of Change	Brief Rationale
Title Page	Addition of EU Clinical Trials number.	Amended to adhere to the EU CTR: Regulation (EU) No 536/2014 requirements.
Section 1.1, Synopsis	Study title corrected.	Correction of typographical error.
Section 1.1, Synopsis, Statistical Methods; Section 4.4.1, Study Stopping Criteria; Section 9.5 Interim Analysis; Section 9.6, Independent Data Monitoring Committee	Updated to clarify that the interim analysis for futility conducted in the predecessor studies may impact PROSPERO.	Clarification.
Section 1.1, Synopsis, Objectives and Endpoints; Section 3, Table 3, Objectives and Endpoints	ADA endpoint wording revised from “incidence” to “presence”.	To align description in the predecessor studies.

Section Number and Name	Description of Change	Brief Rationale
Section 1.1, Synopsis, Intervention Groups and Duration; Section 4.1, Overall Design; Section 4.4, End of Study Definition; Section 6.3.2, Methods for Ensuring Blinding	Updated to note that AstraZeneca team involved with PROSPERO readout will remain separate and blinded from AstraZeneca team involved with reporting the predecessor studies after their unblinding.	To provide additional clarity around maintenance of trial integrity.
Section 1.3, Table 1, Table 2, Schedule of Activities	In the footnote k, a text regarding lab kit was not applicable to PROSPERO and removed.	Clarification.
	All instances of “> 1 day” were revised to “≥ 1 day”.	Correction of typographical error.
Section 1.3, Table 1, Table 2, Schedule of Activities; Section 4.1.2, Treatment Period; Section 7.1.1, Procedures for Discontinuation of Investigational Product	Clarified timing of E/D visit for participants who discontinue IP.	Clarification.
Section 1.3, Table 1, Table 2, Schedule of Activities; Section 8.1.3, Spirometry (Pre- and Post-BD)	Added that post-BD spirometry is optional and may be skipped if a participant cannot tolerate post-BD spirometry, or their condition otherwise contraindicates it.	Post-BD spirometry in participants with severe or very severe COPD may be burdensome and is made option to reduce patient and site burden.
Section 1.3, Table 1, Table 2, Schedule of Activities; Section 8.1.4 Quantitative CT Assessment	CT is optional.	Clarification that the follow-up CT is optional.
Section 1.3, Table 1, Table 2, Schedule of Activities; Section 8.2.5, Echocardiography	Clarified additional echocardiography assessments throughout study and at the end of the treatment period.	Clarification.
Section 2, Introduction	Revised list of conditions for which tozorakimab is being developed.	Update in line with IB.
Section 2.2, Background	Updated description of tozorakimab mechanism of action.	Update in line with IB.
Section 2.3.1, Risk Assessment; Section 2.3.3, Overall Benefit: Risk Conclusion	Updated risks and benefits according to latest data.	Update in line with IB.

Section Number and Name	Description of Change	Brief Rationale
Section 3, Table 3, Objectives and Endpoints; Section 8.7 Healthcare Resource Utilisation	Exploratory endpoint of COPD-related HRU streamlined.	Simplification to focus on relevant HRU events.
Section 3, Table 3, Objectives and Endpoints; Section 9.4.2.3 Exploratory Endpoints	Added that analyses for exploratory endpoints may be reported outside of the CSR.	Exploratory endpoints may be reported outside of the CSR.
Section 4.1.1, Enrolment	Noted that medical history and demographics that are to be collected in the predecessor studies may also be collected in PROSPERO as needed.	Clarification.
Section 4.2.5, Participant Input into Design	Text was revised and updated.	To align with predecessor study protocols.
Section 4.3, Justification for Dose	Included data from recent studies.	Update in line with IB.
Section 4.4, End of Study Definition	Clarified definition of the end of study according to EU and FDA requirements.	For consistency and alignment in terms of posting study results.
Section 5.3.4, Tobacco, E-cigarettes, and Vaping; Section 8.2.2, Smoking Status	Text updated to clarify heated tobacco products are discouraged in the former smoker cohort.	Clarification.
Section 6, Study Intervention	Placebo formulation updated.	Details of excipients used in placebo formulation added.
Section 6.1.2, Investigational Product Administration Rescheduling	Amended that if a participant misses 3 or more consecutive doses of IP, the site must discuss with the study physician whether continuation or discontinuation of IP is in their best interests, instead of requiring permanent discontinuation.	Phase 2 programmes in atopic dermatitis and asthma indicate that tozorakimab has a low immunogenic profile. Hence, the risk of re-instating IP after 3 or more missed doses is now considered low.
Section 6.2.2.1, Preparation	Redundant text not applicable to APFS removed.	Clarification.
Section 6.2.2.3, Optional Self-Administration of IP	Added text that one IP kit will be provided at a time for the subsequent remote visit.	Clarification.
Section 6.3.2, Methods for Ensuring Blinding	Statement added regarding unblinding of AstraZeneca for primary reporting.	To further specify methods to ensure blinding and trial integrity.
	Text added to clarify that the site will know if the patient was recruited before or after Q8W arm was completed at the site.	Clarification.

Section Number and Name	Description of Change	Brief Rationale
	A text regarding unblinded IP records was not applicable to PROSPERO and removed.	Clarification.
Section 6.3.3, Methods for Unblinding	Text updated to clarify that information indicating the treatment randomisation for each randomised participant, will be available to the investigator(s) from the RTSM.	Clarification.
Section 6.5, Concomitant Medications	Added cross-reference to Appendix J Country-Specific Requirements for auxiliary medicinal products as applicable to EU CTR.	Amended to adhere to the EU CTR: Regulation (EU) No 536/2014 requirement.
Section 6.5.2 Maintenance Therapy	Updated to clarify which COPD maintenance therapies other than inhalers should be stable during the study. Updated to clarify that COPD inhaled maintenance therapy compliance will be recorded in the source documentation.	Clarification.
Section 6.5.4, Table 5, Allowed, Restricted, and Prohibited Concomitant Medications	Revised allergen immunotherapy classification to restricted from prohibited.	Consistency with medication instructions.
	Text not applicable to the study was removed in non-biologic IPs and investigational biologics other than tozorakimab.	Clarification.
	Exceptions added to prohibition of marketed biologics and classification changed to restricted from prohibited.	Several marketed biologics are accepted as exceptions in the study to ensure participants can receive adequate medical care for comorbid conditions where no interaction with tozorakimab mechanism of action or confounding effect on the study outcomes is expected.
	Updated xanthines details and classification changed to restricted from prohibited.	Clarification on allowed xanthines doses.
	Guidance for inactivated vaccines added.	Clarification.

Section Number and Name	Description of Change	Brief Rationale
Section 7.1, Discontinuation of Investigational Product	Update text to clarify that in case of safety concerns IP administration must be immediately withheld until the final decision is made. Added “Confirmed Hy’s law event” to list of reasons that participants may be discontinued from IP.	Amended to adhere to the EU CTR Regulation (EU) No 536/2014 requirement.
Section 7.1.1, Procedures for Discontinuation of Investigational Product	Updated text to clarify that SAE and ongoing AEs will be collected and followed after the E/D visit.	Clarification.
Section 8.1.3 Spirometry (Pre-and Post-BD)	Updated wording describing the timing of BD dosing for post-BD spirometry relative to pre-BD spirometry.	Clarification.
Section 8.2.1, Physical Examinations, Weight, and Supplemental Oxygen Status	Height assessment was not applicable and removed.	Clarification.
Section 8.2.4, Electrocardiogram	Updated text to clarify that the ECG assessment “is recommended to be” performed before interventions with the participant but must be prior to IP administration.	Clarification.
Section 8.2.5, Echocardiography	Removed requirement to send the image to a central vendor for assessment.	There will no longer be a central read in addition to the local read. This will reduce complexity and burden to site.
	Removed the sentence “Additional parameters may also be evaluated”.	Correction.
Section 8.3.1, Time Period and Frequency for Collecting AE and SAE Information	Text updated to clarify how AE and SAE will be collected in the predecessor and PROSPERO study eCRFs.	Clarification.
Section 8.3.11, Medication Error, Drug Abuse, and Drug Misuse	Added drug abuse and drug misuse definition.	Update required due to CT-3 regulation and corporate safety CAPA.
Section 8.3.13, Device Constituent and Third Party Medical Device Deficiencies	Added text to clarify how third-party medical device deficiencies will be reported.	To adhere to new guidelines.

Section Number and Name	Description of Change	Brief Rationale
Section 8.4, Human Biological Samples	Updated to confirm that all research on biological samples, including possible future research, is subject to national regulations and will only be conducted in a particular country if approved in that country.	Amended to adhere to the EU CTR Regulation (EU) No 536/2014 requirements.
Section 9.3.1, Table 8, Treatment Groups for Analysis	Description for “All placebo” treatment group was updated to remove groups “G” and “I”.	Correction to align with SAP.
Section 9.4.2.1 Primary Endpoint, Handling of Missing Data	The text regarding handling of missing data has been expanded.	To give further specific detail on how the missing data imputation will be performed.
Section 10, Appendix A1, Regulatory, Ethical, and Study Oversight Considerations	Sub-heading updated and included updates of data protection and regulatory reporting requirements for serious breaches.	Amended to adhere to the EU CTR: Regulation (EU) No 536/2014 requirements and global company requirement.
Section 10, Appendix A4, Data Protection	Updated details regarding protection of personal data.	To comply with GDPR.
Section 10, Appendix A7, Data Quality Assurance	Updated information about retention timelines of records and documents and monitoring details and responsibilities.	Amended to adhere to the EU CTR: Regulation (EU) No 536/2014 requirements and global company requirement.
Section 10, Appendix B4, Medication Error, Drug Abuse, and Drug Misuse	Text regarding medication error as relates to AE definition was moved from Drug Misuse subsection to Medication Error subsection.	Correction of error in location.
Section 10, Appendix D6, Laboratory Tests, Section 10, Appendix J, Country-Specific Requirements	Footnote text regarding laboratory tests in the Hy’S Law Lab Kit was removed from Appendix D6 and moved to Appendix J China.	Correction of error in location.
Section 10, Appendix H2 Re-enrolment Procedures for Participants to Reconfirm Study Eligibility	Updated text regarding enrolment delay.	Clarification.
Section 10, Appendix I, Examples of Inhaled Corticosteroids Equivalent Doses	Revised footnote on the use of fixed dose combinations.	To ensure locally approved ICS doses are used.
Section 10, Appendix J, Device Constituent Deficiencies	Removed Appendix J and the Product Complaint Intake Form.	The Product Complaint Intake Form is included in IP Manual.

Section Number and Name	Description of Change	Brief Rationale
Section 10, Appendix J, Country-Specific Requirements	China: added text to clarify handling of ADA samples.	Clarification.
	Text added for Czech Republic-specific requirement that self-administration at home by the participant or caregiver is not applicable. Text added related to Germany-specific requirement of study termination if National Authority has been revoked and clarification that the maximum sum insured can be increased.	Requested update due to agency feedback.
Section 11, References	New references added.	To support updates in multiple sections.

11 REFERENCES

- Adeloye D, Chua S, Lee C, Basquill C, Papana A, Theodoratou E, et al. Global and regional estimates of COPD prevalence: systematic review and meta-analysis. *J Glob Health*. 2015;5(2):020415.
- Allinne J, Scott G, Lim WK, Birchard D, Erjefält JS, Sandén C, et al. IL-33 blockade affects mediators of persistence and exacerbation in a model of chronic airway inflammation. *J Allergy Clin Immunol*. 2019;144(6):1624-37. e10.
- Alqahtani JS, Oyelade T, Aldhahir AM, Goncalves-Mendes R, Alghamdi SM, Miravittles M, et al. Reduction in hospitalised COPD exacerbations during COVID-19: a systemic review and meta-analysis. *PloS One*. 2021;16(8):e0255659.
- Celli BR, MacNee W, ATS/ERS Task Force. Standards for the diagnosis and treatment of patients with COPD: a summary of the ATS/ERS position paper. *Eur Respir J*. 2004;23(6):932-46.
- Cohen ES, Scott IC, Majithiya JB, Rapley L, Kemp BP, England E, et al. Oxidation of the alarmin IL-33 regulates ST2-dependent inflammation. *Nat Commun*. 2015;6:8327.
- Donaldson GC, Seemungal TAR, Bhowmik A, Wedzicha JA. Relationship between exacerbation frequency and lung function decline in chronic obstructive pulmonary disease. *Thorax*. 2002;57(10):847-52.
- Fleming TR. Addressing missing data in clinical trials. *Ann Intern Med*. 2011;154(2):113-7.
- Global Initiative for Chronic Obstructive Lung Disease. Global strategy for the diagnosis, management, and prevention of chronic obstructive pulmonary disease. 2022 Report [cited 17 June 2022]. Available from: https://goldcopd.org/wp-content/uploads/2021/12/GOLD-REPORT-2022-v1.1-22Nov2021_WMV.pdf.
- Gottlow M, Hollis S, Wan R, Hirsch I, Darilay A, Weissfeld L, et al. A Simulation study of a controlled imputation approach for analyzing missing data in recurrent events due to early discontinuations. *PSI Annual Conference* 2015.
- Graham BL, Steenbruggen I, Miller MR, Barjaktarevic IZ, Cooper BG, Hall GL, et al. Standardization of spirometry 2019 update. An official American Thoracic Society and European Respiratory Society technical statement. *Am J Respir Crit Care Med*. 2019;200(8):e70-e88.
- Guarascio AJ, Ray SM, Finch CK, Self TH. The clinical and economic burden of chronic obstructive pulmonary disease in the USA. *Clinicoecon Outcomes Res*. 2013;5:235-45.

Kearley J, Silver JS, Sanden C, Liu Z, Berlin AA, White N, et al. Cigarette smoke silences innate lymphoid cell function and facilitates an exacerbated type I interleukin-33-dependent response to infection. *Immunity*. 2015;42(3):566-79.

Kemp SF, Lockey RF, Simons FE; World Allergy Organization ad hoc Committee on Epinephrine in Anaphylaxis. Epinephrine: the drug of choice for anaphylaxis. A statement of the World Allergy Organization. *Allergy*. 2008;63(8):1061-70.

Kim V, Crapo J, Zhao H, Jones PW, Silverman EK, Comellas A, et al. Comparison between an alternative and the classic definition of chronic bronchitis in COPD. *Ann Am Thorac Soc*. 2015;12(3):332-9.

Kim V, Zhao H, Regan E, Han MK, Make BJ, Crapo JD, et al. The St. George's respiratory questionnaire definition of chronic bronchitis may be a better predictor of COPD exacerbations compared with the classic definition. *Chest*. 2019;156(4):685-95.

Lahousse L, Seys LJM, Joos GF, Franco OH, Stricker BH, Brusselle GG. Epidemiology and impact of chronic bronchitis in chronic obstructive pulmonary disease. *Eur Respir J*. 2017;50(2):1602470.

Leuppi JD, Schuetz P, Bingisser R, Bodmer M, Briel M, Drescher T, et al. Short-term vs conventional glucocorticoid therapy in acute exacerbations of chronic obstructive pulmonary disease: the REDUCE randomized clinical trial. *JAMA*. 2013;309(21):2223-31.

Mannino DM, Braman S. The epidemiology and economics of chronic obstructive pulmonary disease. *Proc Am Thorac Soc*. 2007;4(7):502-6.

Miravitlles M, Ribera A. Understanding the impact of symptoms on the burden of COPD. *Respir Res*. 2017;18(1):67.

Müllerová H, Maskell J, Meeraus WH, Galkin D, Albers FC, Gait C. Characterization of COPD patients treated with inhaled triple therapy containing inhaled corticosteroids [ICS], long-acting beta2-agonists [LABA], and long-acting muscarinic antagonists [LAMA] in the UK. *Am J Respir Crit Care Med*. 2017;195:A4986.

Myers LC, Faridi MK, Hasegawa K, Hanania NA, Carmago Jr CA. The hospital readmissions reduction program and readmissions for chronic obstructive pulmonary disease, 2006-2015. *Ann Am Thorac Soc*. 2020;17(4):450-6.

Okajima Y, Come CE, Nardelli P, Sonavane SK, Yen A, Nath HP, et al. Luminal plugging on chest CT scan: association with lung function, quality of life, and COPD clinical phenotypes. *Chest*. 2020;158(1):121-30.

Parikh RH, Seliger SL, Christenson R, Gottdiener JS, Psaty BM, deFilippi CR. Soluble ST2 for prediction of heart failure and cardiovascular death in an elderly, community-dwelling population. *J Am Heart Assoc.* 2016;5(8):e003188.

Quanjer PH, Stanojevic S, Cole TJ, Baur X, Hall GL, Culver BH, et al. Multi-ethnic reference values for spirometry for the 3-95-yr age range: the global lung function 2012 equations. *Eur Respir J.* 2012;40(6):1324-43.

Rabe KF, Martinez FJ, Ferguson GT, Wang C, Singh D, Wedzicha JA, et al. Triple inhaled therapy at two glucocorticoid doses in moderate-to-very-severe COPD. *N Engl J Med.* 2020;383(1):35-48.

Rabe KF, Celli BR, Wechsler ME, Abdulai RM, Luo X, Boomsma MM, et al. Safety and efficacy of itepekimab in patients with moderate-to-severe COPD: a genetic association study and randomised, double-blind, phase 2a trial. *Lancet Respir Med.* 2021;9(11):1288-98.

Radicioni G, Ceppe A, Ford AA, Alexis NE, Barr RG, Bleecker ER, et al. Airway mucin MUC5AC and MUC5B concentrations and the initiation and progression of chronic obstructive pulmonary disease: an analysis of the SPIROMICS cohort. *Lancet Respir Med.* 2021;9(11):1241-54.

Ratitch B, O’Kelly M, Tosiello R. Missing data in clinical trials: from clinical assumptions to statistical analysis using pattern mixture models. *Pharm Stat.* 2013;12(6):337-47.

Ritchie AI, Wedzicha JA. Definition, causes, pathogenesis, and consequences of chronic obstructive pulmonary disease exacerbations. *Clin Chest Med.* 2020;41(3):421-38.

Sampson HA, Munoz-Furlong A, Campbell RL, Adkinson NF Jr, Bock SA, Branum A, et al. Second symposium on the definition and management of anaphylaxis: summary report--second National Institute of Allergy and Infectious Disease/Food Allergy and Anaphylaxis Network symposium. *J Allergy Clin Immunol.* 2006;117(2):391-7.

Schmitz J, Owyang A, Oldham E, Song Y, Murphy E, McClanahan TK, et al. IL-33, an interleukin-1-like cytokine that signals via the IL-1 receptor-related protein ST2 and induces T helper type 2-associated cytokines. *Immunity.* 2005;23(5):479-90.

Scott I, England E, Rees D, Erngren T, Huntington CC, Houslay K et al. Tozorakimab: a dual-pharmacology anti-IL-33 antibody that inhibits IL-33 signalling via ST2 and RAGE/EGFR to reduce inflammation and epithelial dysfunction. *Eur Respiratory Soc*; 2022.

Seemungal TA, Donaldson GC, Paul EA, Bestall JC, Jeffries DJ, Wedzicha JA. Effect of exacerbation on quality of life in patients with chronic obstructive pulmonary disease. *Am J Respir Crit Care Med.* 1998;157(5 Pt 1):1418-22.

Smith D, Helgason H, Sulem P, Bjornsdottir US, Lim AC, Sveinbjornsson G, et al. A rare IL33 loss-of-function mutation reduces blood eosinophil counts and protects from asthma. *PLoS Genet.* 2017;13(3):e1006659.

Soler-Cataluna JJ, Martinez-Garcia MA, Roman Sanchez P, Salcedo E, Navarro M, Ochando R. Severe acute exacerbations and mortality in patients with chronic obstructive pulmonary disease. *Thorax.* 2005;60(11):925-31.

Stanczak MA, Sanin DE, Apostolova P, Nerz G, Lampaki D, Hofmann M, et al. IL-33 expression in response to SARS-CoV-2 correlates with seropositivity in COVID-19 convalescent individuals. *Nat Commun.* 2021;12(1):2133.

Stone A. The application of bespoke spending functions in group-sequential designs and the effect of delayed treatment switching in survival trials. *Pharm Stat.* 2010;9(2):151-61.

Stott-Miller M, Mullerova H, Miller B, Tabberer M, El Baou C, Keeley T, et al. Defining chronic mucus hypersecretion using the CAT in the SPIROMICS cohort. *Int J Chron Obstruct Pulmon Dis.* 2020;15:2467-76.

Strickson S, Houslay K, Negri V, Ohne Y, Ottosson T, Dodd R, et al. Oxidised IL-33 signals via RAGE/EGFR to drive a COPD-associated phenotype. European Respiratory Society. Oral presentation. c2022 [cited 2023, July, 27]. Available from: <https://www.ers-education.org/lr/show-details/?idP=269789>

Suissa S, Dell'Aniello S, Ernst P. Long-term natural history of chronic obstructive pulmonary disease: severe exacerbations and mortality. *Thorax.* 2012;67(11):957-63.

Toy EL, Gallagher KF, Stanley EL, Swensen AR, Duh MS. The economic impact of exacerbations of chronic obstructive pulmonary disease and exacerbation definition: a review. *COPD.* 2010;7(3):214-28.

van Manen JG, Bindels PJ, Dekker FW, Bottema BJ, van der Zee JS, Ijzermans CJ, et al. The influence of COPD on health-related quality of life independent of the influence of comorbidity. *J Clin Epidemiol.* 2003;56(12):1177-84.

Vestbo J, Papi A, Corradi M, Blazhko V, Montagna I, Francisco C, et al. Single inhaler extrafine triple therapy versus long-acting muscarinic antagonist therapy for chronic obstructive pulmonary disease (TRINITY): a double-blind, parallel group, randomised controlled trial. *Lancet.* 2017;389(10082):1919-29.

Wan R, Hirsch I, Gottlow M, Hollis S, Darilay A, Weissfeld L, et al. Controlled imputation approach for analysing missing data in recurrent events due to early discontinuations. DIA/FDA Statistics Forum 2015.

Wedzicha JA, Seemungal TA. COPD exacerbations: defining their cause and prevention. *Lancet*. 2007;370(9589):786-96.

Wedzicha JA, Brill SE, Allinson JP, Donaldson GC. Mechanisms and impact of the frequent exacerbator phenotype in chronic obstructive pulmonary disease. *BMC Med*. 2013;11:181.

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